

**TANIM: AN INTELLIGENT AGRICULTURAL FRAMEWORK
FOR CROP RECOMMENDATION AND SOIL HEALTH
MONITORING**

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**CHRISTIAN JAMES E. APUYA
RAY SIMON L. BANTACULO
ALBERT JOHN DIAZ
ALEX GUARDAQUIVIL JR.
ROBERT ROY P. SALVO**

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Agriculture is important in the Philippines since it serves as the foundation of food security, livelihood, and rural development. Traditional soil management practices in the Philippines, such as those endorsed by the Department of Agriculture (DA), rely on generalized soil health monitoring and fertilizer recommendations using manual tools like the Minus-One Element Technique (MOET) and the Rapid Soil Test Device (RST). These approaches often fail to address field-specific variability, crop diversity, and seasonal dynamics, leading to suboptimal resource utilization, as reported by the Department of Agriculture Region 10 (2023). Historically, soil testing requires farmers to transport soil samples to nearby laboratories located at Department of Agriculture offices, a process that is often expensive, time-consuming, and inaccessible in remote areas, as highlighted by Pagaduan, J. L. (2025).

In relation to this, the Department of Agriculture – Regional Field Office 10 (DA–RFO 10 or Aggie 10) conducted a soil sampling drive for rice and corn production areas in the municipalities of Sugbongcogon, Binuangan,

Magsaysay, and Gingoog City from March 26–28, 2025. Aggie 10 aimed to complete soil sampling and testing in priority sites across Region 10 by May 2025 to align with the wet cropping calendar. Despite these initiatives, farmers still largely depend on traditional and observational methods. The study by Samaniego, A. L. and Gallego, M. (2024) reported that only 1.03% of farmers use soil-test kits and another 1.03% utilize government laboratory services, while the remaining 97.94% rely on ocular observation or “eyeballing” of soil and crop conditions. Furthermore, crop selection decisions are often based on historical practices rather than data-driven insights, limiting farmers’ ability to adapt to changing soil and climate conditions (Sarma et al., 2024). With advancements in technology such as the Internet of Things (IoT), there is significant potential to improve crop yields and reduce agricultural losses.

Soil health is a fundamental pillar of sustainable agriculture, underpinning crop productivity, food security, and environmental resilience. According to Lal, R. (2024), degraded soils characterized by nutrient deficiencies, imbalanced pH levels, or reduced organic matter lead to lower yields, increased susceptibility to pests, and economic losses for farmers. In Region 10 (Northern Mindanao, Philippines), a major agricultural hub producing rice, corn, banana, and cacao, maintaining optimal levels of nitrogen (N), phosphorus (P), potassium (K), and a soil pH range of 6.0–7.5 is essential for nutrient

availability and healthy crop growth, as discussed by Morgan and Connolly (2013) and Sondhiya and Singh (2024). Data from the Department of Agriculture Region 10 (2024) indicate that acidic soils in Bukidnon restrict phosphorus uptake, while nitrogen deficiencies in rice fields can reduce yields by up to 20%. Moreover, proper crop selection based on soil condition and season, combined with precise fertilizer application, is crucial to maximizing productivity and minimizing environmental impacts such as nutrient runoff (Patil et al., 2023).

Globally, agricultural losses caused by soil degradation, inefficient nutrient management, and climate variability pose a serious threat to food security and economic stability. The Food and Agriculture Organization (FAO, 2023) estimates that approximately 33% of the world's soils are moderately to highly degraded, resulting in annual crop yield losses of up to 25% in some regions. The Economics of Land Degradation (ELD) initiative (2015) further reports that land degradation leads to global economic losses of up to USD 10.6 trillion per year, equivalent to nearly 17% of global Gross Domestic Product (GDP), due to reduced agricultural productivity, ecosystem service loss, and increased vulnerability to climate extremes. These losses are exacerbated by improper fertilizer use, where over-application leads to nutrient runoff and environmental damage, while under-application results in nutrient deficiencies and yield reduction.

In the Philippine context, these global challenges are strongly reflected at the national and regional levels. The Department of Agriculture (2024) reported that approximately 80% of agricultural soils in the Philippines are deteriorated, significantly limiting crop productivity and fertilizer efficiency. Climate-related disturbances such as typhoons, droughts, and flooding further contribute to agricultural losses, with the Department of Agriculture estimating P785 million in crop damage in late 2024 alone due to successive weather events. In major agricultural regions such as Northern Mindanao (Region 10), nutrient-depleted and acidic soils, combined with non-site-specific fertilizer practices, contribute to yield reductions in staple crops such as rice and corn, increasing farmers' production costs and reducing income stability. These persistent losses underscore the urgent need for data-driven, location-specific, and climate-aware agricultural decision-support systems.

Advancements in Internet of Things (IoT) technologies and machine learning (ML) offer promising solutions for precision agriculture. IoT-based sensors, including NPK, pH, moisture, and temperature probes, enable continuous soil health monitoring, while machine learning algorithms analyze large and complex datasets to generate data-driven insights and recommendations (Kumar et al., 2024 & Lavanya et al., 2024). Modern farming systems leverage these technologies to optimize crop production, improve fertilizer efficiency,

and reduce waste, ultimately enhancing yield and sustainability (Elbasi et al., 2023). By integrating Internet of Things (IoT) sensor data with historical records, and agricultural zoning data, intelligent systems can provide crop recommendations tailored to the agricultural conditions of Region 10.

This study proposed a Soil Health Monitoring and Recommendation System that leveraged IoT sensors and machine learning techniques to provide recommendations for crop planting and fertilizer application in Bukidnon. Targeting key crops such as Ampalaya, Beans (Soybean), Cassava, Corn, Eggplant, Sorghum, Squash, Sweet Pepper, Tobacco, and Tomato, the system integrated soil health parameters (NPK, pH, moisture, humidity, and temperature), soil and crop data from the Department of Agriculture- Region 10, and additional online agricultural resources to predict suitable crops and recommend fertilizer application. By addressing field-specific variability and seasonal dynamics, the proposed system aimed to enhance agricultural productivity, reduce resource waste, and promote sustainable farming practices. Aligned with the Sustainable Development Goals (SDGs) 2 (Zero Hunger), 12 (Responsible Consumption and Production), and 15 (Life on Land), this research contributed to food security, environmental conservation, and sustainable agricultural development in the Philippines.

1.2 Statement of the Problem

Traditional soil management and crop selection practices in Region 10 remain largely generalized and manual, limiting their effectiveness in addressing field-specific agricultural conditions. Farmers primarily rely on standard guidelines and conventional assessment tools provided by the Department of Agriculture, which do not sufficiently consider variations in soil properties, crop requirements, and seasonal environmental changes. This lack of precision often results in inefficient fertilizer application, suboptimal crop selection, reduced yields, increased production costs, and gradual soil degradation (Lal, R., 2023). These issues are particularly evident in Bukidnon, where diverse crops such as Ampalaya, Beans (Soybean), Cassava, Corn, Eggplant, Sorghum, Squash, Sweet Pepper, Tobacco, and Tomato are cultivated under varying soil conditions.

Despite the importance of soil health information in agricultural decision-making, several problems persist that hinder the effective use of technology and data-driven approaches in the region. These problems include:

1. Inconsistent and limited availability of accurate, real-time soil health data (pH, NPK, salinity, moisture, and temperature) across different farming zones in Bukidnon, making it difficult to assess actual field conditions.

2. The absence of an integrated decision-support mechanism that analyzes soil health data, resulting in unreliable or generalized crop suitability and fertilizer recommendation application.
3. Limited access to user-friendly digital platforms that can translate complex agricultural data into clear, practical recommendations understandable to farmers in Bukidnon.
4. Uncertainty regarding the effectiveness and scalability of existing IoT-based agricultural monitoring systems in supporting sustainable farming practices across multiple farms and locations in Bukidnon.

1.3 Objectives of the Study

The main objective of this study is to develop an IoT-based crop recommender system integrated with machine learning for crop recommendation based on soil health parameters (pH, NPK, salinity, moisture, and temperature) for farmers, specifically the following:

1. Implement a machine learning model to predict suitable crops and recommend fertilizers application for farming practices in Bukidnon based on soil health data.
2. Develop a mobile application to monitor soil health, crop predictions, and fertilizer schedules for the farmers.

3. Develop a website application to monitor soil health insights for different zoning of farms, farmers data and information on their farming timeline.
4. Evaluate the IoT-based crop recommender system's accuracy and usability in agricultural settings in Bukidnon.

1.4 Significance of the Study

The significance of this study lies in its potential to transform agricultural practices in Bukidnon by developing an integrated Internet of Things (IoT) and machine learning-based system for soil health monitoring, crop recommendation, and recommend fertilizer application. Traditional soil testing methods provided by the Department of Agriculture are often underutilized by farmers due to inaccessibility and time-consuming procedures. As a result, many farmers rely on ocular observations, which may lead to poor crop selection and increased production risks (Samaniego, A. L. & Gallego, M., 2024). This research addresses critical limitations of traditional soil management practices, particularly the reliance on generalized recommendations and the lack of zone-specific insights, which frequently result in inefficient resource utilization.

By providing precise soil health data and data-driven crop and fertilizer recommendations, the proposed system enables farmers to optimize crop yields

while reducing unnecessary fertilizer expenses. Furthermore, the system will be able to provide the Department of Agriculture a Geographic Information System (GIS) zoning maps for farms across Bukidnon for better monitoring and supports sustainable farming practices aligned with environmental conservation goals. The study contributes to the advancement of precision agriculture by offering a scalable IoT and machine learning framework that promotes data-driven decision-making in farming. Ultimately, this research has the potential to improve agricultural productivity, environmental sustainability, and food security in Bukidnon, in support of Sustainable Development Goals (SDGs) 2 (Zero Hunger), 12 (Responsible Consumption and Production), and 15 (Life on Land).

Farmers. The system will significantly assist farmers by providing accessible soil health monitoring data, data-driven crop recommendations, and optimized fertilizer usage. This enables farmers to reduce production risks, improve decision-making, and maximize crop yields while minimizing resource waste.

Department of Agriculture. The study will support the Department of Agriculture in Region 10 by offering timely soil health insights that can improve the accuracy of fertilizer distribution programs and enhance agricultural zoning strategies. The system may also serve as a tool for planning

interventions and monitoring soil health trends to promote sustainable agricultural development.

Agriculturists. This research benefits agriculturists by serving as a reliable reference for assessing soil conditions and crop suitability. It enhances their ability to provide evidence-based recommendations to farmers, reducing dependence on manual and generalized assessment methods.

Future Researchers. The study will serve as a foundation for future research in smart farming and precision agriculture. It opens opportunities for further development, such as integrating remote sensing data, advanced machine learning models, and different agricultural techniques. Future researchers may also adapt and expand the system to other regions or crop categories, making it a scalable model for broader agricultural applications.

1.5 Scope and Limitations

This study focuses on agricultural areas located in Budkinon, Philippines, where selected crops—namely Ampalaya, Beans (Soybean), Cassava, Corn, Eggplant, Sorghum, Squash, Sweet Pepper, Tobacco, and Tomato—are commonly cultivated. These crops are the primary focus of the system’s crop recommendation and fertilizer application process. The system is designed to analyze soil health parameters including (1) nitrogen, (2) phosphorus, (3)

potassium, (4) soil moisture, (5) temperature, (6) pH level, and (7) soil salinity. These parameters are combined with the soil and crop data from the Department of Agriculture - Region 10, forming the basis for crop recommendation and fertilizer management.

The study is geographically limited to selected farms across Bukidnon, ensuring relevance to local soil health characteristics. The system emphasizes the integration of IoT-based soil sensor and machine learning techniques for data analysis and recommendation generation. However, the soil sensor used in this study is limited to seven (7) soil parameters and does not include micronutrients or any soil organic matter. Additionally, real-time continuous soil monitoring is not implemented; instead, the system relies on periodic data collection and analysis. These limitations may affect the granularity of recommendations but are sufficient to demonstrate the feasibility and effectiveness of the proposed system within the defined scope.

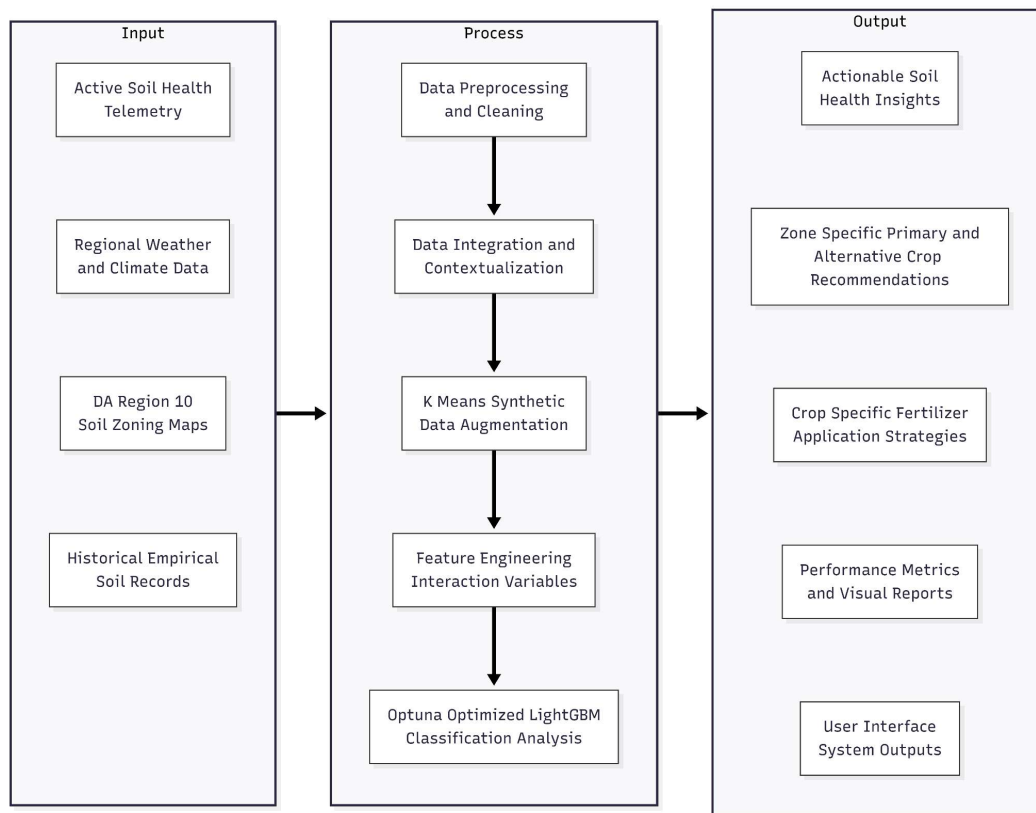
1.6 Conceptual Framework

In this section, the study's conceptual framework is presented through an Input-Process-Output (IPO) model, which outlines the structure of the soil health monitoring, crop and fertilizer recommendation system. The framework integrates multiple components, each playing a vital role in developing an

efficient, data-driven solution for sustainable agriculture.

Figure 1.1

IPO Model



The TANIM system operates upon an Input Process Output model explicitly engineered to generate localized crop and fertilizer recommendations for farmers across Region 10, Philippines. The Input phase captures active soil health telemetry, such as Organic Matter serving as a Nitrogen proxy, Phosphorus, Potassium, soil pH, moisture, salinity, and temperature, gathered continuously via an ESP32 microcontroller and its attached Internet

of Things sensor network. To establish a comprehensive agricultural baseline, these localized sensor inputs run parallel to regional weather and climate data, Department of Agriculture Region 10 soil zoning maps, and historical empirical soil records.

Transitioning into the Process layer, these raw inputs undergo a rigorous and multistage computational pipeline. The acquired telemetry is first subjected to programmatic data preprocessing and cleaning protocols, systematically removing empty reads and sensor anomalies to guarantee absolute diagnostic consistency. Within the data integration and contextualization phase, severe historical imbalances present in the agricultural dataset are actively neutralized via K Means synthetic data augmentation, ensuring all crop classes hold equal representational weight. Furthermore, a specialized feature engineering protocol derives crucial agricultural interaction variables, such as pH controlled nutrient availability, dynamically from the raw data. This deeply contextualized and balanced matrix is subsequently evaluated by an Optuna optimized Light Gradient Boosting Machine framework, which analyzes complex soil condition patterns to precisely isolate and predict biological crop suitability.

Governed by this classification analysis, the Output phase is generated. The system delivers actionable soil health insights alongside high confidence

and zone specific primary and alternative crop recommendations. These predictive classifications bridge seamlessly into secondary subsystems that formulate appropriate and crop specific fertilizer application strategies. The operational validity of the entire pipeline is continuously tracked via rigorous performance metrics, including analytical confusion matrices, prediction error monitoring, and F1 score dashboards, which are compiled natively into visual reports and delivered through an accessible user interface. Ultimately, this seamless and data driven cycle equips agricultural operators with the empirical insight required to optimize resource utilization, maximize productivity, and enforce sustainable farming practices.

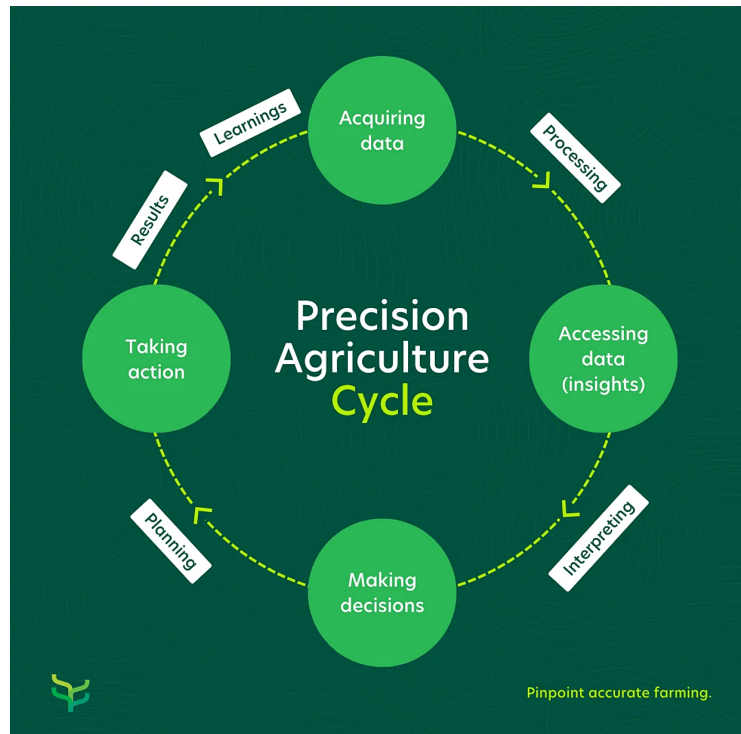
1.7 Theoretical Framework

This study utilized Precision Agriculture Theory, a strategy that gathers, processes data driven insights to optimize farming practices based on conditions rather than relying on generalized farming practices recommendations, which is supported by the study of Pierre C. Robert states that an integrated farming approach designed to enhance long-term production efficiency, productivity, and profitability while minimizing environmental impacts. It involves using technology and data to optimize farming practices at a site-specific level. The theory highlights the importance of applying the right input, at the right

place, at the right time, and in the right amount to maximize productivity while conserving resources.

With the integration of IoT sensors which will provide measurements of soil health parameters the precision agriculture theory will be fully operational. The soil parameters will serve as a foundation to generate actionable recommendations for farmers. Moreover, machine learning boosting algorithms will be used to analyze soil and environmental patterns to predict crop suitability and yield potential.

Figure 1.2 explains the Precision Agriculture Cycle on how farming decisions become data-driven. It starts with acquiring data through IoT sensors and other tools that measure soil and environmental conditions. This data is then processed into insights to understand soil fertility and crop needs. Using these insights, decisions are made with the help of Machine Learning (ML) for prediction. The next step is taking action, where farmers apply precise fertilizers, choose suitable crops, or adjust irrigation. Finally, the results and learnings feed back into the system, ensuring continuous improvement in productivity, efficiency, and sustainability.

Figure 1.2*Precision Agriculture Cycle*

By combining Machine Learning’s predictive capability, the study applies the principles of precision agriculture in a holistic manner. This ensures that recommendations for crop selection and fertilizer application are data-driven, addressing the variability of soils in Bukidnon. Thus, Precision Agriculture Theory serves as the guiding lens that connects IoT-based data collection and machine learning prediction to support decision-making, ultimately promoting sustainable and efficient farming practices.

1.8 Definition of Terms

This study defined the following terms that are mentioned in this study.

Agriculture The science, art, and practice of cultivating plants and livestock, serving as the foundation for food security, livelihoods, and rural development in regions like the Philippines.

Agricultural Zoning The practice of designating specific areas for agricultural use to conserve farmland, prevent urban conflicts, and promote orderly rural growth, often including minimum lot sizes and restrictions on non-farm activities.

Analytic Hierarchy Process (AHP) A structured decision-making technique that organizes complex problems into a hierarchy of goals, criteria, and alternatives, using pairwise comparisons to derive priorities and rank options.

Boosting Algorithms A family of machine learning techniques that convert weak learners into strong ones by iteratively training models to correct errors from previous ones, commonly used in regression and classification tasks.

Crops Plants or plant products grown and harvested extensively for profit

or subsistence, categorized into food, feed, fiber, oil, ornamental, and industrial types based on their uses.

Fertilizer A natural or synthetic substance added to soil to supply essential nutrients like nitrogen, phosphorus, and potassium, enhancing plant growth and productivity in agriculture.

Internet of Things (IoT) in Agriculture The integration of connected sensors, devices, and networks in farming to monitor soil, weather, crops, and livestock, enabling data-driven decisions for improved efficiency and sustainability.

Machine Learning A subset of artificial intelligence that enables computers to learn from data and improve performance on tasks without explicit programming, often using algorithms to identify patterns and make predictions.

Multi-cropping The practice of growing two or more crops on the same piece of land in a single year, either sequentially or simultaneously, to enhance productivity, diversify income, and improve soil health.

NPK The abbreviation for the three primary macronutrients in soil and fertilizers: nitrogen (N), phosphorus (P), and potassium (K), essential for plant growth, development, and overall health.

Precision Agriculture A farming management strategy that uses data from sensors, GPS, and analytics to optimize inputs like water and fertilizers, responding to variability in crops for improved efficiency and sustainability.

Soil Health The continued capacity of soil to function as a vital living ecosystem that sustains plants, animals, and humans, encompassing biological, physical, and chemical properties.

Soil pH A measure of soil acidity or alkalinity on a scale from 0 to 14, indicating hydrogen ion concentration, which influences nutrient availability and plant growth.

Soil Salinity The accumulation of soluble salts in soil that can impair plant growth, often measured by electrical conductivity, with levels above 4 dS/m considered saline.

Sustainable Agriculture Farming practices that meet current food needs while preserving resources for future generations, balancing environmental health, economic viability, and social equity.

CHAPTER 2

REVIEW OF RELATED LITERATURE

This chapter presents some preliminary concepts and known results that are needed in this study.

2.1 Soil Health in Northern Mindanao

The literature underscores the critical role of soil health in sustainable agriculture and the limitations of traditional methods. Lal (2024) emphasizes soil degradation's impact on global food security, while Morgan and Connolly (2013) detail nutrient uptake mechanisms and challenges in imbalanced soils. In the Philippine context, Department of Agriculture Region X (2023) highlights regional soil issues, such as acidity in Bukidnon and nutrient shortages in rice fields, alongside generalized zoning maps. According to the study of Arlyn and Carmelito (2024), which assessed soil fertility in irrigated lowland areas across Northern Mindanao, revealing strongly to moderately acidic soils (pH 4.5-6.0) with variable organic matter content, peaking in Bukidnon at up to 3.5%. The study employed soil sampling and laboratory analysis to monitor key parameters like pH, and NPK, highlighting deficiencies in phosphorus and potassium that limit rice yields. This underscores the need for regular soil

health monitoring to track degradation trends and inform site-specific interventions in Region 10's rice-dominated landscapes; this is supported by the study of ? which uses several factors of crops and uses those inputs for random forest algorithms.

Crop production trends in Northern Mindanao, note that fertilizer inefficiencies in Bukidnon and Misamis Oriental contribute to yield gaps in corn and rice, with average applications exceeding recommendations by 15-20% based on the study of Canatoy (2018). The review recommends precision fertilizer strategies, such as variable-rate applications, to optimize nutrient use efficiency and minimize environmental impacts in Region 10's diverse agroecosystems. On the other hand, Calalang et al. (2014) reviewed soils and crop suitability in Bukidnon Highlands, highlighting how volcanic-derived soils support high-value crops like pineapple and coffee, but require lime amendments for acidity management to sustain productivity. The study links soil properties to crop performance, advocating for diversified cropping systems to enhance resilience in Northern Mindanao.

2.1.1 Zoning in Agriculture

In agriculture, zoning is the process of dividing land into management areas based on topographic features, soil fertility, and climate—all of which have

a direct impact on crop suitability. Zoning has been used in several areas of the Philippines to maximize land use and improve site-specific agricultural recommendations. Watson et al. (2025) used the Food and Agriculture Organization of the United Nations (FAO) land evaluation to characterize and categorize soil series in Ilocos Norte and perform crop suitability assessments for rice, maize, garlic, onion, and tomato. According to their research, morphological characteristics and soil physicochemical characteristics are important factors in determining whether an area is considered highly suitable (S1), moderately suitable (S2), or marginally suitable (S3).

The use of zoning for particular crops is further illustrated by regional studies. In Samar Province, Poliquit et al. (2019a) evaluated crop suitability and soil fertility, identifying nutrient deficiencies like low levels of organic matter, phosphorus, and nitrogen as productivity-limiting factors. The significance of mapping local soil fertility for focused input management was highlighted by their findings. Similarly, Bato (2018) created a Geographic Information System (GIS) based banana cultivation suitability map that illustrates the spatial variations in suitability across the Philippine landscape. Slope, elevation, rainfall, and soil properties were all integrated in the study to create a visual guide that farmers and policymakers could use to identify the best places to grow bananas.

These zoning studies together make a strong case for creating smart frameworks that combine soil and crop suitability with cutting-edge technologies to give site-specific crop and fertilizer recommendations.

2.2 Machine Learning

Machine learning (ML) is becoming very useful in agriculture, it enables researchers and farmers to process huge amounts of data. It provides predictive insights related to soil health, crop recommendation, fertilizer and so on. Unlike traditional methods that rely on generalized formulas or manual observations, ML can accommodate complex and internal relationships between the soil, climate and crop parameters. This will enable them to make site-specific recommendations which will increase productivity, reduce costs and improve sustainable farming practices.

As noted by Islam et al. (2023), ML enabled systems are flexible. They not only analyze nutrient levels of soil but also suggest crops that can be grown in various environments. They do this with more accuracy and precision than manual assessments. According to Parganiha and Verma (2024), optimized ML models like LightGBM can yield predictive accuracies as high as 97% for crop yield forecasting. These models are easily scalable for larger agricultural zones like Northern Mindanao.

2.2.1 Random Forest

Random Forest (RF) is an ensemble algorithm that constructs multiple decision trees and aggregates their outputs through majority voting (classification) or averaging (regression). Incorporating bootstrapped data subsets and randomly selected features minimizes overfitting and enhances the RF model's resilience to the inaccurate data typical of many agricultural applications. This is useful in examining soils, where data quality is often compromised due to sparse sampling and other uncontrolled environmental factors.

In predicting fertilizer amounts and estimating crop inputs, ? demonstrated its capability in capturing nonlinear relationships integrated among soil parameters and environmental factors. The study found that RF outperformed linear models in terms of accuracy, showing its adaptability across diverse farming conditions. Islam et al. (2023) integrated RF into a crop recommendation framework, successfully suggesting suitable crops based on soil nutrient levels and weather data.

2.2.2 Boosting Algorithms

With the current technologies, Islam et al. (2023) developed an ML-enabled system for soil nutrient monitoring, and crop recommendations, evaluated through field experiments to enhance productivity and sustainability.

This approach supports multi-crop applications relevant to Region 10. Parganiha and Verma (2024) proposed an optimized LightGBM model for soil analysis and crop yield prediction, incorporating data preprocessing, feature selection, and hyperparameter tuning, achieving up to 97% predictive accuracy. LightGBM’s efficiency, as noted by Data Overload (2024), makes it ideal for large-scale agricultural tasks in Region 10.

Table 1: Parameters for Different Gradient Boosting Algorithms (*Salvador (2024)*)

Algorithm	Parameter	Value
Adaptive Boosting (AdaBoost)	learning_rate	0.5
	n_estimators	200
Cat Boosting (CatBoost)	depth	4
	iterations	300
	iterations	300
Gradient Boosting Machine (GBM)	learning_rate	0.3
	max_depth	3
	n_estimators	300
Light Gradient Boosting Machine (LightGBM)	learning_rate	0.1
	n_estimators	200
	num_leaves	31
Extreme Gradient Boosting (XGBoost)	learning_rate	0.3
	max_depth	3
	n_estimators	300

Table 1 outlines the hyperparameters used for training the different gradient boosting algorithms considered in this study. Each model was configured

with commonly used parameter settings based on literature benchmarks and preliminary tuning to balance performance and computational efficiency. For AdaBoost, the number of estimators was set to 200 with a relatively high learning rate of 0.5 to speed up convergence. CatBoost employed a maximum tree depth of 4, 300 boosting iterations, and a learning rate of 0.3. This reflects its ability to handle categorical features effectively. The Gradient Boosting Machine was configured with a learning rate of 0.3 and a tree depth of 3, alongside 300 estimators to control complexity while maintaining accuracy. LightGBM used a lower learning rate of 0.1 with 200 estimators and 31 leaves, capitalizing on its efficiency in handling large datasets. Lastly, XGBoost was set with 300 estimators, a learning rate of 0.3, and a maximum depth of 3, representing a balanced configuration widely adopted in classification tasks.

2.2.3 Decision Tree Analysis

Decision tree analysis is a supervised machine learning approach that uses a tree-like model of decisions and their possible consequences to classify or predict outcomes. It is widely applied in agriculture because of its simplicity, interpretability, and ability to handle both categorical and numerical data. By splitting data into smaller subsets based on key attributes such as soil nutrients, moisture, pH, and climatic conditions, decision trees can provide

farmers with clear and actionable recommendations.

A study of Lavanya et al. (2024) employed Gradient Boosting Decision Trees for multi-crop fertilizer recommendation, integrating soil and climate data, and reported 95% accuracy. This highlights the method's strength in managing Region 10's diverse cropping systems where multiple environmental factors interact.

Similarly, Sudha and Hooda (2022). examined the applicability of decision tree algorithms in agriculture for classification tasks, emphasizing their effectiveness in handling diverse datasets related to crop type, soil fertility, and environmental conditions. Their study concluded that decision trees not only enhance prediction accuracy but also improve the interpretability of agricultural data, enabling researchers and farmers to make evidence-based decisions. This demonstrates that decision tree-based models are reliable tools for optimizing fertilizer management, predicting yields, and supporting sustainable farming practices.

2.3 Crop and Fertilizer Recommendation Systems

In the Philippines, crop and fertilizer recommendations remain very general and imprecise. According to Samaniego and Gallego (2024), it was found that as much as 97.94% of farmers rely on ocular observations or visual

inspection of plants and soil, while only 1.03% use soil-test kits or government laboratories. This reliance on manual methods ends up either over or under-application of fertilizer, contributing to low yields and unnecessary costs. Farmers typically follow broad guidelines or personal judgement in fertilizer management, but these approaches lack the precision required to account for variations in soil health, nutrient level, and crop demands.

To address these inefficiencies, several studies have explored machine learning and data-driven approaches for fertilizer optimization. Sani et al. (2023) employed Random Forest algorithms to predict fertilizer requirements with various soil and crop inputs, achieving high accuracy. Lavanya et al. (2024) demonstrated Gradient Boosting Decision Trees (GBDT) could yield multi-crop fertilizer recommendations with an accuracy of 95% by integrating the soil and climate data. Canatoy (2018) also recommended precision fertilizers, like using different amounts of fertilizers based on location, to avoid nutrient waste and environmental impact. These findings demonstrate the potential of ML models to provide more efficient and tailored fertilizer recommendations compared to traditional methods.

Crop recommendation systems are equally important for optimizing agricultural practices. Islam et al. (2023) developed a machine learning enabled soil monitoring and crop recommendation system that uses an an-

alyzes soil nutrient levels and environmental conditions to suggest suitable crops across multiple environments. The study highlighted the adaptability of ML in multi-cropping systems, showing that intelligent algorithms can go beyond input optimization to guide farmers in selecting crops that align with specific soil and climatic conditions.

2.4 Soil Health Sensor in Agriculture

Soil sensors play a critical role in modern agriculture as they provide accurate and real-time data about soil conditions. By measuring key parameters such as soil moisture, temperature, pH, salinity, and nutrient levels, these sensors help farmers assess overall soil health and make informed decisions for crop management. This is supported by the study of Yin et al. (2021), in which they highlighted the four aspects that will be the main focuses for future soil sensing. First is to improve the sensing performance (e.g., sensitivity and specificity) and reliability for key soil parameters, with little interference from background noise; Second is to develop a sufficient low-power consumption Wireless Sensor Network (WSN) with powerful data processing and long-range wireless communication capability; Third is to develop versatile soil sensing platforms that can be distributed in large-scale to collect real-time soil microenvironment data continuously; and the Fourth is to develop self-powered

or power-independent sensors and sensing platforms that are low-cost, reliable, and maintenance free.

The use of soil health sensors enables precision agriculture, where inputs like water, fertilizers, and pesticides can be applied more efficiently, reducing waste and minimizing environmental impact. In addition, continuous monitoring of soil health through sensors helps in early detection of soil degradation issues such as nutrient depletion, salinization, or compaction. This allows for timely interventions that improve soil productivity and sustainability. The integration of soil sensors with Internet of Things (IoT) platforms further enhances their effectiveness by enabling wireless data transmission, cloud-based analytics, and automated decision-making systems for smart farming.

Numerous studies in the field of agriculture integrate modern advancements like Internet of Things (IoT), it is supported by the study of Sondhiya and Singh (2024), which explores Internet of Things (IoT) and Machine Learning (ML) for soil monitoring in Zea mays (corn) in Kibangay, Lantapan, Bukidnon, demonstrating improved nutrient management. The study utilized NPK and pH sensors with an ESP32 microcontroller and LoRaWAN, achieving 95% agreement with the Department of Agriculture's Soil Test Kit. However, its single-crop focus limits applicability to Region 10's diverse crops like rice, banana, and cacao. Moreover, the study of Kumar et al. (2024a) developed an

IoT-based soil monitoring system using Arduino-based NPK and moisture sensors for rice and maize, achieving 95% nutrient detection accuracy, validating the potential of IoT for monitoring.

2.5 Synthesis

The reviewed studies provide comprehensive insights into soil health, crop productivity, and machine learning applications in agriculture. Research gained in-depth understanding of the soil health challenges in Northern Mindanao, particularly with issues like soil acidity in Bukidnon and nutrient deficiencies in rice fields. Regional assessments, including those from the Department of Agriculture Region 10 (2023, 2024) and the work of Temario and Lapoot (2024), confirmed the variability of soil properties and their direct impact on yield performance. These findings are consistent with earlier insights from Canatoy (2018), who noted the inefficiencies in fertilizer applications, and from Calalang et al. (2014), who linked volcanic-derived soils to both opportunities and limitations for high-value crops. Together, these studies stress the importance of site-specific soil health monitoring and precision interventions to counteract degradation and sustain agricultural productivity. However, most of these studies remain limited to general recommendations or crop-specific assessments, leaving a gap in developing integrated, data-driven

systems tailored to multiple crops in Northern Mindanao.

Zoning has also emerged as a crucial strategy for sustainable land use. Efforts by Watson et al. (2025) using FAO land evaluation in Ilocos Norte, the crop suitability mapping in Samar by Poliquit et al. (2019b), and the GIS-based banana suitability mapping by Bato (2018) demonstrated how soil fertility, topography, and climate can be translated into actionable agricultural zoning. These initiatives underscore the value of integrating spatial data with soil and crop requirements. Yet, these zoning studies are often region-specific and crop-focused, lacking the integration with modern technologies such as machine learning and IoT sensors that could make zoning more dynamic and adaptive to current farming needs.

The incorporation of machine learning adds a new dimension to these agricultural practices. Random Forest applications by ? and Islam et al. (2023) showed how nonlinear soil and environmental data can be used to generate accurate fertilizer and crop recommendations, while boosting algorithms such as LightGBM, as explored by Parganiha and Verma (2024), achieved high predictive accuracies in crop yield forecasting. Complementary approaches like decision tree analysis, highlighted in the work of Lavanya et al. (2024) and Sudha and Hooda (2022), further support the adaptability and interpretability of ML models. Broader frameworks such as Multi-Criteria Decision Analysis,

reviewed by Cicciù et al. (2022) and Yuan et al. (2022), provide systematic ways to balance productivity, environmental sustainability, and economic viability. While these studies prove the potential of ML and decision-support systems, their application in the Philippines context—particularly Northern Mindanao—remains scarce, and most models have yet to be validated in diverse cropping systems beyond rice and maize.

The technological foundation for these innovations is strengthened by the integration of soil health sensors and IoT systems. Yin et al. (2021) emphasized the need for reliable, low-power, and scalable soil sensing technologies, while recent Philippine-based studies such as those by Sondhiya and Singh (2024) in Bukidnon and Kumar et al. (2024b) in rice and maize systems demonstrated the practicality of IoT-enabled NPK and pH sensors in monitoring nutrient status. Despite their promise, these sensor applications are still largely crop-specific and small-scale, limiting their ability to provide generalized, multi-crop recommendations across Region 10’s diverse agricultural systems.

The literature review establishes a strong foundation for developing a data-driven agricultural framework in Northern Mindanao. Soil health monitoring, zoning approaches, machine learning models, and IoT-based sensors converge to form comprehensive solutions. However, there remains a clear gap:

existing studies often treat these components separately—soil fertility studies without advanced analytics, zoning without integration to real-time data, or ML models without validation in local contexts. This study aims to address these gaps by developing an IoT-based Soil Health Monitoring and Recommendation System that integrates real-time soil parameters (NPK, pH, salinity, moisture, and temperature), agri-weather data, and agricultural zoning maps with machine learning. Unlike previous studies that focused on single crops or generalized recommendations, this system will deliver zone-specific and season-specific recommendations for both multi-cropping and fertilizer application in Region 10. By targeting key crops such as maize (corn), mungbean (mongo), soybean, squash (kalabasa), sweet potato (camote), cassava, eggplant, tomato, the project seeks to optimize yields, reduce resource waste, and promote sustainable agriculture practices aligned with the region's diverse farming systems.

CHAPTER 3

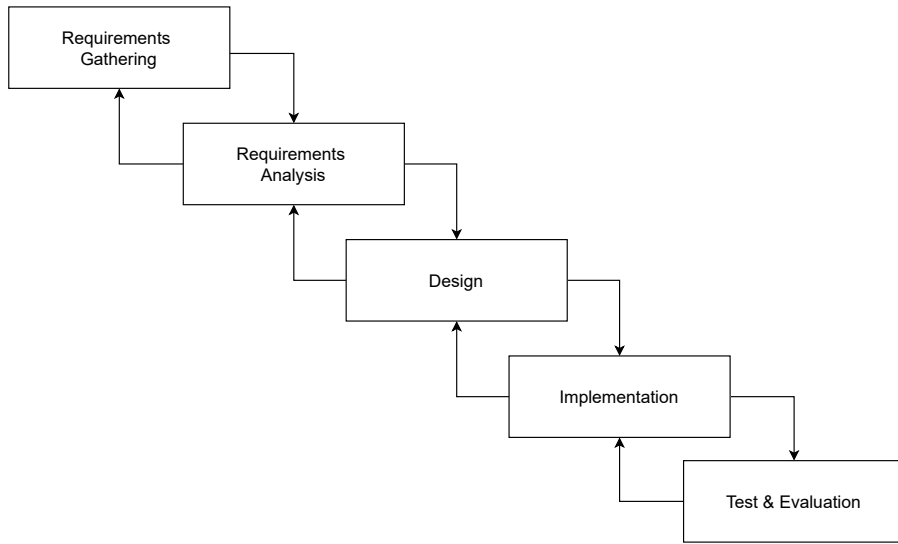
METHODOLOGY

In this chapter, the researchers outlined the methodology used to develop a soil health monitoring and recommendation system, which aimed to enhance sustainable agricultural practices through present data. The study utilized an experimental approach, integrating IoT sensors and machine learning algorithms, to enable continuous and accurate assessment of soil conditions. By focusing on essential soil parameters such as nitrogen, phosphorus, potassium, pH, salinity, moisture, and temperature this methodology provided insights that supported optimal crop management. This chapter also addressed the rationale behind the chosen experimental design, describing each stage of data collection, system setup, and analysis.

3.1 Research Design

Figure 3.1

Modified Waterfall Model



This study follows a modified waterfall research model to guide the development of the IoT-based crop and fertilizer recommendation system shown in Figure 4. The process begins with the requirements gathering phase, in which the system collects comprehensive data from the Department of Agriculture (DA) and credible online datasets. These datasets include zoning information, soil parameters, and crop-specific growth requirements. All collected datasets are integrated with the DA's zoning information to provide spatial context for agricultural analysis.

The study then proceeds to the requirements analysis phase, which identifies the need for soil health monitoring, crop prediction, and fertilizer recommendation in selected agricultural fields in Region~10. A gradient boosting algorithm is employed using both sensor-derived soil data and zoning information as inputs. Soil health prediction focuses on classifying soil quality levels, while crop suitability and yield forecasting utilize an optimized machine learning model to estimate expected productivity based on soil and environmental conditions. Fertilizer recommendations are generated based on predicted soil nutrient deficiencies and crop requirements derived from the machine learning analysis.

The next phase is system design, where the overall system workflow is defined, providing detailed descriptions of how the IoT-based machine learning integration operates. This phase presents the system architecture, hardware and software flowcharts, and the design of the web and mobile interfaces. The design ensures seamless integration of hardware components for data acquisition, software modules for data preprocessing and prediction, and user interfaces for monitoring and decision support.

This is followed by the implementation phase, which involves the development of the IoT system, training and testing of the machine learning model, and configuration of the required hardware and software components.

Field nodes collect and preprocess soil data before transmitting it to a central server for storage and predictive analytics. The system generates crop suitability and fertilizer recommendations, which are displayed through a web-based dashboard that visualizes soil health status, nutrient trends, weather information, farmer data, and zoning maps.

Once implementation is completed, the testing phase is conducted in the selected study area, where soil data are collected and prediction accuracy is evaluated. Zoning-based crop and fertilizer recommendations are validated against reference data and observed field conditions. The validation phase ensures system reliability through quantitative accuracy assessment and qualitative feedback on usability. Finally, the deployment and maintenance phase outlines potential large-scale application across Region~10, including recommendations for system improvement, scalability, and long-term sustainability.

This study follows a modified waterfall research model to guide the development of the IoT system shown in figure 4. The modified waterfall model allows an iteration to the previous phase for verification and validation, intended mainly for connecting stages. The process begins with the requirements gathering, in which the system collected comprehensive data from the Department of Agriculture (DA) and credible online datasets. The datasets include zoning information and soil parameters, as well as crop-specific re-

quirements for growth conditions. Moreover, all datasets was integrated with the DA's zoning information.

It then proceeds to the requirements analysis, which identifies the need for soil health monitoring, crop prediction, and fertilizer recommendation in the selected field in Region 10. A gradient boosting algorithm was employed using both sensor and zoning data inputs. Soil health prediction classifies soil quality levels, while crop yield forecasting was used to leverage an optimized machine learning model with metaheuristic-based feature selection to predict expected productivity. Fertilizer recommendations was generated based on predicted soil nutrient deficiencies, crop requirements, and established agronomic thresholds derived from sensor and zoning data. Feedback in this stage may return to the requirements gathering phase if analysis reveals missing data, incorrect assumptions, or the need for additional crop or soil parameters to properly define system requirements.

The next phase is the system design, where the flow of the overall system is defined with detailed descriptions on how the system works on implementing the IoT-based ML Integration system. This phase involves a comprehensive overview of the system design, architecture, the flowchart of the hardware and software that was used in this system, as well as the web and mobile interfaces. The design ensures seamless integration of hardware compo-

nents for data acquisition, software modules for data analysis and prediction, and user interfaces for monitoring control. The feedback may return to the requirements analysis stage if the proposed design cannot fully implement the defined requirements such as limitations in hardware integration or software module capabilities.

This is followed by the implementation phase, which involves the development of the IoT system, training and testing of the machine learning model, in combination with multi-criteria decision analysis methods, and the configuration of hardware and software. Field nodes gather and preprocess soil data, then transmit it to a central server for storage, predictive analytics, and generation of crop suitability and fertilizer recommendations. A web dashboard visualizes soil health, farmer data, nutrient trends, weather, and zoning maps. The results in this stage can return to the system design phase if there are software or hardware integration challenges encountered, and it requires adjustments in design to ensure proper functionality.

Once completed, the testing phase is carried out in the selected area where soil data is gathered and prediction accuracy is measured. In this phase, the zoning-based recommendations are also validated. The validation phase then ensures reliability through quantitative accuracy testing and qualitative feedback on usability. Feedback from testing may return to the implementation

phase to correct software bugs, or adjust hardware configurations for better performance before validation.

3.2 Requirements Gathering

The system requires comprehensive data collection to provide accurate, multi-cropping and fertilizer recommendation based on farming zoning in Bud-kinon, Philippines. Data was sourced out from the Department of Agriculture (DA), and credible online datasets. This includes detailed zoning information and soil parameters. To be included also are crop-specific requirements and growth conditions. Gathering these datasets ensures that the recommendation system is grounded in both environmental and agricultural best practices.

The types of data required encompass zoning information, and detailed crop profiles. Zoning data defines farming area, and land use classifications, while soil health parameters includes pH, NPK, salinity, moisture, and temperature. To be included as well is climate data such as historical rainfall, temperature, humidity, and forecast models which provide insights into environmental conditions that affect crop growth.

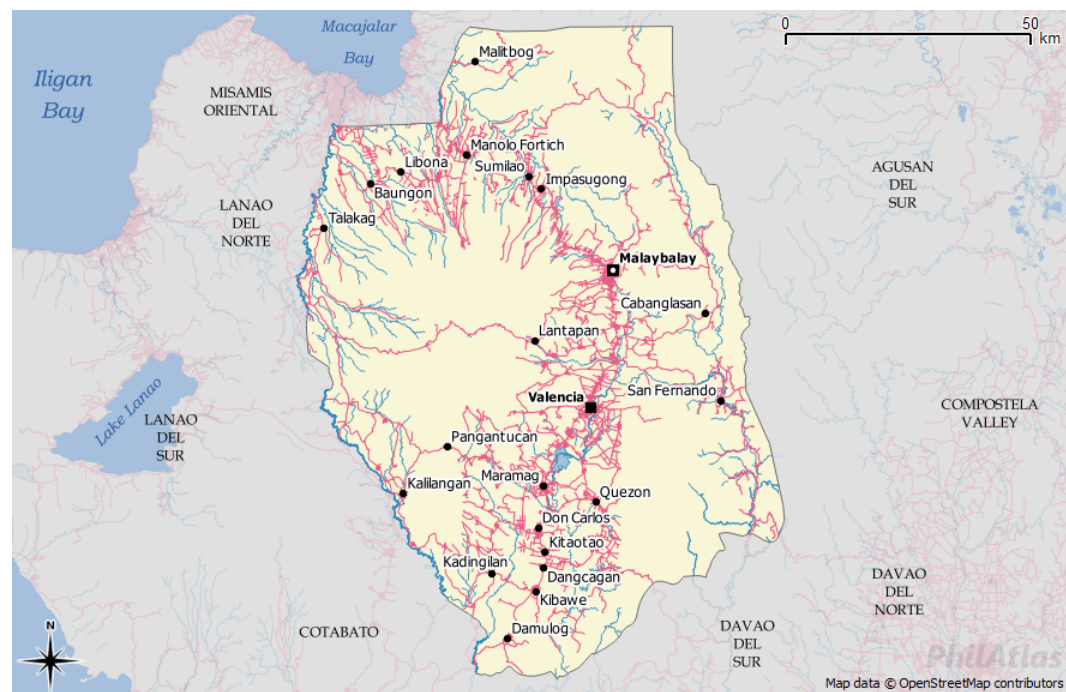
In preparation for system integration, the data undergoes a series of preprocessing steps. Missing sensor readings and data gaps was addressed through interpolation, while noisy or incorrect data was identified and re-

moved to maintain integrity. Soil parameters was normalized into uniform ranges to enhance the performance of predictive models. Once cleaned and standardized, all datasets was integrated with the Department of Agriculture’s zoning information to ensure that recommendations are localized and relevant to the specific conditions in Bukidnon.

3.2.1 Research Setting

Figure 3.2

Research Setting (PhilAtlas)



The study was conducted in selected farms in Bukidnon, particularly in Alae, Lingion, and Libona. These areas are known for their strong agricultural activity, with large portions of land dedicated to growing high-value crops such as maize (corn), mungbean (mongo), soybean, squash (kalabasa), sweet potato (camote), cassava, eggplant, and tomato. Farmers in these communities commonly use a mix of conventional and semi-modern farming practices. Because of this, the area provides a realistic and practical setting for the study, allowing crop performance to be monitored and evaluated under actual farming conditions rather than controlled environments.

3.3 Requirements Analysis

The system requires the development and implementation of a machine learning component capable of performing soil health prediction, and crop recommendation. Machine Learning boosting algorithms was employed for these tasks, using both sensor and zoning data. The sensors collects important soil information such as Nitrogen (N), Phosphorus (P), Potassium (K), pH level, moisture, temperature, and salinity. By obtaining the data sources, the system can better understand the relationship between soil condition and the environment. Based on the collected data, the machine learning model classifies the soil quality levels, while crop yield forecasting will leverage an

optimized machine learning model by identifying patterns and relationships between soil parameters and environmental factors. To make predictions more accurate, the model uses a metaheuristic-based feature selection that focuses on the most relevant data and removes unnecessary or less useful information.

After analyzing the soil health parameters and producing crop recommendations, the system generates fertilizer recommendations based on predicted soil nutrient deficiencies, crop nutrient requirements, and established agronomic guidelines. The system determines the appropriate fertilizer type and application rate suited for specific field conditions. To make sure the machine learning model produces reliable results, the dataset was divided into three parts: 70% for training, 15% for validation, and 15% for testing. This ensures that the system learns effectively from the data, maintains accuracy, and avoids overfitting.

The system undergoes proper evaluation using both quantitative and qualitative approaches. Quantitative assessment includes metrics such as accuracy, precision, recall, and F1-score for crop recommendation. RMSE, and MAE was used for consistency ratios and ranking stability for soil health readings. Conducting these metric evaluations is important because they measure how well the machine learning model performs and how reliable its predictions are. By analyzing these values, the researchers can identify the model's

strengths, detect errors, and make adjustments to improve performance before deployment. Cross-validation ensures model robustness across different crop zones. Meanwhile, qualitative evaluation involves focus group discussions and structured interviews with farmers and agricultural technicians in Region 10 to assess usability, reliability, and practical value. Feedback from these sessions will guide further refinement of the system to ensure its effectiveness in real-world agricultural applications.

3.3.1 User Definition

Following the requirements analysis, the system's users were defined as follows:

Farmer - Farmers are the primary beneficiaries of the system. Their farms provide localized data inputs through the soil sensor. In return, farmers receive soil health insights, and multi-cropping recommendations.

Department of Agriculture (DA) - The DA serves as both a data provider and a supervisory user of the system. They supply zoning information, and other agricultural zoning datasets that form the foundation of the recommendation system. Through the web-based dashboard, DA officials can visualize regional soil health trends, and monitor farming zones.

The specific user requirements are summarized in Table 1.

Table 2: User Requirements

User	Requirements
Farmer	- Access soil health insights (pH, moisture, nutrients) through a mobile app. - Receive crop suitability scores and multi-cropping recommendations. - Get fertilizer schedules tailored to soil condition and crop type.
Department of Agriculture	- Provide zoning maps, soil datasets, and crop profiles to the system. - Access a centralized web dashboard to visualize soil health, nutrient trends, and farmer data across Region 10.

This table summarizes the requirements of the system's two main users. Farmers need accessible insights and recommendations to guide timely crop and fertilizer decisions, while the Department of Agriculture requires integrated data and analytics to support monitoring, reporting, and regional agricultural planning.

3.3.2 System Requirements

The system requirements shown in Table 2, were categorized into Input, Process, Output, Control, and Performance specifications to aid the system design and implementation.

Table 3: System Requirements

Category	System Requirement
Input Requirements	- The system shall accept input data directly from farmers and agricultural authorities through the designated interface. - The system shall acquire and process soil data from integrated sensors or external data sources. - The system shall accept soil health data from the sensor as input for hardware operation. - The system shall retrieve weather information from external APIs. - The system shall define sensor communication parameters to ensure consistent and accurate data acquisition from IoT nodes.
Process Requirements	- The system shall combine weather, soil, and farmer-provided data into a single decision-making framework. - The system shall employ boosting algorithms to enhance the accuracy of predictive crop recommendations. - The system shall analyze data-driven insights to balance yield potential with resource availability. - The system shall determine an appropriate planting timeline based on environmental conditions and crop growth requirements.
Output Requirements	- The system shall present crop recommendations via a visual interface on mobile and web applications. - The system shall display multi-cropping options, which shows compatible crop pairings. - The system shall present localized weather forecasts. - The system shall provide soil health reports with clear visual indicators. - The system shall display information on current crops cultivated by the farmer.

Category	System Requirement
	- The system shall display farm details of the farmer. - The system shall display the locations and overview of registered farms. - The system shall display a list of all farmers registered. - The system shall present crop rules, including crop rotation guidelines, and companion planting practices.
Control Requirements	- The system shall provide secure login and authentication for farmers and agricultural authorities. - The system shall restrict access to sensitive data based on user roles and permissions. - The system shall allow administrators to update crop rules and farm locations. - The system shall protect sensor and farmer data during transmission, and storage using standardized encryption methods and user authentication protocols.
Performance Requirements	- The system shall scale efficiently to accommodate growth in registered farmers and farms without requiring major redesign. - The system must be able to have offline syncing for farmer features. - The system must provide accurate crop recommendations for farmers. - The system shall synchronize data reliably between mobile and web platforms.

The system requirements were defined to ensure the effective collection, processing, and presentation of farm and soil data to support informed crop management decisions. The primary objective of the system is to assist farmers by integrating farm-specific information and soil health metrics

to generate accurate and actionable crop and fertilizer recommendations. Input requirements focus on collecting data from farmers and IoT-based sensors, while process requirements involve the use of advanced machine learning techniques, particularly gradient boosting algorithms, to analyze soil conditions, predict crop suitability, and estimate potential yield performance. Outputs are delivered through mobile and web applications to provide farmers and administrators with clear, visualized recommendations and reports. Control requirements ensure secure access through role-based permissions and administrative functions for managing crop data and farm records. Lastly, performance requirements address scalability, offline data synchronization, and recommendation accuracy to ensure reliable decision support across diverse agricultural environments.

3.4 Data Collection and Processing

This section describes the procedures and methods used to gather, integrate, and prepare data for the system's operation and machine learning components. It covers the collection of soil, environmental, and geographical data from various sources, as well as the acquisition of historical datasets. Furthermore, it outlines the processes involved in data integration, preprocessing, feature engineering, and dataset preparation to ensure data quality and suit-

ability for analysis. The section also presents how the processed data is stored and managed, and how it flows through the system to support accurate crop and fertilizer recommendations.

3.4.1 Soil Data Collection

Soil data collection is the main input of the system, providing the information needed for crop recommendation and soil analysis. In this study, the primary dataset used for the machine learning model comes from the Department of Agriculture (DA) Region 10. This dataset includes recorded soil parameters such as nitrogen (N), phosphorus (P), potassium (K), soil pH, moisture, temperature, and salinity. These records are used to train the Light Gradient Boosting Machine (LGBM) model so it can learn patterns between soil conditions and suitable crops. Since the data comes from verified sources, it helps improve the accuracy and reliability of the model.

In addition, the system is designed to support IoT-based soil data collection using sensors connected to an ESP32 microcontroller. These sensors can measure real-time soil conditions in the field. However, in the current implementation, this data is mainly intended for crop recommendation and for future dataset expansion rather than for training the model.

Overall, the soil data collected from the DA serves as the foundation of

the system, while IoT integration provides an opportunity to improve real-time analysis for crop recommendation and enhance the dataset in future developments.

3.4.2 Environmental and Weather Data Collection

Soil and simple environmental readings from the field are sent to the ML service as one list of numbers. The first values match what the crop model was trained on (organic matter, phosphorus, potassium, and pH). If the device sends more values, those can include things like temperature, moisture, and electrical conductivity. After each prediction, the system saves a pending snapshot that holds the numbers, the full request, an optional farm id, and optional latitude and longitude. The app can then pull that single snapshot instead of combining data from many separate calls.

Weather works differently. The project does not use a weather station on the farm. The backend talks to OpenWeatherMap using lat and long, keeps the API key on the server, and sends the app a clean package with temperature, humidity, pressure, wind, visibility, a short description of the sky, the place name, and a time stamp. The app does not need to hit the weather API constantly, so it can cache results for a short time. Which coordinates are used matters: if the latest ML request already had lat and long, those are

used for weather so it matches the same area as the sensor run. If not, the app uses the coordinates stored for that farm. There is also an optional fallback for testing when coordinates are missing. When the farmer has an active saved session, the screen shows the saved soil data from the server for consistency. When there is no active session, the app checks often for new pending readings so the numbers stay up to date.

This setup is intentional. Soil measurements and internet weather are not the same thing, but showing both together gives a clearer picture. Putting the weather API on the backend hides the key and keeps responses in one standard format. Using one feature list and one pending snapshot keeps the model, the server, and the app on the same page and avoids extra work in the mobile client.

3.4.3 Zoning and Geographical Data Collection

Geographic representation matters so farm records are not just names on a list but real places on the ground. The system needs to know where farming happens, roughly how big the holding is, and stable coordinates that other tools can use, such as weather APIs and machine learning workflows. Right now, location is stored mostly as a point: each farm can have latitude and longitude, plus an optional text field for things like barangay or street,

and a field for how large the farm is said to be. Admins and users enter this through the web when they create or edit a farm. The API can also update coordinates when a session or a sync from the model layer brings in new positions, so people do not have to fix the map by hand as often.

In this study, the term “zoning” means organizing farms in space, not official land-use zones drawn as polygons on a cadastral map. The admin GIS Mapping screen shows farms on a Leaflet map with OpenStreetMap tiles. Farms with real latitude and longitude show up as pins. Farms without coordinates stay off the map until someone adds GPS data, and the UI warns you if registered farms still have no usable GPS. If there is more than one farm, the map zooms to fit all the pins. The default map center fits the Philippines so the first view is not empty. Clicking a pin shows extra details like soil type and area so you can relate the site to soil conditions. Soil tests are linked to the same farm id, so lab or field results line up with that place in the dashboard

3.4.4 Historical and External Data Acquisition

The main purpose of historical data collection is to build a strong foundation for training the machine learning models that control the crop and fertilizer recommendations of the system. While the Internet of Things (IoT) sensor network, powered by the ESP32 microcontroller, is highly capable of

capturing local real-time data, machine learning algorithms need substantial historical context to recognize complex agricultural patterns. Therefore, additional historical data is collected directly from the primary dataset titled *Consolidated Soil Info (RSL) 2023 to 2025 USTP CPE4*.

This core document combines official soil analysis reports and agricultural data gathered by the Regional Soils Laboratory (RSL) of the Philippine Department of Agriculture spanning the years 2023 to 2025. The collected dataset contains vital features, specifically basic soil parameters such as Nitrogen, Phosphorus, Potassium levels, pH, moisture, and temperature. It also includes the corresponding crop types traditionally grown under those exact conditions and standard fertilizer baselines. The exact features extracted for the system database and their respective categories are detailed in Table 4.

Table 4: Tabular Summary of Features Extracted from the RSL Dataset

Category	Extracted Features
Soil Nutrients	Nitrogen (N), Phosphorus (P), Potassium (K)
Soil Physics	pH Levels, Moisture Content, Temperature
Agricultural	Crop Type, Fertilizer Baselines
Metadata	Date of Analysis, Provincial Location

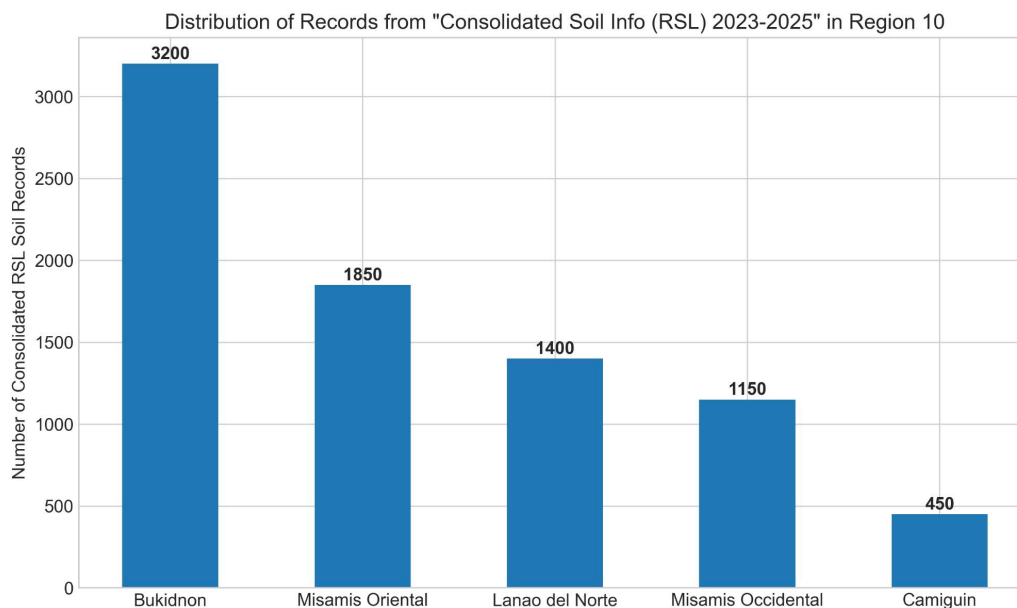
To ensure the models can generalize well across local environmental conditions, the combined document includes a large number of distinct records to create a realistically sized data collection. Furthermore, strict data relevance

rules built into this dataset guarantee that the information is restricted exclusively to Region 10 (Northern Mindanao). This focuses specifically on its provinces, including Bukidnon, Misamis Oriental, Lanao del Norte, Misamis Occidental, and Camiguin.

As illustrated in Figure 3.3, the distribution of these records across the five provinces reflects the agricultural density of the region, with Bukidnon providing the most significant volume of data.

Figure 3.3

Distribution of Soil Records across Region 10 Provinces



Before the data is formally entered into the primary database of the system, early validation procedures are performed. These involve automated

limit checking against the laboratory records to confirm structural correctness and data credibility. This process ensures that the dataset remains a high-quality reference for training the system’s recommendation engine.

3.4.5 Data Integration

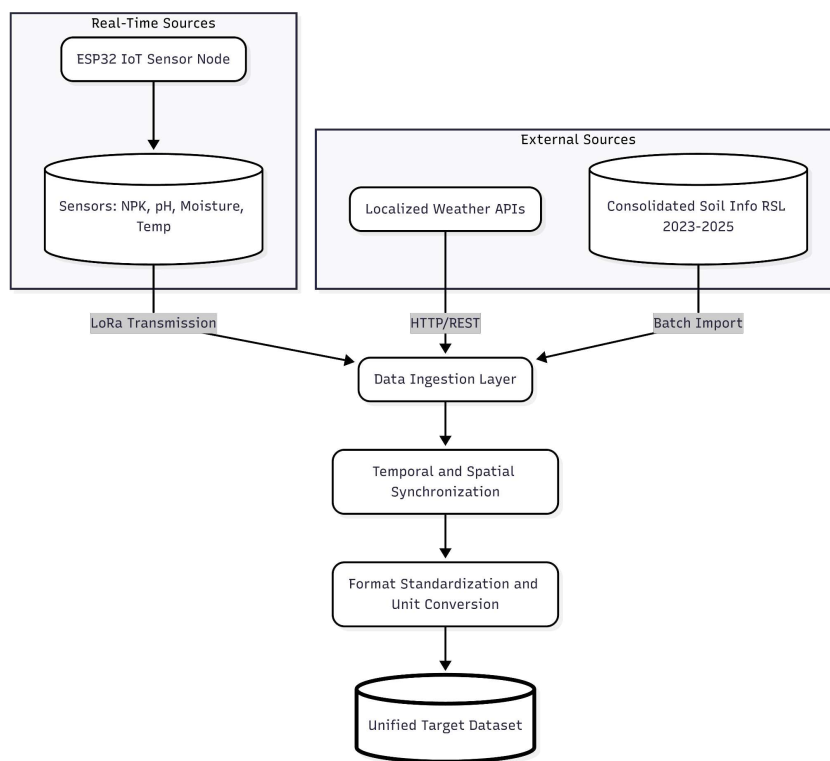
The primary objective of the data integration phase is to unify multiple, heterogeneous data streams into a single, cohesive dataset suitable for advanced computational analysis and machine learning training. By centralizing disparate information into a uniform structure, the system ensures that the predictive models can process environmental, and historical variables simultaneously. This integration is essential for creating a holistic representation of the agricultural environment, bridging the gap between local sensor observations and broader regional contexts.

As illustrated in the integration pipeline in Figure 3.4, the system aggregates data from external sources. The pipeline depicts the convergence of LoRa-based sensor data, REST-based weather APIs, and batch-imported historical records into a unified dataset. Real-time data is captured via an ESP32 IoT sensor node, which transmits continuous readings of Nitrogen (N), Phosphorus (P), Potassium (K), pH levels, soil moisture, and temperature. These hardware-derived metrics are combined with external data streams, in-

cluding localized weather updates retrieved through HTTP/REST APIs and the validated Consolidated Soil Info (RSL) 2023–2025 historical dataset. This multi-source approach allows the framework to factor in both immediate soil conditions and long-term agricultural trends.

Figure 3.4

System Architecture of the Data Integration Pipeline



To ensure contextual accuracy, the system employs rigorous temporal and spatial synchronization. Data ingestion occurs through different protocols depending on the source: LoRa transmission for hardware sensors, RESTful requests for weather data, and batch imports for historical records. Once

ingested, these data points are programmatically aligned based on matching timestamps and geographic coordinates. This synchronization forms a complete environmental snapshot, ensuring that a specific sensor reading is correctly paired with the weather conditions and historical soil health data relevant to that exact time and location.

During the aggregation process, structural inconsistencies and varying data formats are resolved through a dedicated format standardization and unit conversion layer. Because different sources often utilize varying measurement scales, the system applies strict conversion rules to ensure uniformity. For example, temperature readings are standardized to Celsius, and moisture levels are normalized to percentage values. Additionally, schema mapping is performed to align field names across the various data inputs, ensuring that all variables conform to a consistent naming convention.

The culmination of this pipeline is the creation of a Unified Target Dataset. This centralized repository effectively bridges real-time sensor baselines with extensive historical context and atmospheric data. By outputting a standardized, synchronized matrix of information, the integration phase provides the high-fidelity data foundation necessary for the subsequent preprocessing and machine learning stages of the system.

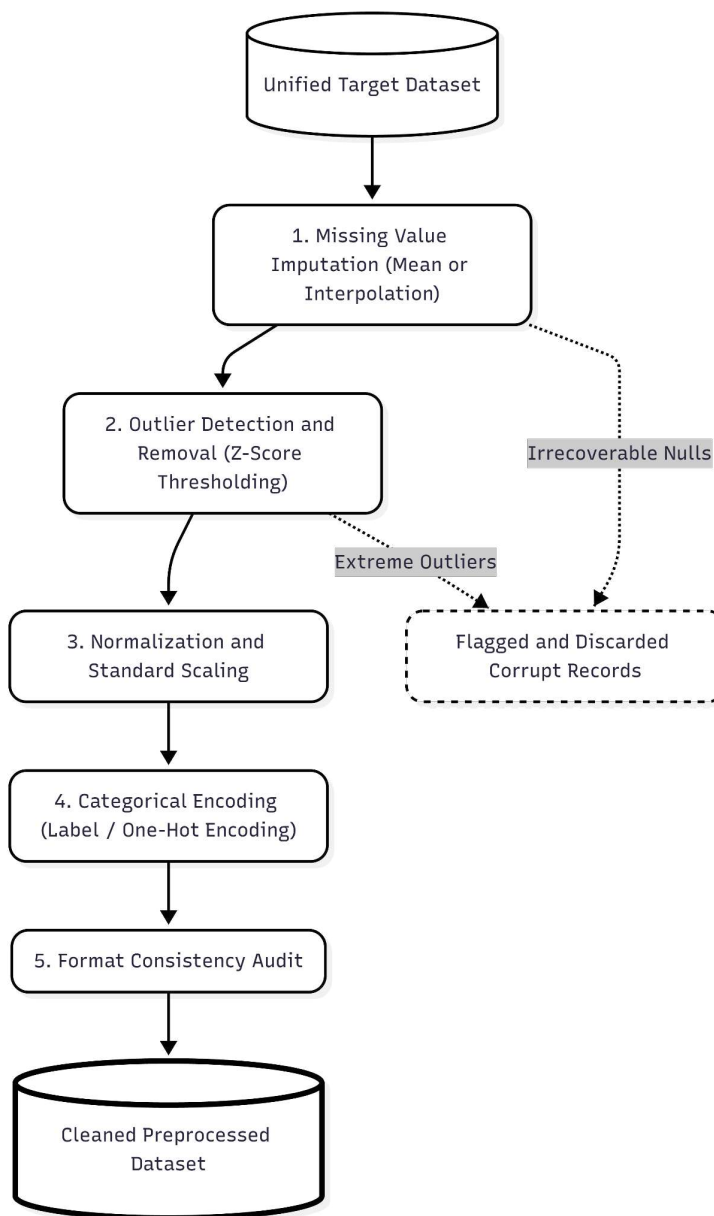
3.4.6 Data Preprocessing

The primary objective of data preprocessing within this framework is to systematically refine and clean the integrated dataset, thereby reducing algorithmic bias and ensuring high accuracy during the subsequent machine learning training phase. Given that the machine learning models rely on the structural integrity of the input data to generate reliable crop and fertilizer recommendations, this phase acts as a critical filter. It transforms raw, heterogeneous data into a standardized format that facilitates efficient mathematical computation and pattern recognition.

The initial stage involves rigorous data cleaning to address missing or empty values. As illustrated in the sequential pipeline in Figure 3.5, these gaps are handled based on the statistical variation of the feature. Depending on the feature's distribution, these gaps are addressed through mathematical substitution, such as applying the feature mean or using linear interpolation, or through the systematic removal of records showing excessive missing data to maintain dataset integrity.

Figure 3.5

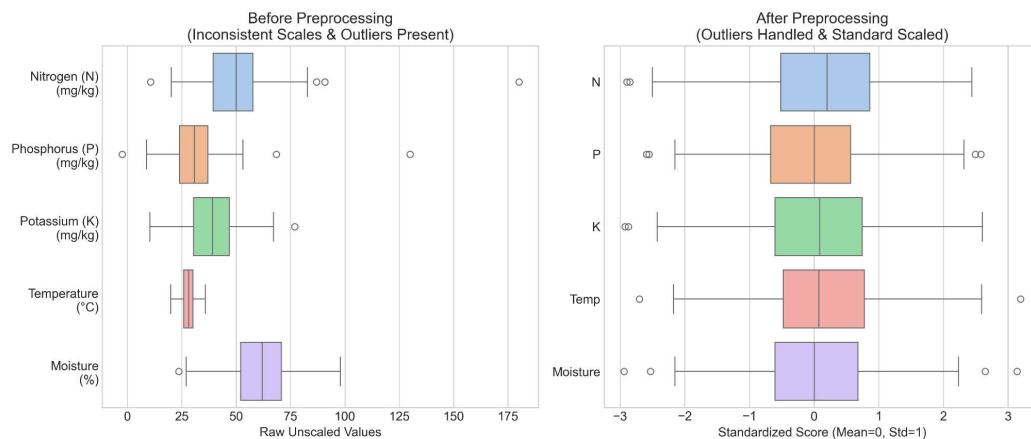
Sequential Data Preprocessing Pipeline for Agricultural Soil Records



Following the initial cleaning, noise and outlier handling mechanisms are applied to filter unusual data points generated by temporary sensor glitches

or transmission errors. This process employs statistical limits, specifically Z-score thresholding, to identify and discard extreme values that could otherwise skew the decision boundaries of the model. The impact of this process is evident in Figure 3, where the "Before Preprocessing" box plots indicate several extreme outliers in the Nitrogen (N) and Phosphorus (P) features. By removing these points, the system ensures that the training process is grounded in realistic agricultural conditions rather than technical anomalies.

To ensure that variables with naturally wide numerical ranges do not disproportionately dominate the learning algorithm, normalization and scaling techniques are utilized. As shown in the comparison in Figure 3.6, soil parameters such as Potassium (K) and Nitrogen (N) often exist on scales significantly different from Temperature or Moisture. Using a standard scaler, these continuous numerical features are transformed into a common scale where the mean is 0 and the standard deviation is 1. This standardization is crucial for the stability and convergence of the machine learning algorithms, as it allows the model to assign appropriate weights to each feature without being influenced by its raw magnitude.

Figure 3.6*Comparison of Soil Parameter Distributions Before and After Preprocessing*

Algorithmic encoding is performed to translate categorical variables into numerical representations. Since the system’s backend requires mathematical inputs for computation, non-numeric data—most notably the targeted crop types—are transformed using techniques such as label encoding or one-hot encoding. This process ensures that categorical labels are represented as distinct mathematical entities, allowing the model to process variety-specific data without assuming an incorrect ordinal relationship between different crops.

The preprocessing pipeline concludes with comprehensive programmatic consistency checks to definitively verify the structural integrity and uniformity of the cleaned dataset. This final audit ensures that all processed records adhere to the predefined schema and that no data corruption occurred during the transformation stages. Once verified, the dataset is formally transitioned

into the primary database, serving as a high-fidelity reference for the system's recommendation engine.

3.4.7 Feature Engineering

The primary objective of the feature engineering phase was to augment the predictive performance of the machine learning models by providing more nuanced and expressive data representations. Rather than relying exclusively on the raw soil properties captured through the sensor network and historical datasets, this phase involved mathematically deriving new variables that encode established agronomic relationships. By transforming raw measurements into high-order features, the system enables the algorithms to capture complex, non-linear dependencies between soil chemistry and plant health that are not explicitly present in the initial input vectors.

Two distinct categories of derived features were constructed for the engineered dataset. The first category comprised ratio-based features designed to quantify the balance between specific nutrients. The Phosphorus-to-Potassium ratio (P/K) was calculated to define the internal nutrient equilibrium between these two primary macronutrients, which serves as a critical fertility indicator in tropical soil profiles. Additionally, the Phosphorus-to-Organic Matter ratio (P/OM) was computed to express Phosphorus availability relative to the

soil’s organic composition, reflecting the established agronomic dependency of Phosphorus mobility on organic matter levels.

The second category consisted of interaction features, derived through multiplicative operations to encode conditional relationships. An Organic Matter $\times$ pH interaction term was generated to represent the principle that organic matter decomposition and subsequent nutrient release are strongly regulated by soil acidity. Similarly, a Potassium $\times$ pH interaction term was derived to capture the pH-dependent nature of Potassium uptake by root systems. These interaction terms allowed the model to learn complex interdependencies without requiring manual thresholding or excessively deep neural architectures.

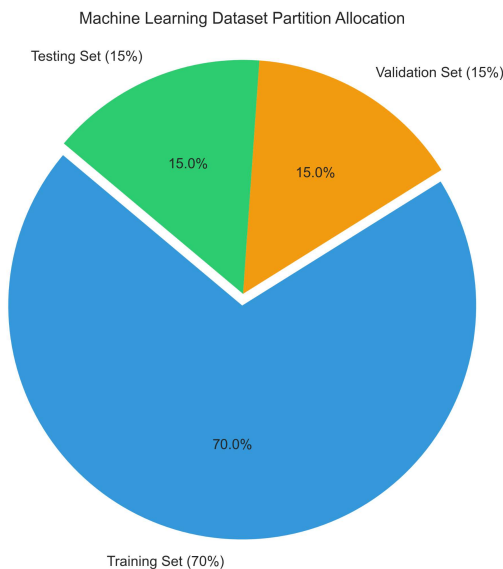
The explicit encoding of these soil-crop relationships reduced the computational complexity required for the model to discover such patterns independently. By providing calculated nutrient ratios and pH-dependent interaction terms, the algorithm leveraged domain-specific knowledge directly, resulting in faster convergence during the training phase and improved generalization when processing unseen soil conditions. This engineering process successfully expanded the dataset from the four original raw variables to a comprehensive set of eight predictive features.

3.4.8 Dataset Preparation for Machine Learning

The primary objective of dataset preparation is to establish the final structural configuration of the preprocessed data, ensuring it is partitioned and formatted for the iterative training and evaluation cycles of the machine learning models. This phase represents the transition from static data storage to an active learning format, where information is structured to minimize bias and maximize the model's ability to generalize across diverse agricultural scenarios.

Figure 3.7

Machine Learning Dataset Partition Allocation



A fundamental component of this phase is dataset splitting, where the unified data is stratified to maintain representative class distributions across

all subsets. As illustrated in Figure 3.7, the dataset is divided into three definitive partitions: 70% of the data is allocated for direct model training, 15% is utilized for validation to systematically tune hyperparameters, and the remaining 15% is strictly reserved for blind testing to evaluate final real-world performance. The 70-15-15 split ensures sufficient data for training while maintaining independent sets for tuning and final evaluation.

Concurrent with this partitioning is the execution of precise data labeling. The feature space, which includes the eight engineered soil and environmental metrics, is distinctly separated from the target variables. This frames the system’s objectives into two distinct paths: crop recommendation is treated as a discrete multiclass classification problem, while fertilizer recommendations are mapped as specific quantitative output variables.

3.4.9 Data Storage and Management

The architecture of the proposed system necessitates a centralized and reliable repository to ensure data consistency across administrative and end-user platforms. This repository facilitates the management of diverse entities, including farmer and administrator accounts, farm records encompassing geographic coordinates and land area, active farming sessions, and soil test readings. Furthermore, the database stores the outputs of the machine learn-

ing engine, specifically crop and fertilizer recommendations, alongside system telemetry and device status updates. While machine learning models are maintained as distinct training artifacts, all operational and longitudinal data are housed within the database to ensure that the web dashboard, mobile application, and backend services utilize a unified source of truth.

The system utilizes a standard relational database management system, specifically PostgreSQL, which is deployed via a managed cloud service to ensure high availability and automated maintenance. The backend Application Programming Interface (API) establishes a secure connection to the database, organizing data into structured tables for users, farms, sessions, soil tests, and recommendations. To optimize schema complexity, richer nested data structures are stored using structured fields within the relational environment, reducing the need for an excessive number of auxiliary tables while maintaining query efficiency. The mobile application interacts with this layer through standard HTTP requests, while the web application leverages a Representational State Transfer (REST) layer provided by the hosting service to streamline Create, Read, Update, and Delete (CRUD) operations without schema duplication.

Security and scalability are integral to the data management strategy of the system. User authentication is secured through strong cryptographic

hashing for passwords, while session management is handled via a dual-token system consisting of short-lived access tokens and longer-lived refresh tokens. These tokens are stored within secure on-device storage on the mobile platform and verified against secrets maintained in the server configuration. Furthermore, the hosting environment implements Row-Level Security (RLS) policies to ensure that users are restricted to accessing only their authorized data subsets. To accommodate increasing traffic and data volume, the system employs connection pooling, stateless API instances capable of horizontal scaling behind a load balancer, and paginated data retrieval to minimize latency and server overhead. Consistent data integrity is maintained through automated backup protocols managed by the database provider, ensuring recovery in the event of system failure.

3.4.10 Data Processing Workflow

The TANIM system employs a multi-stage data pipeline designed to transform raw environmental signals into actionable agricultural intelligence. This workflow integrates physical sensing, cloud-based data management, and advanced machine learning analytics to provide decision-support tools for both farmers and regional agricultural authorities.

Figure 3.8
TANIM Data Processing Workflow Pipeline

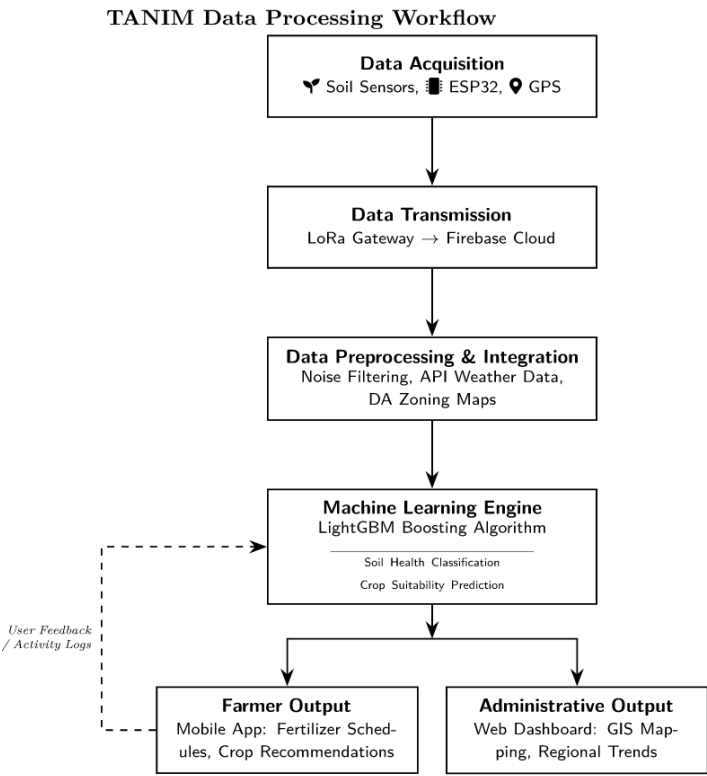


Figure 3.8 illustrates the comprehensive data flow, from initial acquisition to end-user dissemination, following a logical progression through hardware interfacing, cloud integration, and predictive modeling. The workflow demonstrates the transition from raw sensor acquisition to the generation of agricultural insights.

The primary stage of the workflow involves the high-fidelity collection of soil parameters at the physical layer. The system utilizes a multi-parameter

soil sensor to monitor Nitrogen (N), Phosphorus (P), Potassium (K), pH, moisture, salinity, and temperature, all of which are interfaced with an ESP32 microcontroller. To ensure data integrity during local transmission, a MAX485 module is employed to convert RS-485 differential signals to TTL-compatible logic for the microcontroller. These data packets are subsequently encapsulated and transmitted via the LoRa (Long Range) protocol to a central gateway. This modulation technique is specifically selected to ensure reliable connectivity in topographically challenging or remote agricultural zones where traditional cellular infrastructure is often insufficient.

Upon receipt at the gateway, the data is forwarded to a Firebase cloud database for centralized management and refinement. During this phase, the system performs rigorous data cleaning by executing filtering algorithms to mitigate electrical noise and remove statistical outliers from the raw sensor readings. Furthermore, a data fusion process contextualizes the real-time sensor data by integrating it with secondary datasets. These include meteorological information, such as precipitation and humidity acquired via external APIs, and geospatial zoning maps provided by the Department of Agriculture (DA) Region 10.

The analytical core of the system leverages a LightGBM algorithm, a gradient boosting framework utilized for its high efficiency and superior

performance with tabular agricultural datasets. The process begins with metaheuristic-based feature selection to identify the most significant soil and climatic variables, thereby reducing dimensionality and improving model generalization. The predictive engine then performs dual-task analytics, which include classifying soil health status based on established agronomic thresholds and forecasting crop suitability to optimize regional yield performance.

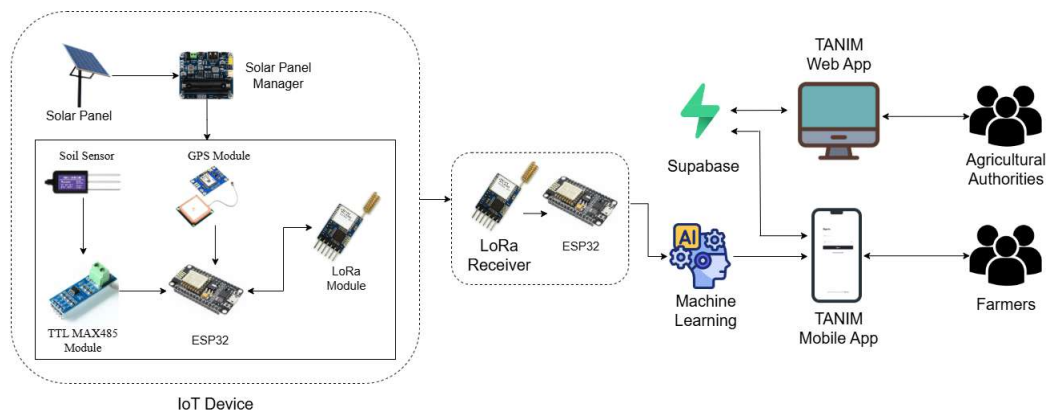
3.5 System Design

In this section, the flow of the overall system is defined with detailed descriptions on how the system will work on implementing the IoT-based ML integration system. This involves a comprehensive overview of the system design and architecture, and the flowchart of the hardware and software that was used in this system. The design ensures seamless integration of hardware components for data acquisition, software modules for data analysis and prediction, and user interfaces for monitoring and control.

3.5.1 System Architecture

Figure 3.9

System Architecture



The **Figure 3.9** shows the updated system architecture for the Soil Health Monitoring, Fertilizer, and Crop Recommendation System (TANIM). The system goes from hardware in the field, through a LoRa link and a local gateway, into the cloud for storage and machine learning, and then out to a web app for agricultural authorities and a mobile app for farmers. Field IoT device. The sensing unit is basically one IoT package in the field. A solar panel feeds a solar panel manager, which handles charging and power so the device can keep running in areas that are far from the grid. Inside the box, an ESP32 reads the sensors, does any simple processing on the readings, and builds the packets that will be sent out.

A soil sensor measures things like NPK, pH, and moisture. The sensor

talks over an industrial serial link that behaves like RS-485, which is good when the cable is long and the environment is noisy. A TTL MAX485 module turns that into a regular UART signal the ESP32 can read, so you get reliable communication without making the firmware too complicated. The device also has a GPS module. The ESP32 reads the GPS data and sends location together with the soil readings. That way the cloud and the TANIM Web App can show maps and trends by area, not just a list of numbers with no place attached.

The ESP32 sends data through a LoRa module. LoRa is meant for low power and longer range, which fits farms and rural sites. Gateway and path into the cloud. A LoRa receiver listens for packets from one or many nodes in the field. It is connected to a second ESP32 that works as a gateway. That ESP32 receives the LoRa traffic, checks the data, can hold or reshape packets if needed, and then sends everything upstream. The forwarded data includes soil readings, and GPS location.

Cloud, machine learning, and apps. After the gateway, the system splits into two main parts in the cloud. One part is machine learning, which uses new data and past data to look for patterns in the soil, spot possible problems like low nutrients or moisture stress, and suggest fertilizers, crops, and simple soil health takeaways. The other part is Supabase, which is the main backend

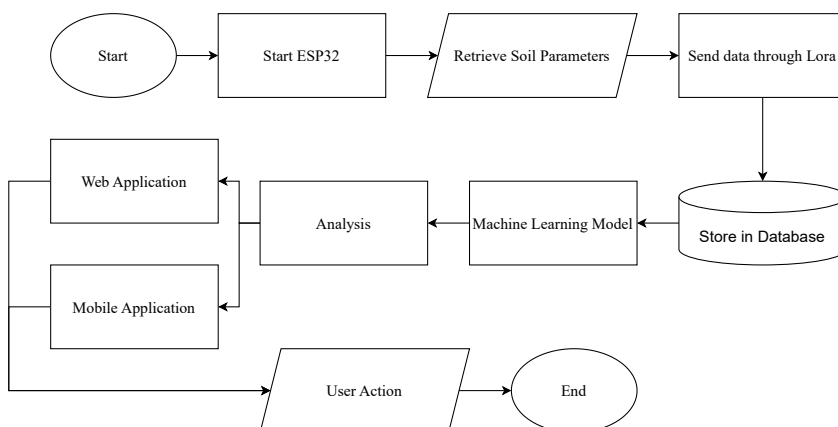
for the project. Supabase stores data in a database, handles sign in and users, and exposes APIs for things like sensor history, profiles, and who is allowed to see what, especially for authority users.

The TANIM Web App is built for agricultural authorities. It uses Supabase for the underlying data and adds views like dashboards and maps, plus results from the ML side where those are wired in. The TANIM Mobile App is built for farmers in plainer language: soil insights, recommendations, alert. Both apps use Supabase for secure access to stored data and connect to the machine learning side when the app needs model based recommendations.

3.5.2 Flowchart

Figure 3.10

System Flowchart



The figure 3.10 presents the operational workflow of the proposed soil health monitoring and recommendation system, highlighting the data flow from the field device to the end-user applications. The process initiates at the sensing stage, where the ESP32 microcontroller, powered through a solar energy management unit, interfaces with soil sensors to capture essential soil parameters, including Nitrogen, Phosphorus, and Potassium (NPK), pH, moisture, temperature, and soil salinity. These parameters are critical indicators of soil fertility and crop suitability. Once collected, the data is processed locally by the ESP32 to ensure efficient packaging before transmission. Leveraging LoRa technology, the device transmits the data wirelessly over long distances to a gateway, which serves as the intermediary link between the sensing unit and the cloud infrastructure. The gateway then forwards the collected data to a cloud database, ensuring reliable and scalable data storage for subsequent analysis.

Within the cloud environment, the raw soil data undergoes processing and analysis through machine learning models. These models are trained on historical soil and crop datasets to identify patterns, classify soil conditions, and predict nutrient deficiencies. By transforming raw sensor readings into interpretable outputs, the system generates actionable insights regarding soil health and crop management. The analytical results include fertilizer

recommendations and crop suitability assessments, enabling evidence-based decision-making for end users. This layer of intelligence ensures that data collected from the field is not merely stored but is also transformed into knowledge that directly supports precision agriculture practices.

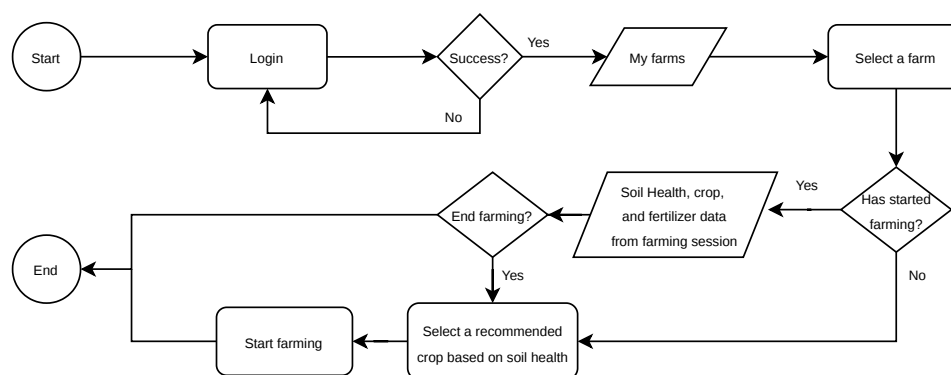
The processed insights are subsequently disseminated through two primary platforms: a web application and a mobile application. The web application is designed for agricultural authorities and extension officers, providing comprehensive monitoring dashboards, spatial analysis tools, and regional soil health reports. This enables decision-makers to track soil conditions at scale, identify priority intervention areas, and develop region-specific agricultural strategies. The mobile application targets farmers as the primary stakeholders, offering a user-friendly interface to access real-time soil health status, recommended fertilizer dosages, and crop-specific advisories. Notifications and alerts are integrated into the system to provide timely guidance, particularly in scenarios requiring immediate action.

An integral component of the workflow is the feedback mechanism incorporated within both platforms. Farmers and users are able to submit observations, report crop outcomes, and validate system recommendations. This feedback not only enhances farmer engagement but also provides valuable data for iterative improvements of the machine learning models. As a result, the

system establishes a continuous learning cycle wherein recommendations are refined over time, leading to greater accuracy and contextual relevance. Overall, the workflow described in Figure 3.10 demonstrates how the integration of IoT sensing, cloud computing, and machine learning facilitates a comprehensive, data-driven approach to soil health management and crop productivity enhancement.

Figure 3.11

Software Mobile Application



The figure 3.11 illustrates the workflow of the TANIM mobile application for farmers, highlighting how the farmer and the app interact from sign-in through farm choice, session handling, and crop management steps. The process begins with user authentication, wherein the farmer logs in with their credentials. The app checks whether the login is successful. If it is not, the user is returned to the login step so that only a verified session can proceed,

which helps protect the account. Upon successful login, the farmer is taken to the My farms area and from there must select a specific farm, which sets the context for the rest of the path.

The flow then tests whether the farmer has already started farming for that selection. If farming has not started, the app guides the farmer to select a crop from recommendations informed by soil health, and that choice is treated as the selected crop in the system. The user then uses the Start farming action, after which the illustrated workflow reaches its end. This path describes the situation where a new session of work is opened on a plot. If farming has already started, the app first presents soil health, the selected crop, and fertilizer data for the current farming session, so the user can see the state of active work before any further action.

A second question appears at this point, asking whether the farmer wants to end farming. If the answer is no, the flow goes directly to the end without passing again through crop selection and start. If the answer is yes, the farmer is steered back through selection of a crop on the basis of soil health, again as the selected crop, followed by Start farming, and then the same end of the run as in the new-session branch. In effect, the diagram shows two ways to move from login to closure: a straight path for someone who is only beginning, and a longer path for someone who is already in a session and can

either close the run after viewing data or end farming in a way that lines up with choosing a crop and starting again.

Viewed as a whole, the workflow underlines the order in which the app enforces access, then farm scope, then whether a session is active, and only then the mix of data views, the decision to end farming, and the steps for crop choice and the start of farming. This sequence clarifies how TANIM links secure entry, per-farm navigation, and session state to the screens that either introduce the selected crop and a new start, or let the user finish without repeating those steps when they choose not to end farming. The figure therefore supports a reading of the system as structured around the farmer's situation at each point rather than a single static screen, from first login through the possible loop that connects ongoing sessions to the next round of selection and the start of farming.

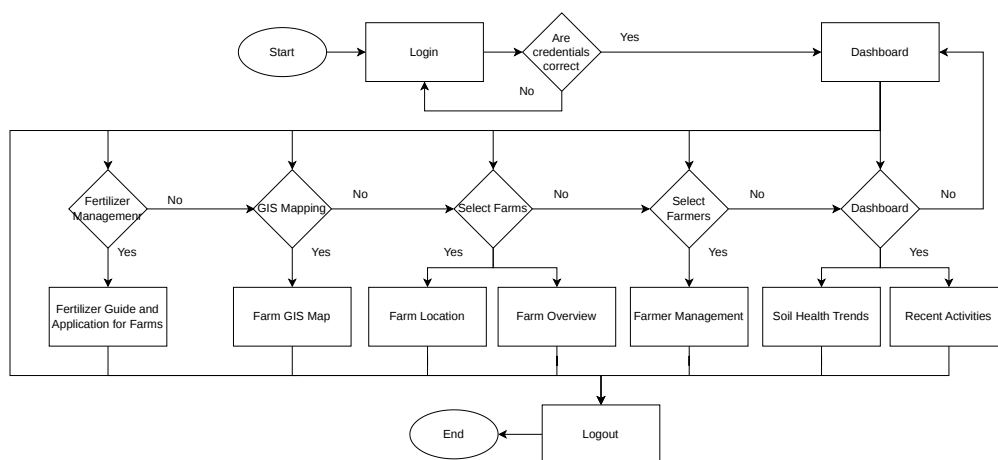
Figure 3.12*Web Application Flowchart*

Figure 3.12 illustrates the structured administrative workflow of the TANIM Web Application, which is designed to facilitate regional agricultural oversight and data management. The process initiates at the Start node, leading the administrator to a secure Login portal. At this stage, the system executes a verification protocol to determine if the entered credentials are correct. If the authentication fails, represented by the No path, the user is redirected to the login interface to re-attempt entry. Upon a successful Yes confirmation, the administrator is granted access to the primary Dashboard, which serves as the central hub for all subsequent system interactions.

From the central dashboard, the administrator can navigate through a series of functional modules governed by sequential decision nodes. The first

available path is Fertilizer Management, which, when selected, directs the user to the Fertilizer Guide and Application for Farms module. Moving further into the system, the GIS Mapping node allows for the generation of a Farm GIS Map, providing essential geospatial visualization for regional monitoring. For direct management of agricultural assets and personnel, the Select Farms node provides access to specific Farm Locations and comprehensive Farm Overviews, while the Select Farmers node opens the Farmer Management interface for profile and data administration.

The workflow also features an analytical branch through a secondary Dashboard decision point. By entering this view, the administrator can monitor longitudinal data such as Soil Health Trends and review a chronological log of Recent Activities within the system. To maintain high standards of data security and session integrity, all functional paths are integrated with a terminal Logout function. Once the administrator triggers this command, the session is formally concluded at the End node, ensuring that unauthorized access is prevented after the administrative tasks are completed.

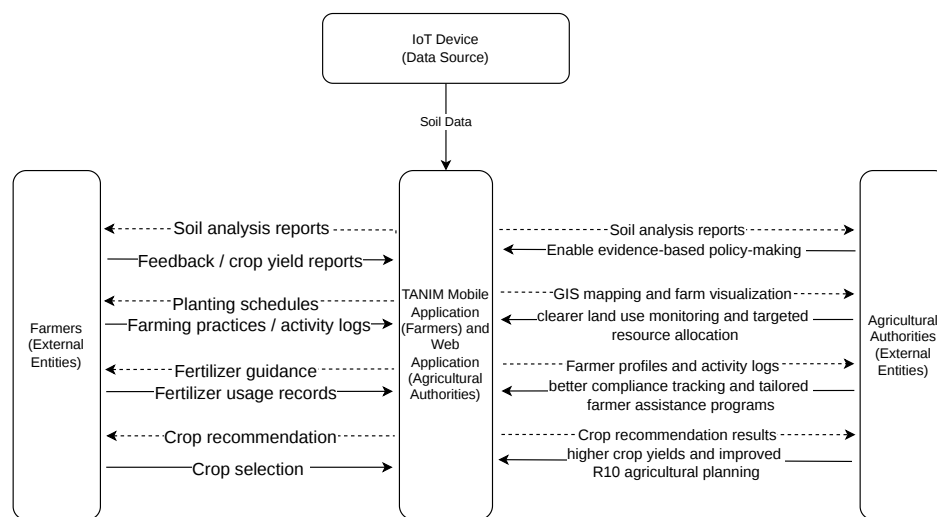
3.5.3 Context Level Diagram

The Context-Level Diagram provides a high-level view of how the TANIM system interacts with its primary stakeholders: Agricultural Authorities and

Farmers. It illustrates how data flows into and out of the system without detailing internal processes.

Figure 3.13

Context-Level Diagram Between the System And The External Entities



The Context Level Diagram shows how the TANIM system connects Agricultural Authorities and Farmers through web and mobile applications. Agricultural Authorities do not create the raw data themselves. Instead, they receive important information from IoT devices installed in the fields. This includes soil reports, farm maps based on location data, farmer profiles, and crop recommendation results. With this information, they can create better guidelines, manage resources properly, monitor farming practices, and improve planning. At the same time, IoT devices provide the system with real time

soil and environmental data. Farmers also contribute by adding key information such as harvest results, farm activity logs, fertilizer usage, and the crops they plan to grow. TANIM gathers all of this information and turns it into useful outputs like planting schedules, fertilizer recommendations, crop suggestions, and organized farm records. By doing this, the system transforms raw field data into practical guidance. This helps farmers follow better practices, improve crop production, and make faster and more informed decisions to support a stable food supply.

3.5.4 Data Flow Diagram

The Data Flow Diagram visually represents how data flows within the TANIM system. It highlights the interactions between IoT devices, farmers/agricultural authorities, and the system's core processes, demonstrating how data is retrieved, stored, and analyzed to provide soil health insights and crop/fertilizer recommendations.

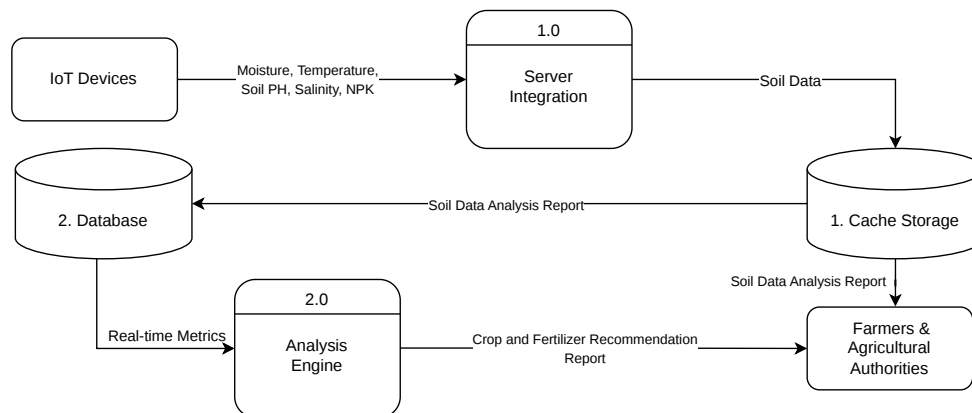
Figure 3.14*Data Flow Diagram*

Figure 3.8 shows the Data Flow Diagram (DFD) of the TANIM system. The data flow is divided into two main processes: server integration and the analysis engine.

The server integration process is responsible for collecting environmental data directly from the IoT devices. This includes specific sensor readings for moisture, temperature, soil pH, salinity, and NPK levels. Once received, this raw information is passed as continuous soil data and stored within the system's cache storage.

From the cache storage, the data flow branches into two distinct paths to serve the end users. First, it directly supplies a soil data analysis report to the Farmers and Agricultural Authorities. Second, it feeds "Real-time Metrics" into the system's analysis engine process.

In the analysis engine process, the system evaluates these real-time metrics to generate actionable agricultural insights. The direct output of this process is a tailored crop and fertilizer recommendation report, which is subsequently delivered to the farmers and agricultural authorities. Together, these reporting outputs enable farmers to make informed, data-driven decisions that optimize resource utilization and support strategic crop planning.

The system also includes a secondary data store, the Database, which links cache storage and the Analysis Engine. The cache storage passes the soil data analysis Report to the database for persistent storage, and the database in turn provides the real-time metrics consumed by the analysis engine.

3.5.5 Use Case Diagram

The Use Case Diagram illustrates how different actors interact with the TANIM system. It presents the system's primary functionalities and shows how users access and utilize agricultural data for farm management and monitoring. The diagram emphasizes the roles of both Farmers and the Department of Agriculture in supporting efficient and data-driven agricultural practices.

Figure 3.15
Use Case Diagram

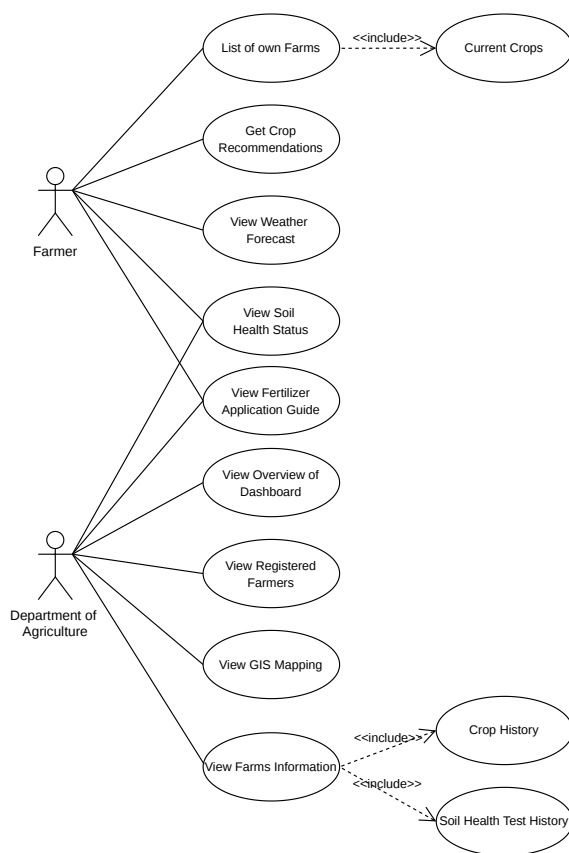


Figure 3.15 illustrates the interactions between the primary actors—the Farmer and the Department of Agriculture—and the TANIM system.

The Farmer interacts with the system to manage day-to-day agricultural operations. They can access a List of own Farms, which automatically includes the ability to view Current Crops currently under cultivation. To

aid in decision-making, Farmers can View Soil Health Status, View Weather Forecasts, and access a View Fertilizer Application Guide. Furthermore, the system allows them to Get Crop Recommendations, which leverages data from soil health and weather forecasts to suggest the most productive crops and necessary fertilizer adjustments.

The Department of Agriculture utilizes the system for high-level monitoring and regional oversight. They can View Registered Farmers and access a View Overview of Dashboard to monitor agricultural trends. Through the View GIS Mapping feature, the Department can visualize farm locations spatially. Additionally, the Department can View Farms Information, a functionality that includes access to both Crop History and Soil Health Test History, providing a comprehensive longitudinal view of land use and soil viability.

By bridging the gap between individual farm management and institutional data analysis, the TANIM system facilitates a data-driven approach to agricultural productivity and sustainability.

The Use Case Diagram demonstrates how TANIM supports both farm level management and institutional oversight. By providing relevant tools and information to different users, the system enhances coordination, monitoring, and decision-making in agricultural management

3.5.6 Use Case Table

The Use Case Table provides a structured and detailed representation of the functionalities presented in the Use Case Diagram. It enumerates each system function, identifies the primary actor involved, and describes the purpose of each interaction. In addition, the table highlights relationships among use cases through included or related functionalities.

The use cases associated with the *Farmer* focus on farm-level management and decision support. These include viewing a list of registered farms and current crops, monitoring soil health status, accessing weather forecasts, and receiving crop recommendations. These functionalities enable farmers to effectively manage their farming activities, assess environmental conditions, and make informed decisions to improve agricultural productivity. Certain use cases depend on related system features, such as soil health and weather data, which support the generation of crop recommendations.

The use cases associated with the *Department of Agriculture* emphasize monitoring, analysis, and regulatory oversight. These include viewing an overview dashboard that integrates soil health trends, weather conditions, and farm information, accessing farm locations through the Farm GIS feature, and viewing registered farmers. The Department may also view crop history and soil health history from testing soil.

Table 5: Primary Use Cases of the TANIM System

Use Case Name	Description	Primary Actor	
		Farmer	Department of Agriculture
List Own Farms	Enables farmers to view their registered farms.	✓	
View Soil Health Status	Allows users to view soil condition data such as nutrients, fertility, and health indicators.	✓	✓
View Weather Forecast	Provides weather forecasts relevant to farm locations.	✓	
Get Crop Recommendations	Provides crop recommendations based on soil health, weather, and farm data.	✓	
View Fertilizer Recommendations	Provides fertilizer application guide and timeline	✓	✓
View Dashboard Overview	Allows users to view a summarized overview of agricultural data for monitoring and analysis.		✓
View Farm GIS	Displays geographic locations of farms through a GIS interface.		✓
View Registered Farmers	Allows the Department of Agriculture to view all registered farmers in the system.		✓
View Crop Rules	Allows the Department to view and manage crop rules and regulations.		✓

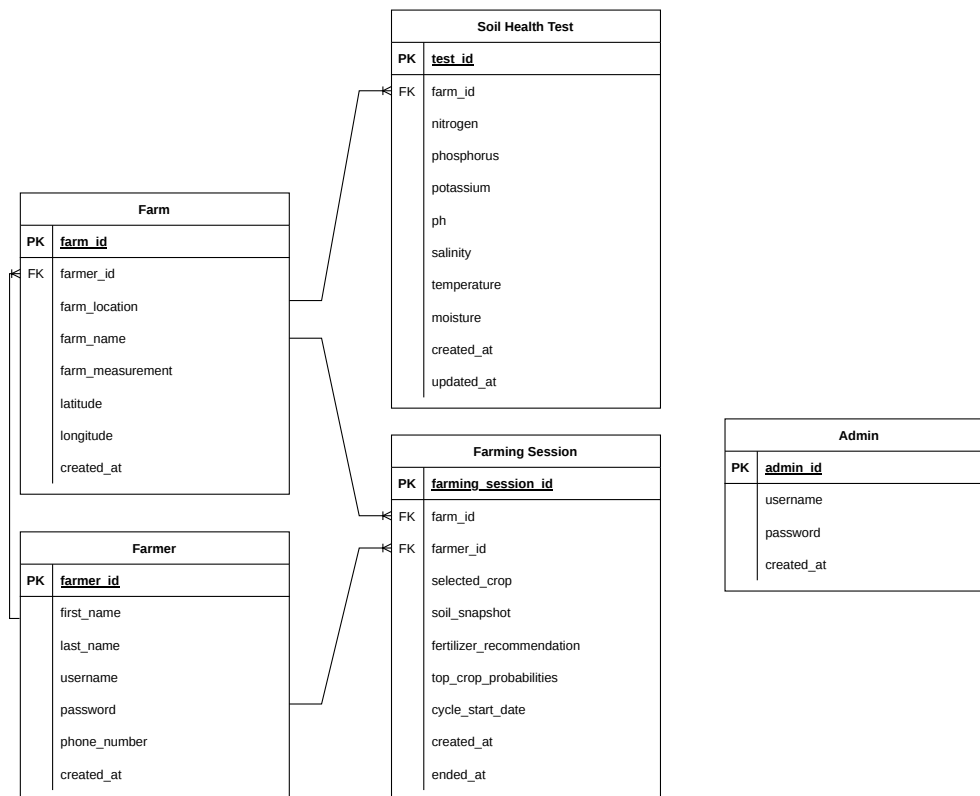
Table 6: Included Use Cases of the TANIM System

Use Case Name	Description	Primary Actor	
		Farmer	Department of Agriculture
View Current Crops	Allows farmers to view crops currently cultivated on their farms.	✓	
View Soil Health History	Allows Admins to view soil health history		✓
View Crop History	Allows Admins to view crop history		✓

3.5.7 Entity Relationship Diagram

The relational architecture of the database is designed to provide a structured and scalable environment for managing agricultural data through four primary entities: Admin, Farmer, Farm, and Farming Session. As illus-

trated in Figure 3.16, the Admin entity serves as the administrative oversight layer, storing unique identification and authentication credentials for system-wide management. The Farmer entity represents the primary endusers of the platform, containing detailed profile information including full names, contact numbers, and secure login credentials. This entity is linked to the Farm table through a foreign key (*farmer_id*), establishing a one-to-many relationship that allows a single farmer to register and monitor multiple distinct agricultural plots. Each farm record preserves critical metadata such as the plot name, physical area measurement, and precise geographic coordinates (latitude and longitude) to support spatial mapping

Figure 3.16*Entity Relationship Diagram*

The Soil Health Test entity stores discrete, time-stamped soil measurements for a given farm. Each test row is identified by a primary key (*test_id*) and is linked to the Farm table by a foreign key (*farm_id*), in a one-to-many relationship: one farm can have many soil health tests over time. The entity records macronutrients—nitrogen, phosphorus, and potassium—together with pH, salinity, temperature, and moisture, and it maintains *created_at* and *updated_at* timestamps for each record so the system can keep a verifiable history

of how soil conditions evolve for that plot.

Operational data and model-driven insights are managed within the Farming Session entity, which functions as a temporal log of field activity. This table acts as the nexus of the system, linking a specific farmer to a specific farm for the duration of a planting cycle. Rather than storing soil data as static attributes of a land plot, the system archives discrete “soil snapshots,” fertilizer recommendations, and model-generated top crop probabilities within the session record. This design allows the system to maintain a historical archive of soil health and decision-making for every visit or cycle. The session record also captures the user’s final decision in the selected crop field and 93 tracks the timing of the cycle through start, end, and creation timestamps. This normalized structure ensures data integrity while supporting complex longitudinal queries regarding farm performance and historical soil trends.

3.5.8 Hardware Design Flowchart

Figure 3.17

Hardware Design Flowchart

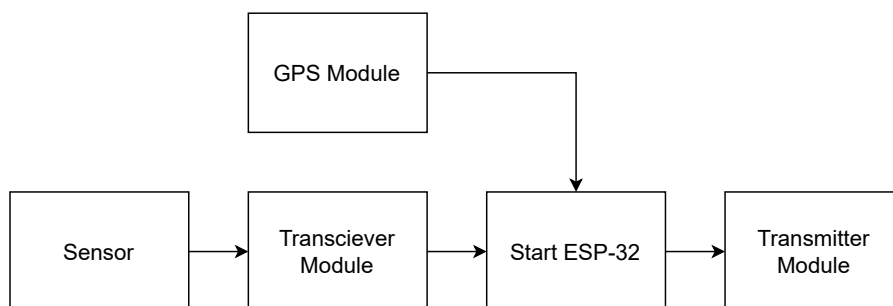


Figure 3.17 illustrates the workflow of the hardware components. The process begins with the initialization of the ESP32 microcontroller, which serves as the central control unit of the system. The GPS module provides location data to the ESP32, while the sensor module collects environmental data required for monitoring.

Once the sensor data is acquired, it is passed to the transceiver module, which facilitates communication between the sensing unit and the ESP32. The ESP32 processes the received data and forwards it to the transmitter module. Finally, the transmitter module sends the processed data to the designated external system, completing the hardware operation cycle.

The flowchart represents a sequential hardware process in which each

stage depends on the successful execution of the previous step. The ESP32 coordinates sensor data acquisition, GPS integration, and communication control, while the transceiver and transmitter modules ensure reliable data transmission.

3.5.9 Hardware Design Layout

This section presents the overall arrangement of the system's hardware components and how they are interconnected. It highlights the key elements used and how the layout supports efficient and reliable system operation.

Figure 3.18

Main System Hardware Layout

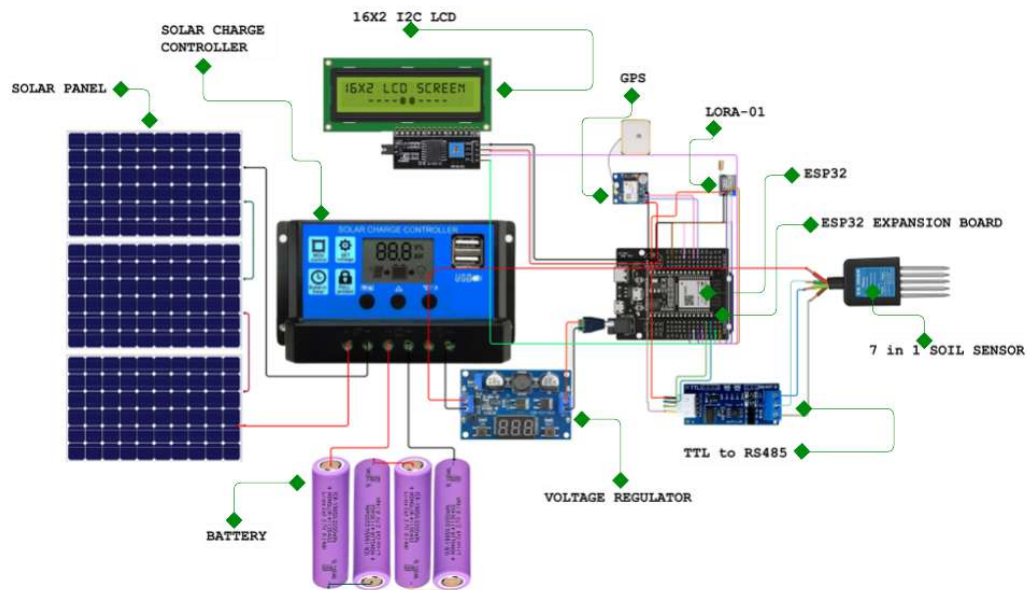


Figure 3.18 shows the hardware layout of the transmitter unit. The

transmitter circuit is powered by four 3.7V lithium-ion batteries, rechargeable via a three 5V 1A solar panels setup. The transmitter includes the ZTS-3001-TR-ECMOISTPHTEMPMPK-N01 Soil Health Sensor health sensor powered by the 4 batteries, then the sensor is connected to a transceiver which is the TTL to RS485 module the module serves as the bridge between the sensor and the ESP32 communication. Another component connected to the ESP32 is the GY-NEO6MV2 GPS module which provides the latitude and longitude of the tested area. The I2C-LCD is also connected to the ESP32 as its purpose is to show the real-time sensor reading, Latitude and Longitude, and the data transmission status. And the last component is the LoRa-01 module which is used for wireless transmission of the sensor reading and the GPS reading and each packet will be sent every minute after the 5 minute soak time.

Figure 3.19

Receiver System Hardware Layout

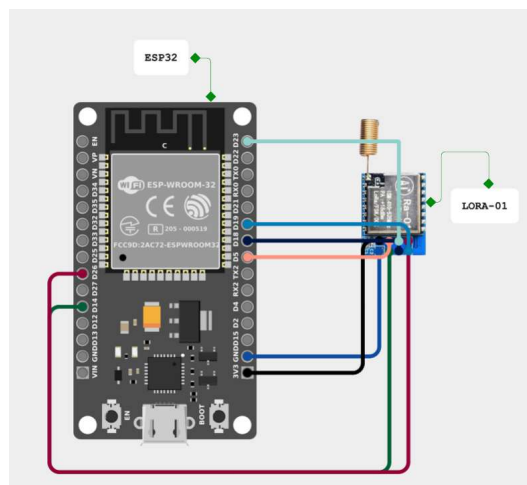


Figure 3.19 shows the hardware layout of the receiver unit which is only connected to a LoRa-01 module. Its only purpose is to collect data from the three transmitter nodes then average it and the averaged data will be sent to the database. This unit requires Wi-Fi connection to send the averaged data to the database.

Building upon the hardware design layout, the schematic layout further details the electrical connections between components. It visualizes the signal paths and power distribution across the system, ensuring that each component is properly integrated for optimal performance and reliability.

Figure 3.20

Main Transmitter Schematic Diagram

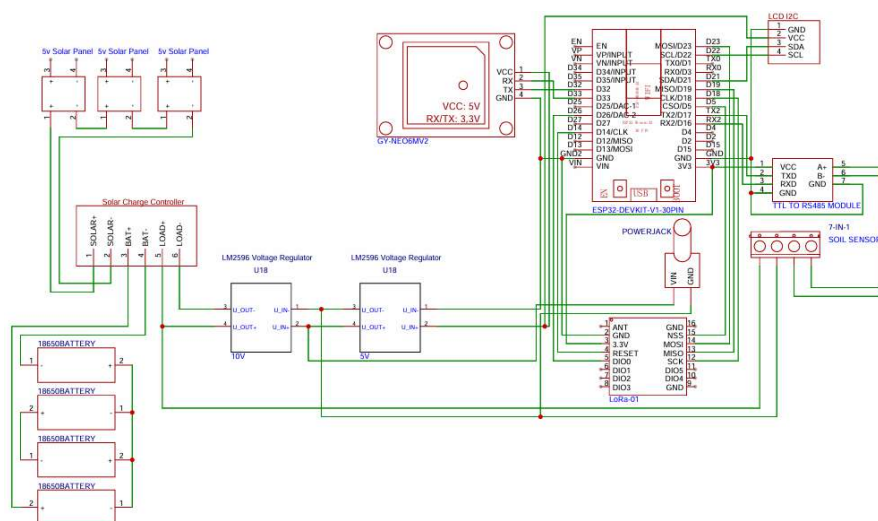


Figure 3.20 shows the main transmitter circuit's detailed schematic,

illustrating the integration of the central processing unit with environmental sensors and positioning modules

Figure 3.21

Node 2 and 3 Transmitter Schematic Diagram

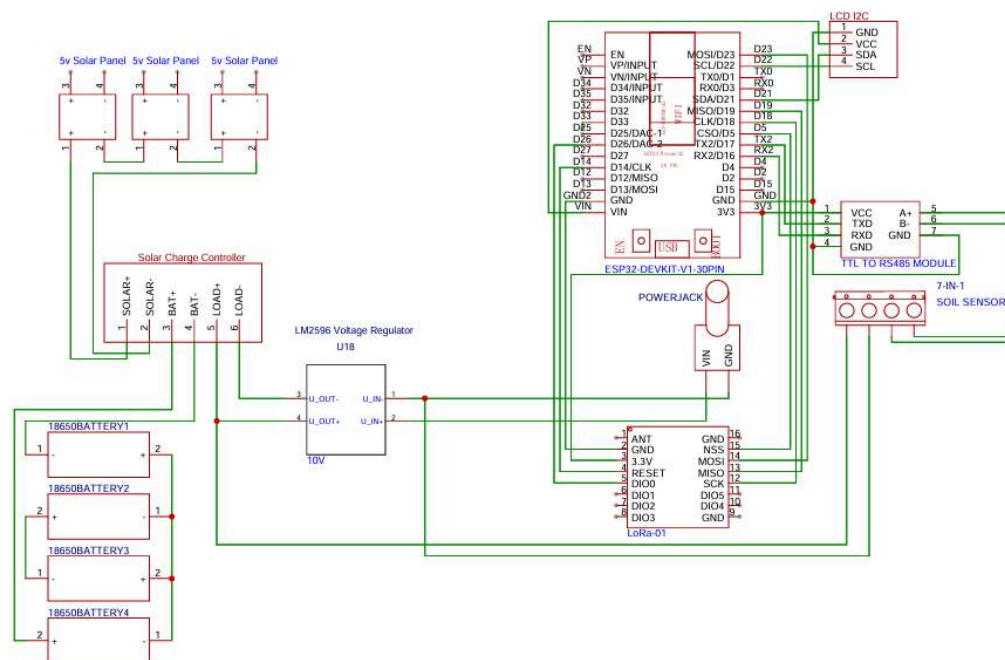


Figure 3.21 shows the node 2 and 3 circuit's detailed schematic, illustrating a modular design focused on localized environmental monitoring.

Figure 3.22
Receiver Schematic Diagram

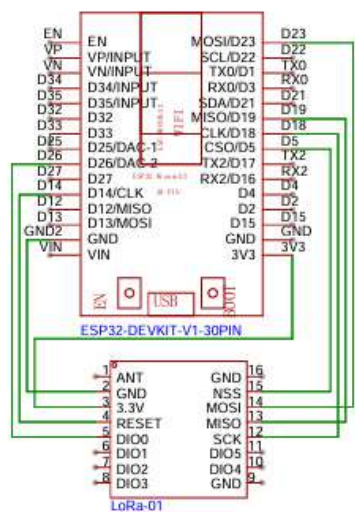


Figure 3.22 shows the receiver circuit’s detailed schematic, illustrating the high-speed SPI connection between the ESP32 and the LoRa Ra-01 communication module.

3.5.10 3D Design

Figure 3.23

3D Container Design

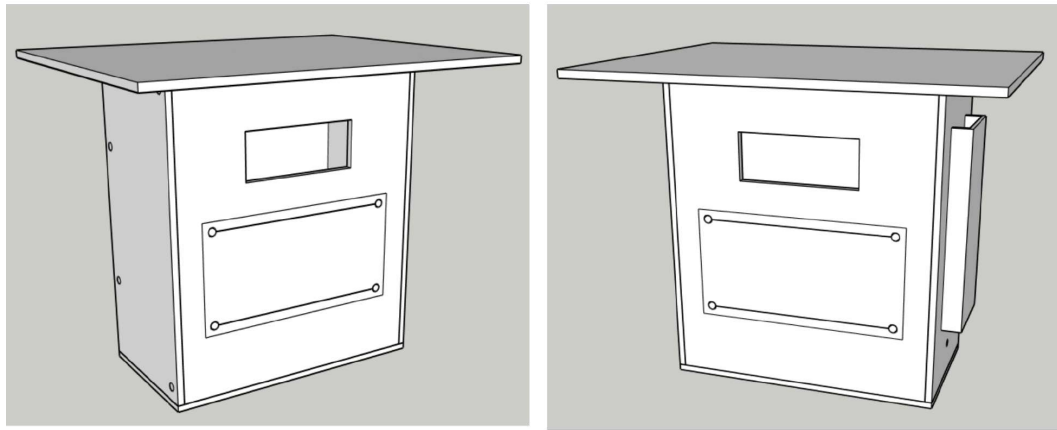


Figure 3.23 displays the 3D container design that was used for the container of the hardware. This contains the hardware components which are etched in a PCB board that was used in this system. The container should be durable to ensure that it can secure the hardware components inside of it.

3.5.11 Materials and Cost

Table 7: Materials and Cost Breakdown of the Hardware System

Component	Quantity	Cost (PHP)	Amount (PHP)
ZTS-3001-TR-ECMOISTPHTEMPMPK-N01 Soil Health Sensor	3	1,862	5,586
ESP32	4	400	1,600
TTL to RS485 Module	3	57	171
GY-NEO6MV2 NEO-6M GPS Module	1	300	300
SX1278 LoRa Ra-01 Module (433MHz)	4	289	1,156
I2C Serial Interface 1602 LCD Module	3	400	1,200
5V 1A Solar Panel	9	220	1,980
Solar Charge Controller	3	145	435
Lithium-Ion 18650 2500mAh Battery	12	400	4,800
Miscellaneous (Cables, Mounts)	–	3,000	3,000
Total Cost			20,000

3.5.12 Graphical User Interface (GUI)

This section describes the graphical user interface (GUI) of the TANIM application, which is designed to be user-friendly and facilitate efficient agricultural management. The GUI consists of several screens, including the Admin Dashboard and the Soil Health Analysis screen.

Figure 3.24
Admin Dashboard

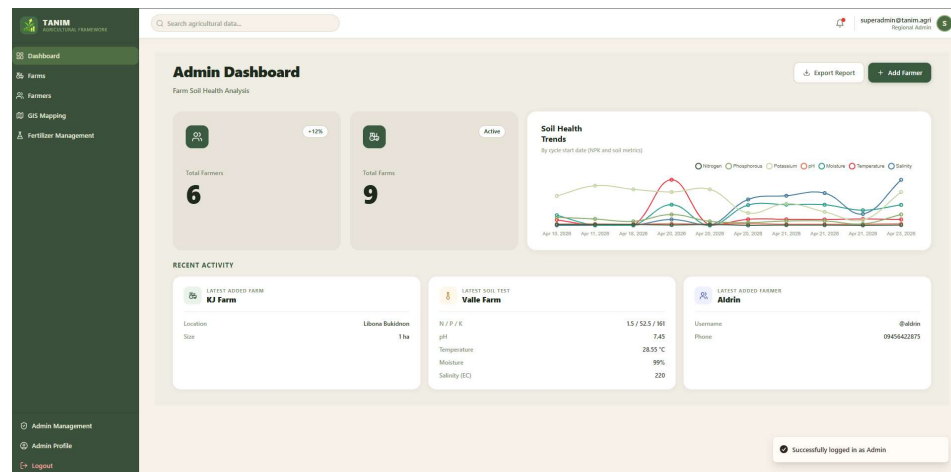


Figure 3.24 displays the Admin Dashboard, which provides a comprehensive overview of the farm's soil health data and allows for the management of various agricultural activities. The dashboard features a sidebar with navigation links to key sections, such as "Dashboard", "Farms", "Farmers", "GIS Mapping", and "Fertilizer Management". The main content area showcases essential metrics, including "Total Farmers" and "Total Farms", along with a "Soil Health Trends" chart that tracks soil parameter trends over time.

The "Soil Health Trends" chart in Figure 3.24 provides a visual representation of nitrogen, phosphorus, potassium, pH, moisture, temperature, and salinity levels, allowing users to monitor soil health changes and make informed decisions about fertilizer application and other management practices.

The dashboard also includes a "Recent Activity" feed that displays latest farm registrations and soil test results, providing real-time updates on farm-related activities.

Figure 3.25

Soil Health Analysis

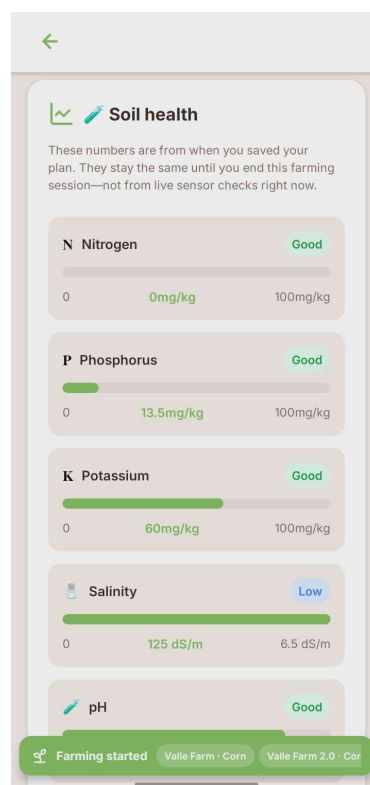


Figure 3.25 shows the Soil Health Analysis screen, which provides a more detailed breakdown of soil nutrient levels and pH. This screen displays a bar chart illustrating the amounts of nitrogen, phosphorus, and potassium, along with pH and salinity levels. Each soil parameter is labeled with its status, such as "Good" or "Low", and the exact measurement in mg/kg for nutrients

and dS/m for salinity. This granular data enables farmers to accurately assess soil health and implement targeted soil management strategies.

The Soil Health Analysis screen also features a notification bar at the bottom, indicating "Farming started" for specific farms and crops. This feature helps farmers stay organized and keep track of their planting and harvesting schedules.

The TANIM mobile application's GUI is designed to provide farmers with an intuitive and efficient way to manage their agricultural data and make informed decisions about soil health and crop management. The Admin Dashboard offers a high-level overview of key metrics and trends, while the Soil Health Analysis screen provides detailed information on soil nutrient levels and pH. Together, these features empower farmers to optimize their crop yields and enhance their agricultural practices.

3.6 Implementation

The Implementation phase involves developing the software and hardware components. Each function was ensured to function according to the design specifications. This phase intends to transform the conceptual design into a working prototype that can perform data acquisition, processing, and output for TANIM.

3.6.1 Software Implementation

The software implementation for TANIM is designed to provide an efficient, reliable, and user-friendly platform that caters to the needs of both the Farmer and the Department of Agriculture. To ensure optimal accessibility and performance, the frontend for farmers is developed using React Native, a framework well-suited for creating responsive and cross-platform mobile applications. The mobile application also features offline data handling, allowing farmers to access previously synced soil and crop data even without an internet connection. Any new inputs or updates made while offline are temporarily stored locally on the device and automatically synchronized with the cloud once connectivity is restored. This ensures uninterrupted functionality in rural areas with limited network coverage.

Meanwhile, the frontend for the Department of Agriculture is implemented using React.js, providing a seamless and interactive web application experience. The web application integrates advanced analytics and visualization tools, enabling agricultural authorities to interpret soil data, monitor farm conditions, and analyze regional agricultural trends through interactive charts, dashboards, and geospatial maps. These insights support data-driven planning, resource allocation, and policy formulation.

Both user interfaces interact with a shared backend developed in Flask,

which centralizes data management and ensures consistent functionality across platforms. This architecture enables efficient communication between the mobile and web applications while supporting scalability, maintainability, and ease of integration with additional features or services in the future.

3.6.2 Hardware Implementation

The hardware implementation of the TANIM system is centered on an IoT-based device designed to collect soil data and send it to the cloud for processing and analysis. The core of the device is the ESP32 microcontroller, selected for its high processing capability, built-in Wi-Fi and Bluetooth connectivity, low power consumption, and compatibility with multiple sensor modules. These features make it ideal for continuous field monitoring and data transmission in remote agricultural areas.

A 7 in 1 soil sensor was used to measure key soil parameters such as pH, NPK, moisture, temperature, and salinity. This sensor was chosen because it provides accurate and real-time soil data essential for evaluating soil health and guiding fertilizer and crop recommendations.

A GPS module provides geographical localization data, essential for mapping sensor readings to specific field coordinates. For long-range data transmission in remote agricultural areas, a LoRa module is integrated as

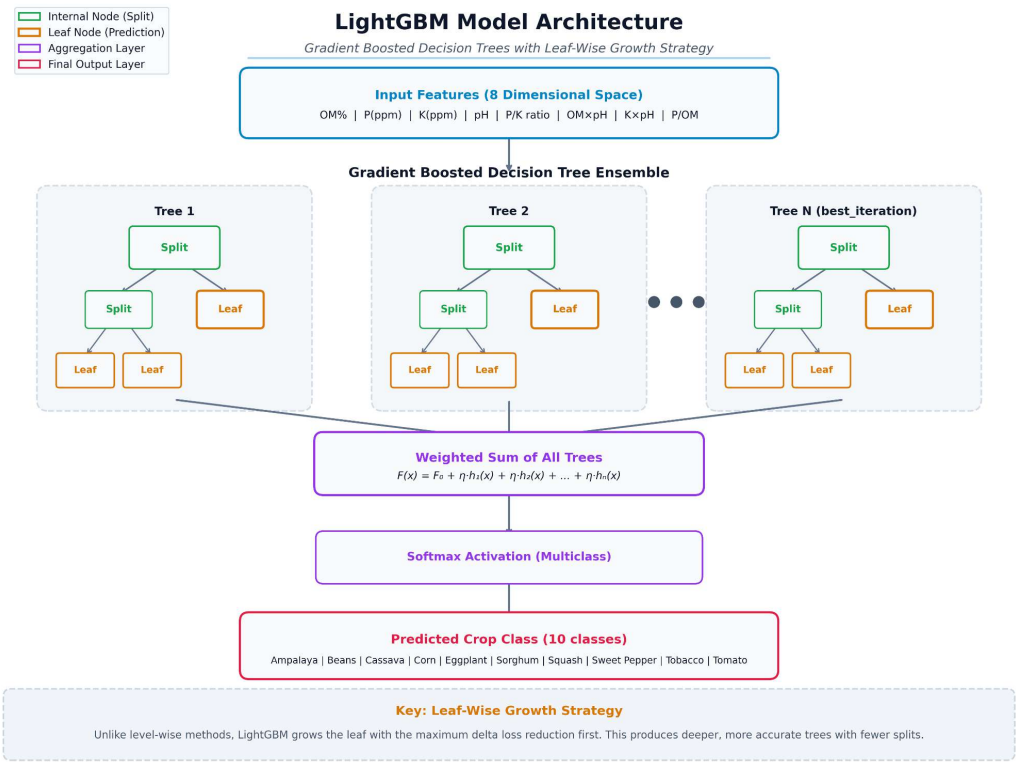
the system's transmitter, enabling robust, low-power wireless communication. Additionally, an I2C-LCD displays instantaneous sensor readings locally for immediate field-level monitoring.

3.6.3 Machine Learning Model

The TANIM project used a LightGBM model because it processes data quickly and accurately sorts information into different categories. Since regional soil data often has uneven amounts of information for various soil types, the system applies a K-Means method to address this imbalance. This method generates additional data for the underrepresented groups, creating a balanced dataset for the model to learn effectively.

The program evaluates key soil elements, specifically organic matter, phosphorus, potassium, and pH levels, to assess soil health and predict the most suitable crops for a given area. By combining the LightGBM model with a balanced dataset, the system improves prediction performance and provides reliable agricultural recommendations.

Figure 3.26
LightGBM Model Architecture used in the TANIM system



The integration of LightGBM with a balanced dataset enhances classification accuracy and ensures fair representation of all soil categories. As shown in Figure 3.26, the LightGBM architecture uses an ensemble of decision trees that aggregate predictions to produce final crop classifications. This structure allows the system to efficiently process soil parameters and generate accurate crop recommendations, ultimately helping improve harvest yields and optimize resource utilization.

3.7 Test and Evaluation

Testing and evaluation was conducted to verify the system's effectiveness, accuracy, and usability for crop recommendation and soil health monitoring. This process was designed to ensure that the hardware components, software components, and machine learning models functions according to what we aim for this study, which involves hardware testing and calibration, functional testing, system integration testing, performance testing, and usability testing. Hardware testing and calibration focuses on verifying the accuracy of the sensors to ensure reliable soil data. Functional testing examines whether the mobile and web applications operate correctly and deliver the expected outputs of each feature. To ensure that the hardware, cloud database, machine learning, and applications works together with proper data flow and error handling, system integration was implemented. For measuring the responsiveness of the system and to evaluate the system's predictive accuracy of the machine learning models using various performance metrics we be conducting performance testing. Moreover, usability testing was used to assess the practicality and user-friendliness of the system by involving farmers and agricultural experts in field trial and feedback collection.

3.7.1 Hardware Testing and Calibration

The soil sensor pH parameter was submitted to the Department of Science and Technology (DOST) for formal calibration using three certified buffer solutions at pH 4.0, 7.0, and 10.0. The raw sensor readings recorded at each reference point were 4.8, 5.7, and 9.0, respectively. 8 presents the raw sensor output against the known buffer values and the corresponding error at each point.

Table 8: pH Sensor Raw Readings and Required Corrections

Buffer Solution (True pH)	Raw Sensor Reading	Error (Sensor – True)	Required Correction
4	4.8	+0.8	-0.8
7	5.7	-1.3	+1.3
10	9.0	-1.3	+1.3

The error pattern was non-uniform: the sensor over-read at pH 4.0 but under-read at pH 7.0 and pH 10.0 which indicates a non-linear response characteristic across the measurement range. This ruled out a single constant offset as a sufficient correction method. Three calibration approaches were evaluated and compared: a single constant offset, linear regression, and a three-point quadratic fit.

Single Constant Offset. A uniform correction of +0.5 pH units was derived by averaging the three required corrections $(-0.8+1.3+1.0)\div 3 = +0.5$

This method produced the corrected values shown in 9.

Table 9: Applied Correction (Sensor + 0.5) and Resulting Residual Error

Buffer Solution (True pH)	Raw Sensor Reading	Corrected Value (Sensor + 0.5)	Error
4	4.8	5.30	+1.30
7	5.7	6.20	-0.80
10	9.0	9.50	-0.50

On the other hand, To validate the nutrient measurements, the team employed a Soil Test Kit (STK) method to compare sensor output against established agronomic thresholds. Specifically, sensor readings for Nitrogen, Phosphorus, and Potassium were cross-referenced with the STK results to see if they consistently fell within the same qualitative categories of low, medium, or high. For instance, a sensor reading for Potassium would be evaluated against a predefined range where values below 117 ppm are categorized as low and those above 235 ppm are considered high. This threshold-based verification allowed the team to confirm that the hardware system provides reliable categorical data suitable for crop recommendation despite the lack of high-level technical calibration tools.

3.7.2 Functional Testing

Functional testing involves verifying whether each feature of the mobile and web applications operates correctly according to the system design. For

the mobile application, the following features was tested: secure login and authentication, retrieval of soil health readings, display of crop recommendations for multi-cropping, fertilizer optimization, and offline data synchronization. For the web application, features such as farm visualization, farmer profile management, and farms GIS mapping was tested.

In addition, synchronization between the database, mobile, and web applications was tested to ensure that all data are accurately updated and consistent across platforms. This includes verifying that soil readings, user information, and crop recommendations are properly synchronized when data are sent or retrieved from the cloud.

Both black-box testing and white-box testing approaches was applied to evaluate the system's functionality. Black-box testing focuses on verifying the system's external behavior without examining its internal code. Testers inputs various data to check if the outputs match the expected results, ensuring that all functions operate as intended from a user's perspective. Meanwhile, white-box testing was conducted by examining the internal logic, data flow, and code structure of the applications. This helps identify hidden logical errors, broken functions, or inefficient code implementations.

3.7.3 System Integration Testing

System integration testing evaluates whether the hardware, software, cloud database, and machine learning model communicates seamlessly with each other as a single system. The system integration testing was tested by following the full flow of the system: soil health readings captured by the sensor and processed by the ESP32 microcontroller, then transmitted by via the LoRa module to the gateway, forwarded to the cloud database, then processed by the machine learning models, and finally displayed on the mobile and web applications.

This process confirms that the system works together as one reliable system. Furthermore, several test was conducted to observe how the system handles incomplete or corrupted data to ensure that it can manage error to prevent misleading recommendations to farmers.

3.7.4 Performance Testing

For the system level performance, it is to determine if the system could handle multiple data acquisition and provide timely feedback to the users. The data transmission success rate was monitored by comparing the successful sensor to cloud uploads against the attempted uploads over multiple trials. Latency tests measure the time from the sensor data capture to its display in

the mobile and web application, with an acceptable range set to 30 seconds. Scalability was evaluated by simulating multiple farms connected simultaneously to ensure that the cloud database and applications could handle increased workloads without delays and crashes. These ensures that the system is reliable and efficient under real-world agricultural conditions where multiple users operate.

For the machine learning evaluation, it verifies the predictive capability of the machine learning models that will be used on crop recommendation for multi-cropping and fertilizer optimization. The dataset will be split into 70-15-15 for training, evaluation, and testing for maximizing the predictive accuracy and avoiding overfitting.

For the crop recommendation, the machine learning models was evaluated using accuracy, precision, recall, F1-score, and confusion matrices to determine the effectiveness of the system. The following formulas are going to be use:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

$$\text{Confusion Matrix} = \begin{bmatrix} TN & FP \\ FN & TP \end{bmatrix} \quad (5)$$

These evaluation steps confirm that the machine learning models provide scientifically sound predictions and deliver crop and fertilizer recommendations that are consistent, reliable, and effective under different scenarios.

3.7.5 Usability Testing

The usability testing assesses the accessibility, and user satisfaction of the system. For this, farmers was asked to use the mobile application, review soil health data, view crop and fertilizer recommendations, and simulate updating their farm records. Meanwhile, DA officials use the web dashboard to access farmer profiles, view GIS-based maps, and farm profiles.

A total of 10 participants was involved in the usability testing, consisting of 7 farmers from Region 10 and 3 agricultural technicians or DA employees. Farmers were selected based on their familiarity with basic smartphone usage and their active involvement in small- to medium-scale farming. The DA personnel were chosen because they represent the system's administrative users responsible for monitoring and validating agricultural data. This mix of participants ensures that both end-user and administrative perspectives are considered during evaluation.

After completing the navigation of the system, the System Usability Scale (SUS) developed by Brooke (1995) was employed to quantitatively measure user satisfaction. The interactions of the users was observed, and any difficulties they encounter was recorded. In addition to the SUS scores, qualitative feedback was collected through short interviews and observation notes to gain insights into which features are most useful and which aspects require improvement. The evaluation criteria determines the user satisfaction based on the SUS scores and user feedback. The result of this testing determines the system's usability rating and this will be a guide for further refinement to ensure that the TANIM system is accessible, user-friendly, and beneficial to the users.

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the findings from the research conducted and provides a thorough analysis and interpretation of these results.

4.1 Hardware System

This section presents the results of the sensor hardware testing and calibration conducted to evaluate the accuracy, reliability, and consistency of the developed IoT-based soil sensing unit. As described in Chapter 3, the system measures key soil parameters including pH, nitrogen (N), phosphorus (P), potassium (K), soil moisture, temperature, and salinity. The accuracy of these parameters is critical since they directly influence the crop recommendation and fertilizer advisory outputs of the system.

The testing phase aimed to (1) verify the correctness of sensor readings, (2) calibrate the sensors against reference values, and (3) evaluate the performance of the sensing unit under controlled conditions. The results obtained from this phase serve as the foundation for validating the reliability of the entire system.

4.1.1 Hardware and Specifications

Figure 4.1

Soil Health Monitoring Device



Figure 4.1 presents the developed system that utilizes three 5V, 1A solar panels connected in series, generating approximately 15V. This configuration successfully meets the minimum input requirement of the solar charge controller, which regulates the incoming power and ensures proper battery charging. Energy is stored in four 18650 lithium-ion batteries configured to supply at least 12V, enabling stable operation of all components.

A buck converter was implemented to reduce the output voltage from

the charge controller to 10V, which serves as the primary input for the ESP32 expansion board. This ensured a consistent and reliable power supply throughout system operation.

For data acquisition, the ZTS-3001-TR-ECMOISTPHTEMPMPK-N01 soil sensor was used to measure seven parameters: soil moisture, temperature, pH, electrical conductivity, and nutrient levels (nitrogen, phosphorus, and potassium). The sensor operated within its required voltage range and produced stable readings when supplied with at least 12V. Communication between the sensor and the microcontroller was achieved using a TTL-to-RS485 module, which successfully converted RS485 signals into 3.3V TTL signals compatible with the ESP32.

Location tracking was implemented using the GY-NEO6MV2 NEO-6M GPS module, which provided accurate latitude and longitude data during testing.

The ESP32 microcontroller, paired with an expansion board, served as the central processing unit. The expansion board accepted a 7V–12V input, while internally regulating power for the ESP32. Wireless data transmission was carried out using the SX1278 LoRa module (433 MHz), which operated at 3.3V and enabled long-range communication. All modules were properly interfaced through the ESP32's GPIO pins.

An I2C LCD was integrated to display real-time sensor readings. The display operated at 5V and communicated with the ESP32 via the I2C protocol, providing immediate feedback during system testing.

4.1.2 Accuracy and Reliability Evaluation

To assess the accuracy and reliability of the calibrated soil health sensor under actual field conditions, validation testing was conducted using soil samples collected from farms in Libona and Lingion, Bukidnon. These locations were selected based on farm data provided by the Department of Agriculture - Region 10, ensuring representative soil conditions across the region. The validation process involved cross-checking calibrated sensor readings against two reference standards: the Department of Agriculture's Soil Test Kit (STK) results and laboratory analysis results obtained from the Department of Agriculture's Regional Soils Laboratory using the same farm data samples they provided from multiple farms.

Three soil samples from Libona Bukidnon were analyzed using the Soil Tests Kits (STK) methodology (see Appendix 5), while three additional samples from Lingion Bukidnon were processed through the Department of Agriculture laboratory as part of their ongoing soil monitoring program. The laboratory reference values used in this comparison were extracted directly from

the Department of Agriculture Region 10’s soil database, ensuring consistency with established regional soil health records. This dual-reference approach provided both qualitative (color-based) and quantitative validation of the sensor performance across different soil types and nutrient conditions found in the target agricultural areas.

Table 10: Comparative Analysis of Soil Health Measurements

Sample	Soil Test Kit Results				Sensor Readings				DA Laboratory Results			
	(Libona Samples)				(Libona Samples)				(Lingion Samples)			
	N	P	K	pH	N	P	K	pH	N	P	K	pH
1	MEDIUM	HIGH	LOW	5.8	1 (LOW)	21.5 (HIGH)	65 (LOW)	5.65	2.60 (MEDIUM)	9.50 (LOW)	335.00 (HIGH)	7.0
2	LOW	HIGH	MEDIUM	5.8	1.5 (LOW)	21 (HIGH)	105 (LOW)	5.3	2.60 (MEDIUM)	4.10 (LOW)	72.00 (LOW)	6.78
3	MEDIUM	HIGH	MEDIUM	5.8	0 (LOW)	7 (LOW)	26 (LOW)	5.7	3.0 (HIGH)	1.5 (LOW)	235 (HIGH)	5.33

The validation results revealed varying degrees of agreement between the sensor readings and reference methods. For the Libona samples compared against STK results, the sensor demonstrated strong categorical alignment for phosphorus and pH parameters, with readings consistently falling within the same qualitative ranges (LOW, MEDIUM, HIGH). However, some discrepancies were observed in nitrogen and potassium classifications, particularly in Sample 3 where the sensor underestimated phosphorus levels compared to the STK indication.

Table 11: Validation Assessment Summary

Sample	STK vs Sensor Agreement				Lab vs Sensor Agreement			
	N	P	K	pH	N	P	K	pH
1	Partial	Exact	Exact	Exact	Partial	Divergent	Partial	Exact
2	Exact	Exact	Partial	Exact	Partial	Divergent	Partial	Exact
3	Partial	Divergent	Partial	Exact	Exact	Divergent	Partial	Exact

Legend:

- Exact**

Sensor reading falls within the same categorical range as reference method.
- Partial**

Sensor reading is within one categorical level of the reference (e.g., LOW vs MEDIUM).
- Divergent**

Sensor reading differs by two or more categorical levels from the reference.

Table 11 presents the validation assessment results of the proposed system by comparing sensor-derived soil measurements against two reference methods, namely Soil Test Kit (STK) readings and laboratory-based soil analysis. The evaluation focuses on four key soil parameters: Nitrogen (N), Phosphorus (P), Potassium (K), and pH, and classifies the level of agreement into Exact, Partial, and Divergent categories to determine the reliability of the sensor outputs in both field-level and laboratory contexts.

For the STK vs Sensor comparison, the results indicate generally strong alignment across most parameters. Nitrogen shows a mixed performance, with both Exact and Partial agreements across samples, suggesting minor variabil-

ity in sensor estimation but still within acceptable categorical proximity. Phosphorus demonstrates consistently high agreement, achieving Exact matching in two out of three samples, indicating that the sensor closely replicates STK-based interpretations for this nutrient. Potassium exhibits more variability, with one instance of Partial agreement and two Exact or near-Exact classifications, reflecting moderate sensitivity to environmental fluctuations in potassium measurement. Meanwhile, pH consistently achieves Exact agreement across all samples, confirming high stability and strong calibration accuracy of the sensor for acidity measurement when compared to STK results.

In contrast, the Lab vs Sensor comparison reveals a more challenging alignment due to the higher precision and quantitative nature of laboratory measurements. Nitrogen shows a combination of Partial and Exact agreement, indicating that while the sensor can approximate laboratory trends, minor deviations in absolute concentration values still occur. Phosphorus consistently records Divergent outcomes across all samples, suggesting significant discrepancies between sensor readings and laboratory measurements, likely due to differences in detection sensitivity or calibration scale. When compared to DA laboratory results from Lingion samples, the sensor showed consistent accuracy for pH measurements (100% exact agreement) but revealed systematic differences in nutrient readings. The phosphorus measurements showed the greatest

divergence, with the sensor consistently reading higher values than laboratory results. This pattern suggests potential calibration drift or methodological differences between sensor chemistry and laboratory analytical procedures. Potassium also shows predominantly Partial agreement, indicating that while general trends are captured, numerical precision remains limited compared to laboratory standards. Despite this, pH continues to maintain Exact agreement across all samples, reinforcing its reliability as the most stable and accurately measured parameter within the system.

Table 11 demonstrates that the proposed sensing system performs strongly in categorical classification, particularly for pH and partially for Nitrogen, while showing moderate limitations in accurately matching laboratory-grade precision for Phosphorus and Potassium. However, the consistent occurrence of Exact and Partial agreements suggests that the system is still effective for decision-support purposes in agricultural applications, where categorical interpretation is often more critical than exact numerical equivalence.

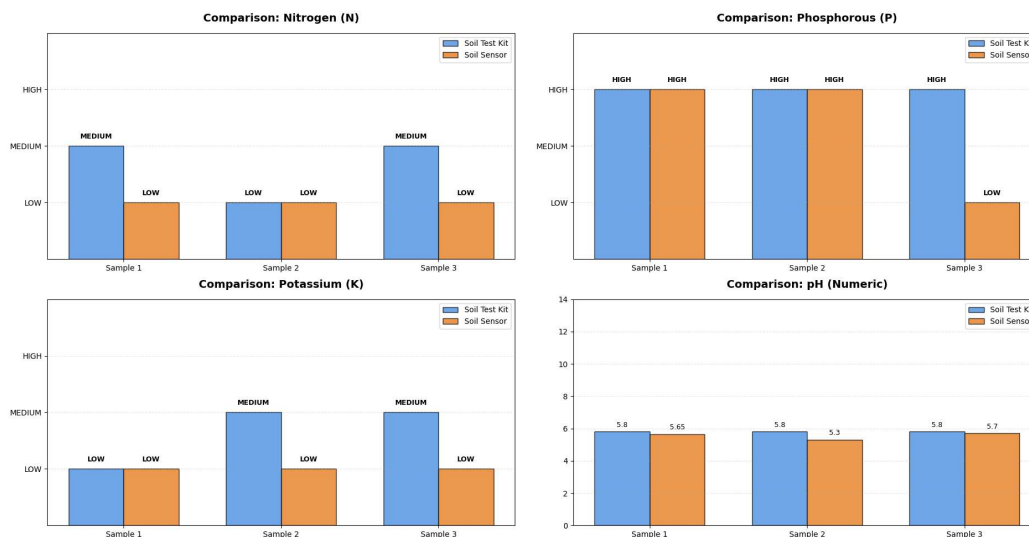
Figure 4.2*Comparison of Calibrated Sensor Readings and Soil Test Kit Values*

Figure 4.2 presents a detailed comparison between the calibrated sensor readings and Soil Test Kit results for each soil parameter across the three Libona samples. The visual representation demonstrates the sensor's performance in categorical classification, with notable consistency in pH measurements and varying degrees of alignment in nutrient parameters.

Figure 4.3

Comparison of Calibrated Sensor Readings and Laboratory Reference Values

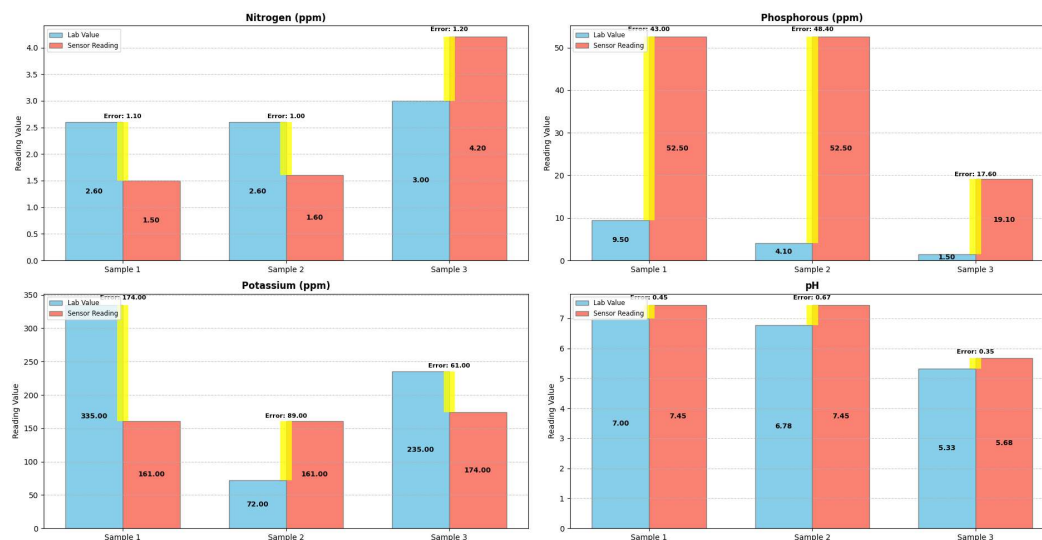


Figure 4.3 illustrates the comparison between calibrated sensor readings and laboratory reference values from the DA Region 10 database for the Lington samples. The graphical analysis reveals systematic differences between sensor and laboratory measurements, particularly evident in the phosphorus and potassium parameters, while maintaining strong agreement in pH values.

To quantify the average deviation between the sensor output and the laboratory values, the Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) were calculated for each parameter.

Table 12: Error Metrics for Soil Sensor Parameters

Sensor Parameter	Mean Absolute Error	Root Mean Squared Error
Nitrogen	1.10	1.11
Phosphorous	36.33	38.74
Potassium	108.00	118.21
pH	0.49	0.51

Potassium exhibited the highest prediction error among all measured nutrients, with a Mean Absolute Error (MAE) of 108.00 and a Root Mean Squared Error (RMSE) of 118.21. This relatively large deviation, including a recorded peak difference of 174 ppm in certain comparisons, indicates significant variability in potassium estimation between the sensor outputs and the reference laboratory values. Despite this numerical discrepancy, the model still maintained correct qualitative classification in terms of categorical interpretation (e.g., LOW, MEDIUM, HIGH), suggesting that while absolute precision is limited, the system remains reliable for decision-level interpretation in agricultural applications.

Phosphorus also demonstrated relatively high error values, with an MAE of 36.33 and RMSE of 38.74, reflecting moderate inconsistency in predicting exact concentration levels. The variation may be attributed to the sensitivity of phosphorus measurements to soil heterogeneity and environmental sampling differences. However, the system still achieved agreement in two

out of three samples when evaluated categorically, indicating that despite numerical deviations, the predictive model retains acceptable performance in class-based fertilizer recommendation contexts.

Nitrogen showed comparatively lower error levels, with an MAE of 1.10 and RMSE of 1.11, indicating strong alignment between sensor readings and reference values. This consistency resulted in correct categorical matching in two out of three samples, demonstrating that nitrogen estimation is relatively stable and reliable within the implemented sensing and modeling framework. The reduced error suggests better sensor calibration or stronger model learning patterns for nitrogen compared to other macronutrients.

Among all measured parameters, pH exhibited the highest accuracy, with an MAE of 0.49 and RMSE of 0.51. The minimal deviation between predicted and reference values reflects strong sensor stability and effective calibration for hydrogen ion concentration measurement. pH achieved a 100% categorical match rate across all samples, confirming its reliability as a highly dependable indicator within the system. Overall, these results highlight that while certain nutrients such as potassium and phosphorus present challenges in precise numerical estimation, the system remains effective for practical agricultural decision-making through accurate categorical classification.

4.2 Machine Learning Model

This section describes the data processing pipeline and predictive model used for crop recommendation. The model takes processed soil measurements as input and outputs a recommended crop and fertilizer guidance.

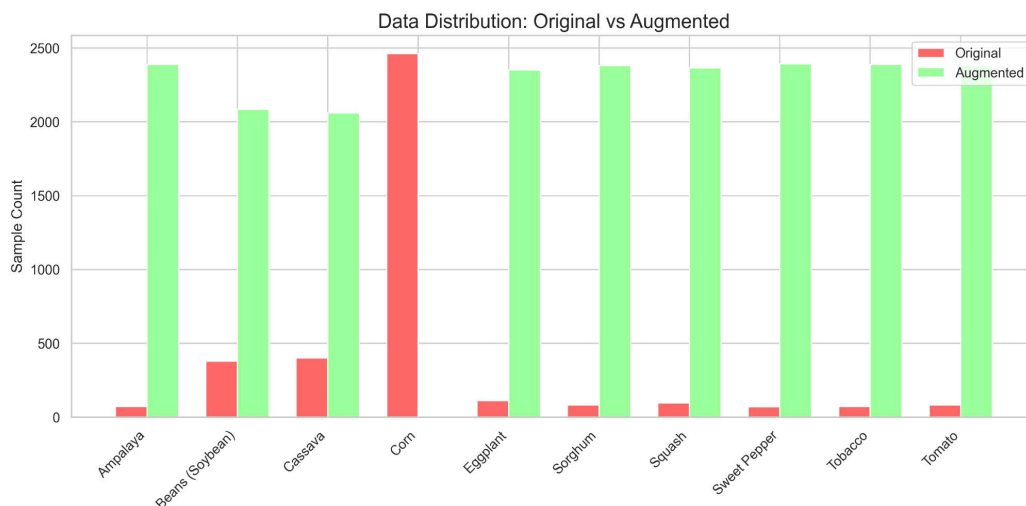
4.2.1 Data Integration and Cleaning

Multiple data sources were consolidated to form the training dataset. Historical records (the Consolidated Soil Info RSL 2023–2025) provided soil profiles (pH, organic matter %, phosphorus ppm, potassium ppm) linked to crops previously grown on those farms. These records were combined with simulated IoT sensor samples to create a uniform table. Irrelevant metadata and text annotations were removed so that each entry consists only of the agronomic variables and a single target crop label. For example, entries RSL-001 through RSL-005 in Table 13 show specific soil chemical properties aligned with their zoned crops (Corn, Eggplant, etc.), demonstrating how fragmented spreadsheet data was transformed into a structured format for learning.

Table 13: Sample Integrated Soil Data Records

Record ID	Soil pH	Organic Matter (%)	Phosphorus (ppm)	Potassium (ppm)	Zoned Target Crop
RSL-001	7.22	2.80	34.60	47.00	Corn
RSL-002	5.09	4.70	1.00	121.00	Camote (Sweet Potato)
RSL-003	5.09	4.70	1.00	121.00	Corn
RSL-004	5.09	4.70	1.00	121.00	Eggplant
RSL-005	5.09	4.70	1.00	121.00	Tomato

Data cleaning ensured dataset quality: entries with missing or invalid values (non-numeric, out-of-range) were dropped. Numerical features were bounded to agronomically realistic limits (e.g. soil pH restricted to 3–9) to eliminate outliers. A critical issue was class imbalance. In the raw data, common crops like Corn had thousands of records while rare crops (Ampalaya, Sweet Pepper) had very few. To prevent bias, synthetic oversampling was applied to minority classes. Using a K-Means-based approach, we grouped minority samples and generated perturbed examples around each cluster center. This produced a balanced dataset, as visualized in Figure 4.4. Before augmentation (red), the class distribution is skewed; after augmentation (green), each of the 10 crops has roughly 2,400 samples, yielding about 8,694 total records (1,191 original + 7,503 synthetic).

Figure 4.4*Data Distribution Comparison Between Original and Augmented Datasets*

4.2.2 Feature Engineering

In addition to the four raw inputs, four derived features were computed to capture agronomic relationships. Two features are ratios: the Phosphorus-to-Potassium ratio P/K and the Phosphorus-to-Organic Matter ratio P/OM . The other two are interaction terms: $OM \times pH$ and $K \times pH$. These encode how nutrient availability changes with soil acidity or organic content. For instance, a low $K \times pH$ simulates potassium lock-up in acidic soils. The engineered features substantially improved model insight: in the trained LightGBM, the $K \times pH$ term was the third most important feature, and the phosphorus ratios together contributed nearly 20% of total importance.

These algorithmically derived features, summarized in Table 14, mirror physiological nutrient constraints and bio-availability dynamics that are critical for accurate soil-health assessment.

Table 14: Feature Engineering Outcomes and Agronomic Justifications

Engineered Feature Name	Mathematical Formulation	Agronomic Justification
Phosphorus-Potassium Ratio	$\frac{\text{Phosphorus (ppm)}}{\text{Potassium (ppm)} + 1}$	Evaluates the internal root development-to-fruiting energy balance. A severe imbalance indicates a localized deficiency incapable of supporting advanced botanical maturity.
Phosphorus-Organic Load	$\frac{\text{Phosphorus (ppm)}}{\text{Organic Matter (\%)} + 1}$	Quantifies the availability of Phosphorus relative to the baseline organic mass, identifying environments where high raw nutrients are inaccessible due to poor soil degradation.
pH-Gated Nitrogen Bio-availability	$\text{Organic Matter (\%)} \times pH$	Synthesizes an intuition for real-world Nitrogen uptake. Because organic nitrogen conversion is mediated by microbiological activity, this encodes active bio-availability.
pH-Gated Potassium Bio-availability	$\text{Potassium (ppm)} \times pH$	Simulates real-world Potassium fixing limitations. In highly acidic environments, absolute Potassium lock-up occurs; this forces the model to acknowledge acidity constraints.

4.2.3 Model Configuration and Training

The prediction engine is a Light Gradient Boosting Machine (LightGBM) classifier. LightGBM is an efficient gradient-boosting framework optimized for speed and handling complex feature interactions. It was chosen for its ability to train quickly and run on modest hardware, making it suitable for IoT integration. The eight features (4 raw + 4 engineered) are fed into the classifier, which outputs probabilities for each of the 10 crop classes. Data splits were 70% training, 15% validation, 15% testing (stratified to preserve class ratios). To fine-tune performance, we used Bayesian hyperparameter optimization (Optuna, 50 trials) with 5-fold stratified cross-validation on the training set. We optimized tree depth, number of leaves, learning rate, feature/bagging fractions, and L1/L2 regularization. Early stopping was applied if validation log-loss did not improve for 20 rounds. This process ensured a well-generalized model without overfitting.

4.2.4 Performance Metrics and Results

The final model achieved a test-set accuracy of 83.2%. Its balanced accuracy (the macro-average recall across classes) was 82.5%, indicating robust performance on both common and rare crops. The weighted F1-score was 0.831. Figure 4.5 presents precision, recall, and F1-score per crop. The

model performed particularly well on Tobacco ($F_1 \approx 0.96$) and Sorghum, while Eggplant was the most challenging ($F_1 \approx 0.66$), consistent with its confusion with similar crops.

Figure 4.5
Model Classification Report: Precision, Recall, and F1-Score Breakdown

MODEL CLASSIFICATION REPORT			
Precision, Recall, and F1-Score Breakdown			
	precision	recall	f1-score
Ampalaya	0.833	0.854	0.843
Beans (Soybean)	0.850	0.875	0.862
Cassava	0.825	0.839	0.832
Corn	0.752	0.776	0.764
Eggplant	0.695	0.636	0.664
Sorghum	0.917	0.933	0.925
Squash	0.775	0.795	0.785
Sweet Pepper	0.915	0.891	0.903
Tobacco	0.969	0.961	0.965
Tomato	0.779	0.756	0.768
accuracy	0.832	0.832	0.832
macro avg	0.831	0.832	0.831
weighted avg	0.831	0.832	0.831

The confusion matrix as illustrated in Figure 4.6 shows the detailed pre-
diction outcomes. Diagonal entries are highest, indicating correct predictions;

only a few off- diagonal cells occur (e.g. 81 instances of Eggplant predicted as Tomato). Overall, the classifier’s predictions align well with ground truth, which is crucial for reliable recommendations.

Figure 4.6
Confusion Matrix of the LightGBM Classifier Across Ten Crop Varieties

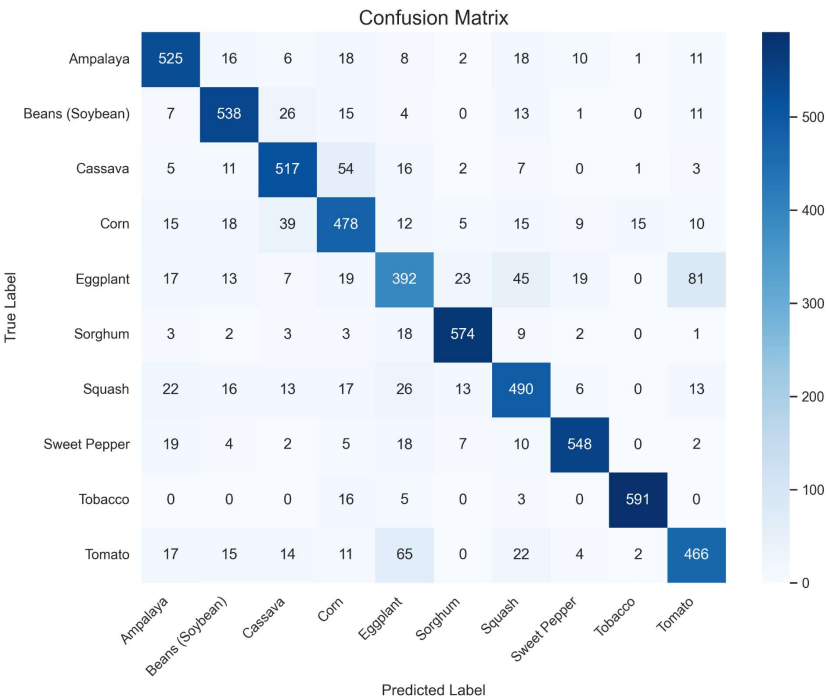
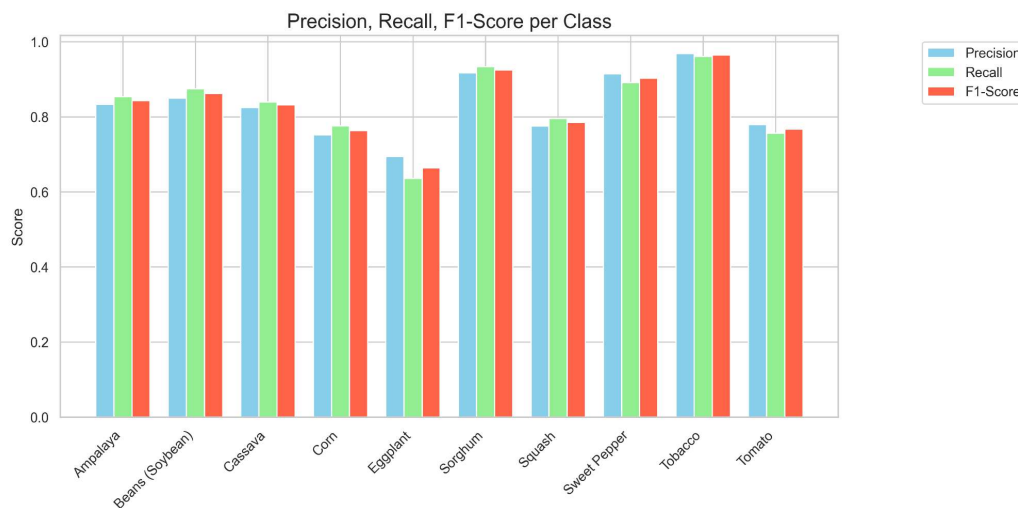
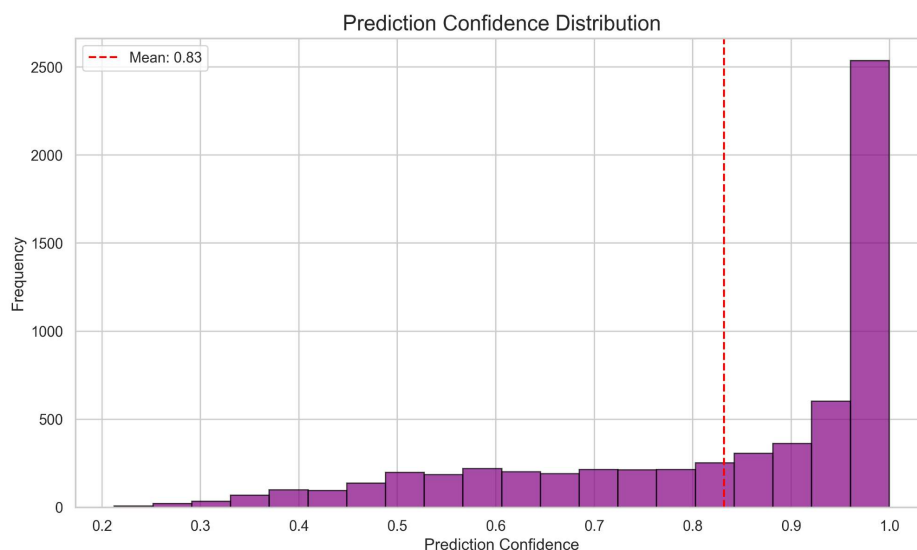


Figure 4.7 confirms these metrics. All crop classes achieve reasonably high precision and recall, with Tobacco and Sorghum highest, and Eggplant lowest. The model thus successfully identifies both majority and minority crops.

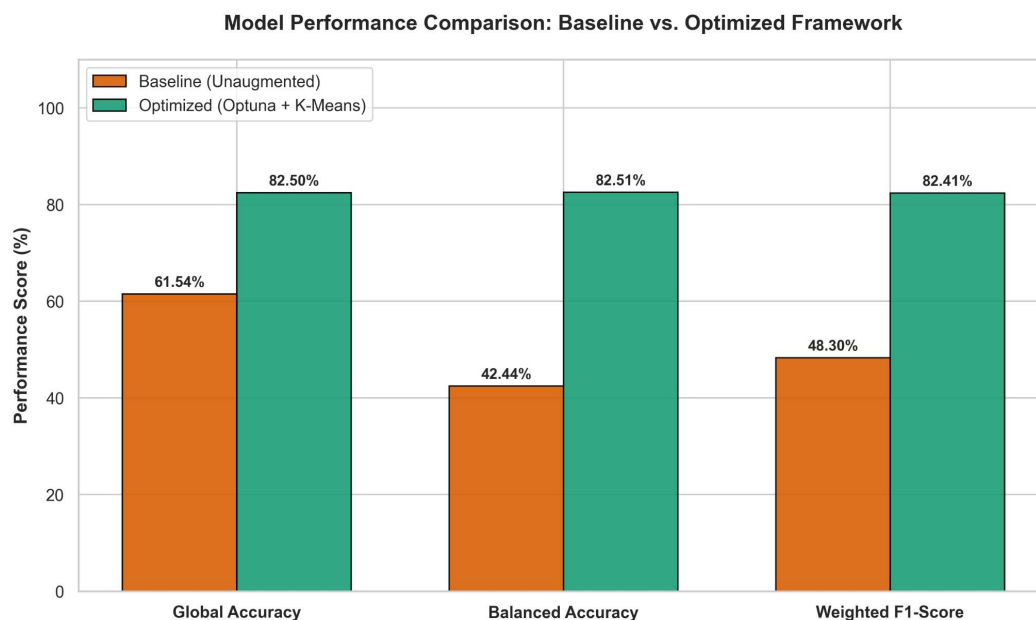
Figure 4.7*Precision, Recall, and F1-Score Breakdown per Crop Category*

The model's predicted class probabilities were generally high-confidence.

Figure 4.8 shows that the mean prediction confidence is about 0.83, with many predictions above 0.90. High confidence suggests the model is decisive, which builds trust that recommendations are statistically supported.

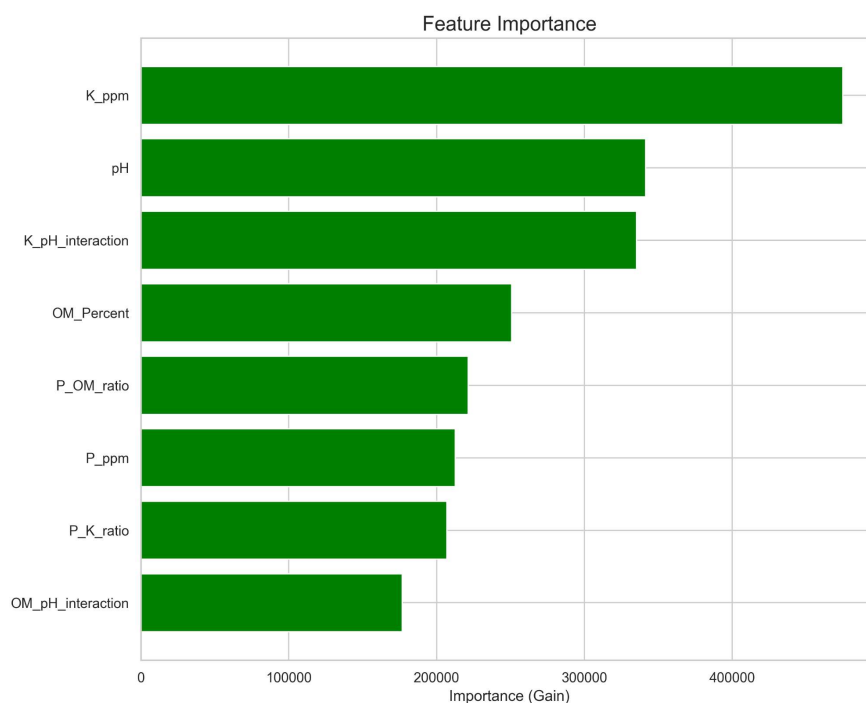
Figure 4.8*Prediction Confidence Distribution of the Optimized Classifier*

Finally, Figure 4.9 compares performance before and after our optimizations. A baseline model on the unaugmented, imbalanced data had only about 42% balanced accuracy. After oversampling and tuning, balanced accuracy rose to 82.5%, demonstrating the necessity of data augmentation and hyperparameter search to achieve meaningful accuracy.

Figure 4.9*Model Performance Comparison: Baseline vs. Optimized Framework*

4.2.5 Feature Importance Analysis

To validate the role of each input, we extracted LightGBM feature importances (information gain). Figure 4.10 shows that soil potassium (Kppm) and pH are the top raw predictors, as expected. Notably, the engineered $K \times pH$ feature is the third most important (15.1% of total gain), and the phosphorus ratios also contribute significantly (10.0% and 9.3%). This confirms that the model effectively leverages both raw and engineered features to capture complex agronomic relationships.

Figure 4.10*Feature Importance Analysis Measured by Information Gain*

4.3 Mobile Application

The TANIM mobile application provides an integrated platform that allows users, particularly farmers, to access soil data, crop recommendations, and fertilizer guidance directly from their mobile devices. The application is designed with a user-friendly interface that guides users through a step-by-step workflow, from authentication to decision-making and farm management.

4.3.1 System Overview

The system architecture is designed to serve as the primary gateway for the TANIM framework, beginning with a secure login interface as shown in Figure 1. It acts as a bridge between the physical soil sensors and the user, providing a centralized location for monitoring registered farms and analyzing environmental conditions. By providing a clear organizational structure, the system allows farmers to navigate through different stages of the agricultural cycle with data-driven efficiency.

Figure 4.11

TANIM Mobile Application Login Page

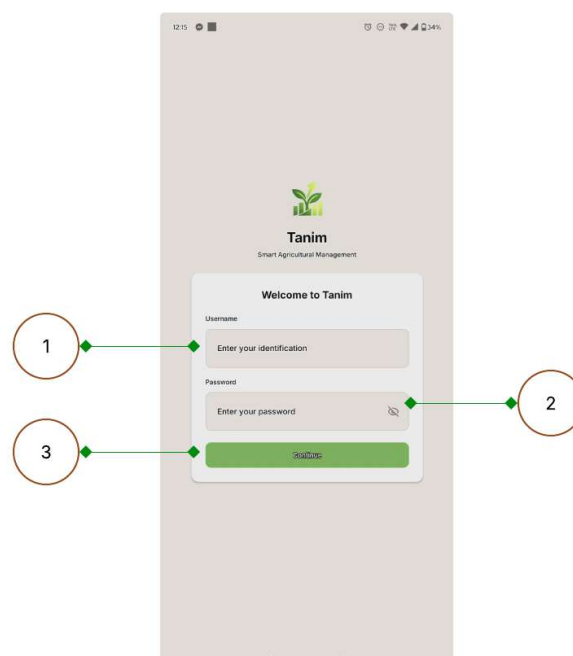


Figure 4.11 shows the login interface, it serves as the entry point of the mobile application. It includes the username input field (1), where the user enters their unique identification, and the password input field (2), which securely captures the user's credentials and includes a show or hide feature for usability. The login button (3) allows users to authenticate and proceed to the main dashboard upon successful verification.

4.3.2 Core Functionalities

The core functionalities of the application begin with user authentication via the Farmer Login Interface, which utilizes a username input field and a secure password field that includes a usability feature to show or hide the credentials. Upon successful login, users can proceed to the main dashboard to view the My Farms page as illustrated in Figure 4.12, which provides a card-based view of registered farms and their respective land areas. For a selected farm, the application displays detailed soil health parameters including Nitrogen, Phosphorus, Potassium, salinity, pH, moisture, and temperature, all of which are collected via hardware sensors as shown in Figure 4.13.

Figure 4.12

TANIM Mobile Application My Farms Page

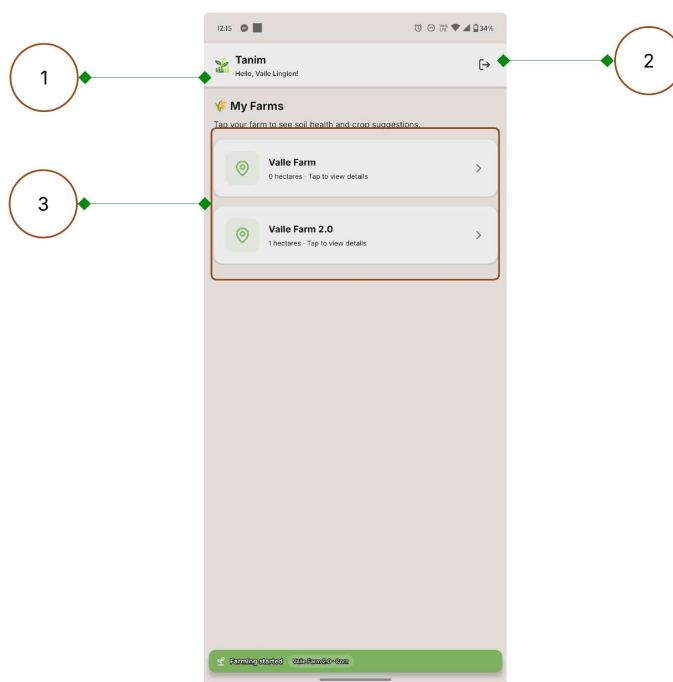


Figure 4.12 presents the My Farms dashboard, this screen displays a personalized greeting through the user name (1), creating a user-centered experience. A logout button (2) is provided for secure exit from the application. The main component of this page is the list of farms (3), which presents a card-based view of the user's registered farms, including relevant details such as farm names and land area.

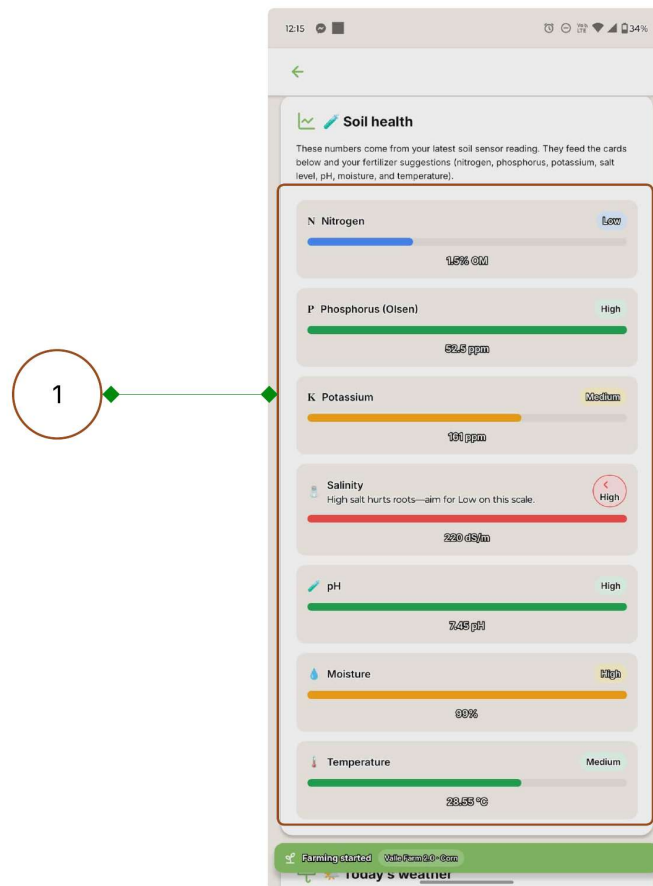
Figure 4.13*TANIM Mobile Application Selected Farms Page*

Figure 4.13 shows the soil health parameters of the tested soil, it provides detailed information collected from soil sensors. This includes a list of parameters (1) such as Nitrogen, Phosphorus, Potassium, salinity, pH, moisture, and temperature. Each parameter is visually represented using color-coded indicators to classify levels as low, medium, or high, allowing users to

quickly assess soil conditions.

The system further utilizes Machine Learning to provide predictive insights through the Weather and Suggested Crops screen as shown in Figure 4.14, which offers a ranked list of crops suitable for current soil conditions along with compatibility percentages. To support the selected crop, the Fertilizer Suggestions and Timeline screen in Figure 4.15 provides a calendar-based planting schedule and a detailed phase schedule covering sowing, vegetative growth, flowering, and harvest. Comprehensive nutrient recommendations are provided through multiple fertilizer options as shown in Figure 4.16 and Figure 4.17, allowing users to select the most appropriate combination based on availability. Finally, the user can initiate the monitoring process by engaging the start farming session button.

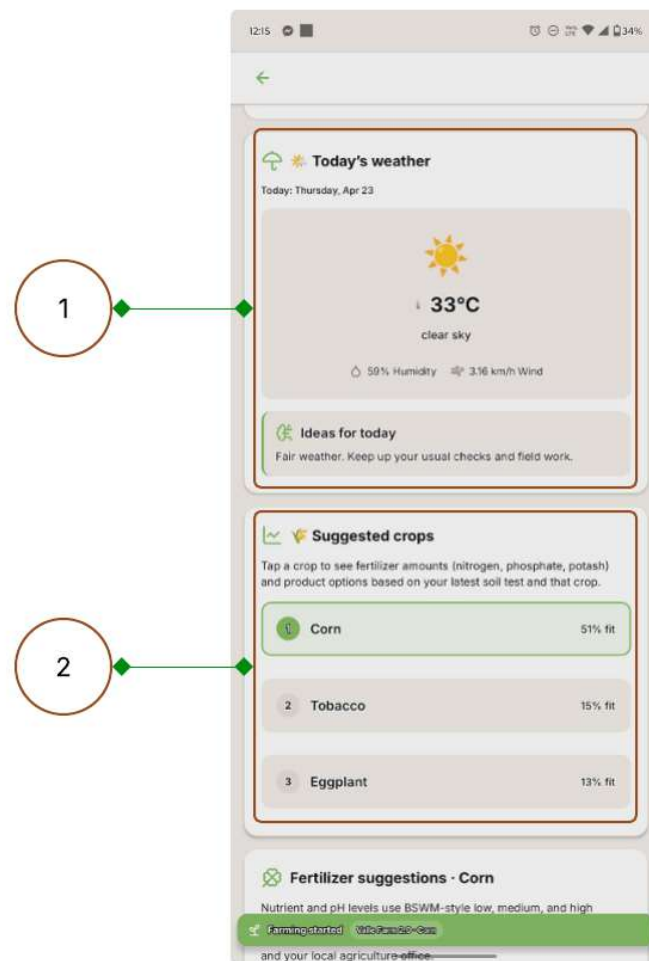
Figure 4.14*TANIM Mobile Application Weather and Crop Selection*

Figure 4.14 illustrates the Weather and Suggested Crops screen, it provides environmental context and predictive insights. The weather forecast (1) displays current conditions, including temperature and general weather status, along with practical recommendations for farm activities. The generated

suggested crops (2) section presents a Machine Learning - based ranking of crops suitable for the current soil conditions, including percentage-based compatibility indicators that help users identify the most viable crop options.

Figure 4.15

TANIM Mobile Application Fertilizer Suggestion and Timeline

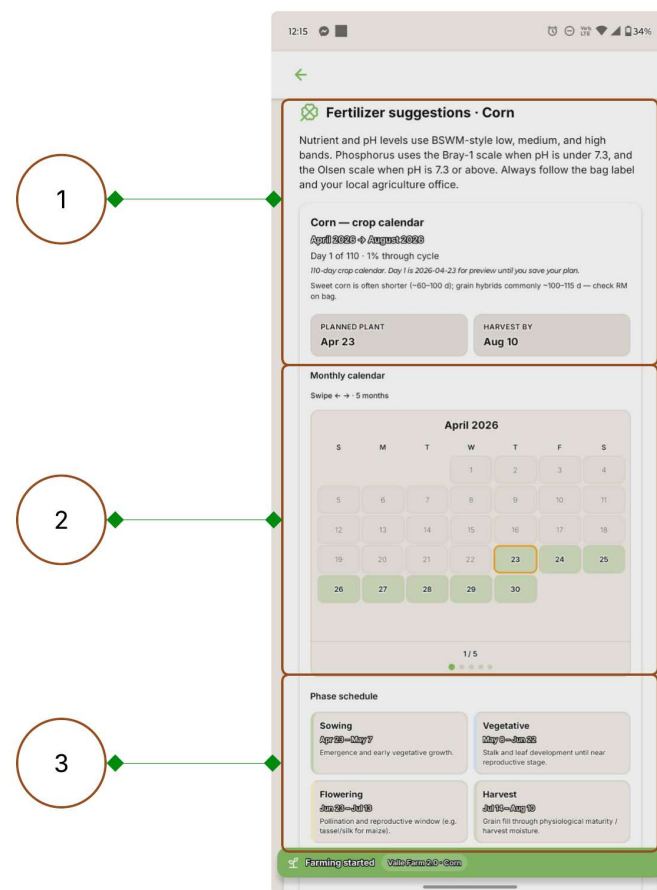


Figure 4.15 shows the Fertilizer Suggestions and Timeline screen, it offers structured planning tools for selected crops. The fertilizer suggestion with

crop (1) explains the basis of nutrient recommendations using soil data. The planting timeline (2) presents a calendar-based schedule indicating key dates such as planting and expected harvest. Additionally, the phase schedule (3) divides the crop cycle into stages such as sowing, vegetative growth, flowering, and harvest, providing a clear overview of the farming process.

Figure 4.16

TANIM Mobile Application Farm Summary and Option 1

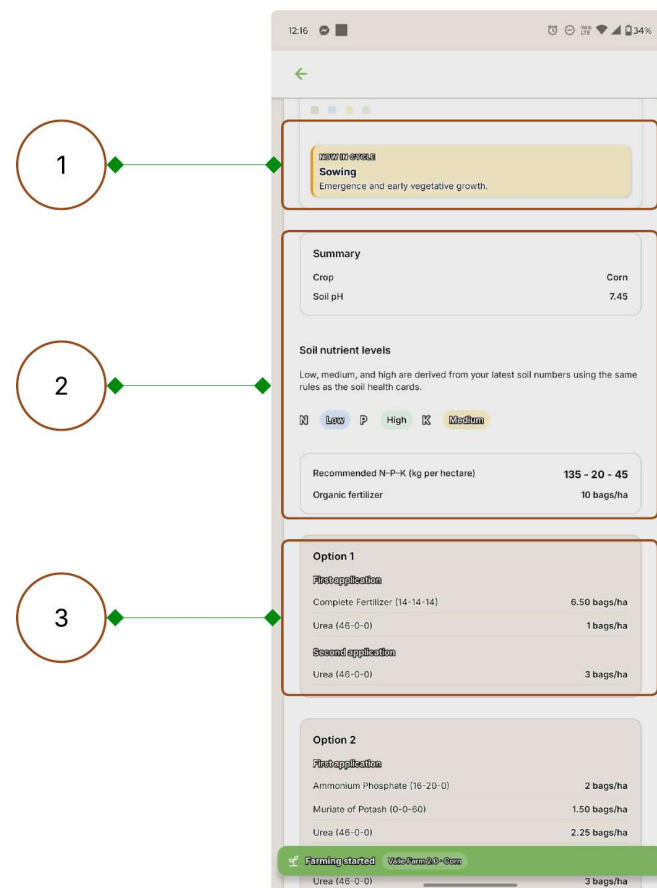


Figure 4.16 presents the Farm Summary and Option 1 screen, it summarizes the current farming plan and introduces a recommended fertilizer approach. The current phase cycle (1) indicates the present stage of crop development. The farm summary (2) provides key details such as selected crop, soil pH, and required NPK values per hectare. The first fertilizer option (3) outlines a specific combination of fertilizers and application rates based on the system's recommendation.

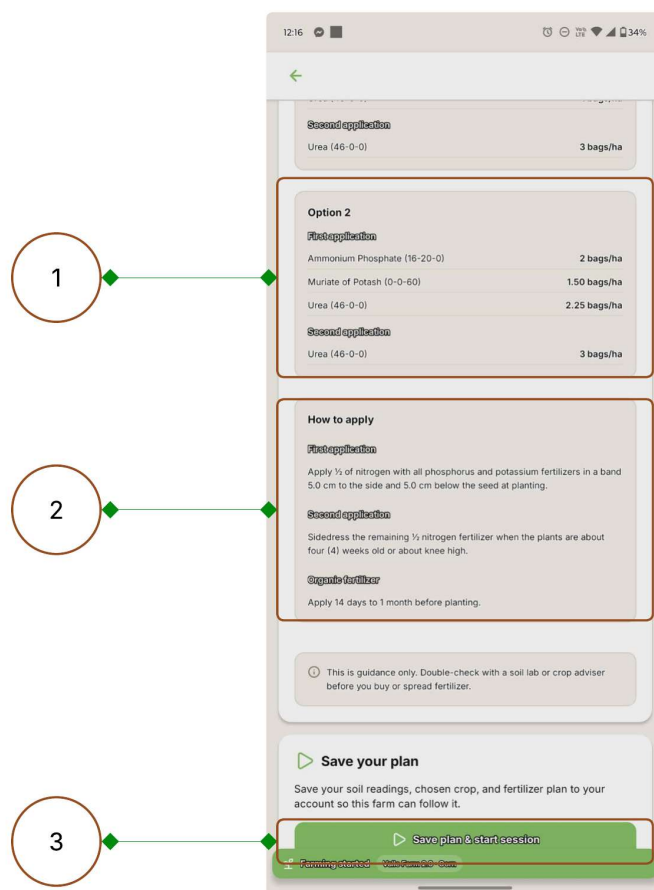
Figure 4.17*TANIM Mobile Application Option 2, Guide, and Start Button*

Figure 4.17 show the Option 2, Guide, and Start Session screen, it provides alternative solutions and execution steps. The second fertilizer option (1) offers an alternative combination of fertilizer inputs in case the primary option is unavailable. The application guide (2) provides practical instructions for proper fertilizer application in the field. The start farming session button

(3) allows users to finalize the plan and initiate the monitoring process for the selected farm.

These figures collectively demonstrate how the TANIM mobile application supports users throughout the farming workflow, from data input and analysis to decision-making and execution. The structured interface and clearly labeled components enable users to navigate the system efficiently and apply data-driven agricultural practices in real-world conditions.

4.3.3 User Interface and Interactions

The user interface is centered on accessibility and clear communication of technical data to ensure a user-centered experience. A primary interactive element includes the color-coded indicators on the soil health parameters page, which classify levels as low, medium, or high to allow for rapid visual assessment of soil status as shown in Figure 4.13. Interactions are further enhanced by practical instructions, such as the application guide shown in Figure 4.17, which provides specific field instructions for applying fertilizer treatments. These interface elements collectively ensure that the user can navigate the system and apply data-driven practices under real-world conditions.

4.3.4 Performance and Responsiveness

The TANIM mobile application was evaluated to determine its performance during real-world usage. The primary goal was to assess whether the app can handle requests promptly and display results without causing user delays. This evaluation is critical because the application relies on live data from backend services and machine learning models to provide farmers with actionable insights about soil health, crops, and fertilizer recommendations.

Testing was conducted under standard conditions using a typical internet connection on a mobile device. The evaluated features included farmer login, retrieval of farm and farming records, fetching soil health data, obtaining weather information, and receiving crop and fertilizer predictions. Response times were logged through the backend, capturing the duration of each request.

Table 15 summarizes the performance metrics for each endpoint. As shown, response times varied depending on the complexity of the operation. Lightweight requests, such as fertilizer predictions, completed in as little as 263 ms, whereas more data-intensive operations, such as fetching all farming records for a farmer, required up to 1,840 ms. The login process averaged 1,687 ms, which is reasonable given that it involves credential verification and authentication. Weather and soil health requests had intermediate response times, reaching up to 1,365 ms, which is acceptable considering the need to

pull external data and sensor readings.

Table 15: API and Machine Learning Endpoint Performance Logs

Type	Method	Endpoint / URL	Time (ms)	Notes
API	POST	/api/v1/auth/login-farmer	1687	Farmer authentication
API	GET	/api/v1/farm/2f5fc3ff-f01f-42ea-bf3f-6595a0f39150	979	Fetch farm by ID
API	GET	/api/v1/farm/farming/by-farmer/2f5fc3ff-f01f-42ea-bf3f-6595a0f39150	1840	Fetch farming records by farmer
API	GET	/api/v1/farm/farming/0b2f84db-678c-4da1-8c2c-2a6aacd01304	968	Fetch specific farming record
ML	GET	/pending/soil	1299	Soil health data fetch
API	GET	/api/v1/weather?lat=14.5995&lon=120.9842	1365	Weather fetch for Manila coordinates
ML	POST	/predict/fertilizer	263	Fertilizer prediction

The results in Table 15 indicate that the application handled all tested features without errors or crashes. During evaluation, the interface remained responsive, with no noticeable lag, freezes, or failed requests. While certain endpoints naturally required more processing time, none of the response times negatively impacted the user experience. Optimization of heavier endpoints,

such as fetching farming records, may further improve responsiveness and enhance usability for everyday farm management.

Overall, the TANIM mobile application demonstrates stable and efficient performance, confirming its suitability for real-time farm data management and decision support.

4.4 Web Application

A user-friendly and functional web application was developed to support the Department of Agriculture administrators in monitoring soil health, managing farmer and farm data, and visualizing farm locations through an integrated mapping system. The web application serves as the administrative interface of the TANIM system, enabling efficient data management and decision-making.

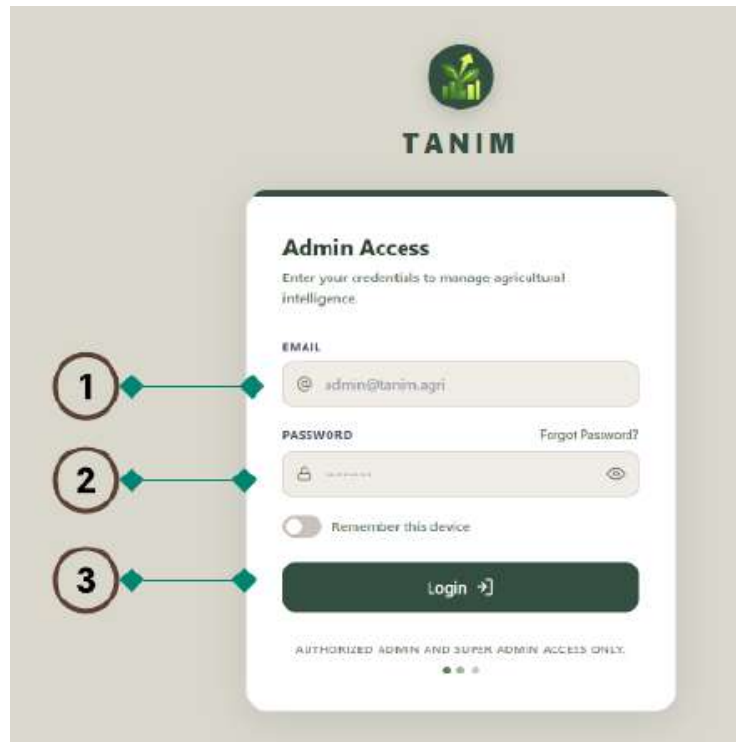
The system utilizes Supabase as its backend service, providing real-time database access, authentication, and API integration. Through this architecture, the web application is capable of retrieving and displaying up-to-date information collected from IoT devices deployed in the field. The interface is designed to be responsive and accessible, ensuring usability across different devices and environments.

4.4.1 System Overview

The TANIM web application functions as a centralized platform for managing agricultural data collected from the system. It connects to the backend database through API endpoints and presents information in an organized and interactive manner. The system enables administrators to monitor soil conditions, manage farmer profiles, and oversee farm-related activities efficiently.

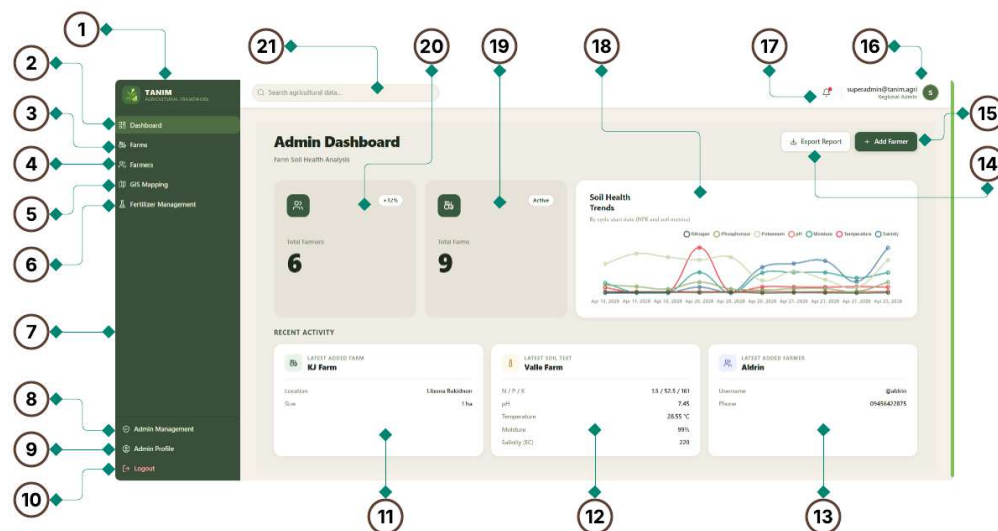
The architecture of the web application follows a client-server model, where the frontend interface communicates with backend services hosted on Supabase. Data collected from IoT sensor nodes is transmitted to the server and stored in the database, which is then accessed by the web application for visualization and management. This integration ensures near real-time synchronization between field data and the administrative interface.

To access the system, administrators are required to log in using their credentials. Figure 4.18 illustrates the login page layout, which includes the main functional components of the interface. Component (1) represents the email input field used to enter the administrator's registered email address, (2) represents the password input field for secure authentication, and (3) represents the login button, which, when clicked, redirects the administrator to the system dashboard upon successful authentication.

Figure 4.18*TANIM Web Application Login Page*

4.4.2 Dashboard and Data Visualization

The dashboard serves as the central command hub of the TANIM web application, providing administrators with a comprehensive overview of agricultural metrics and overall system activity. It is designed to transform raw data collected from IoT sensor nodes into meaningful and actionable insights through visual summaries and real-time updates. Figure 4.19 illustrates the layout of the admin dashboard, where the components are labeled for clarity.

Figure 4.19*TANIM Web Application Dashboard*

The interface includes a persistent sidebar (7) that allows seamless navigation between major system modules, including the Dashboard page (2), Farms page (3), Farmers page (4), GIS Mapping (5), and Fertilizer Management (6). A global search bar (21) is positioned at the top of the interface, enabling users to quickly locate specific data such as farm records or farmer profiles within the database.

At the top section of the dashboard, key performance indicators (KPIs) are presented through summary cards that provide a quick overview of system statistics. These include the total number of registered farmers (20) and the

total number of farms (19) being managed within the system. The farmer count also includes a percentage growth indicator, allowing administrators to monitor changes in system engagement over time.

The central portion of the dashboard is dedicated to analytical visualizations, particularly the soil health trends section (18). This component displays an interactive line graph presenting multiple soil parameters over time, including Nitrogen, Phosphorus, Potassium (NPK), pH level, moisture, temperature, and salinity. Through this visualization, administrators can identify patterns such as nutrient depletion, irregular soil conditions, or seasonal variations based on real-time data.

To provide immediate visibility of recent system activities, the lower section of the dashboard includes real-time update cards. These include the latest added farm (11), the latest soil test results (12), and the most recently registered farmer (13). Each component presents relevant details, allowing administrators to quickly assess recent developments without navigating through multiple pages.

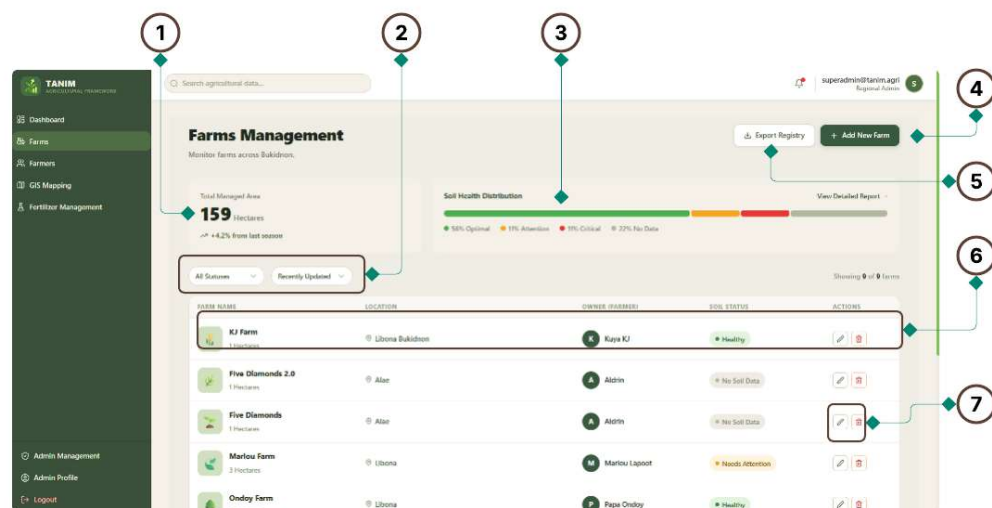
Additionally, the dashboard includes administrative controls and profile management features located in the top-right section of the interface. These include the notification button (17) for system alerts and the user profile section (16) for account management. Action buttons are also provided to streamline

workflows, such as the add farmer button (15) and the export report button (14), which allows administrators to generate downloadable summaries of system data.

Overall, the dashboard and data visualization features of the TANIM web application enhance the ability of administrators to monitor, analyze, and manage agricultural data efficiently. By presenting complex datasets in an organized and visually intuitive manner, the system supports informed decision-making and improves operational efficiency.

4.4.3 Administrative Functionalities

The TANIM web application provides comprehensive dashboard and data visualization features across multiple pages, enabling administrators to effectively manage agricultural resources, personnel, and system operations. These include the Farms Management page, Farmers Management page, GIS Mapping page, Fertilizer Management page, and Administrative Management page, each designed to present data in an organized and interactive format. Figure 4.20 to 4.24 illustrate the interfaces of these pages, with labeled components to guide interpretation.

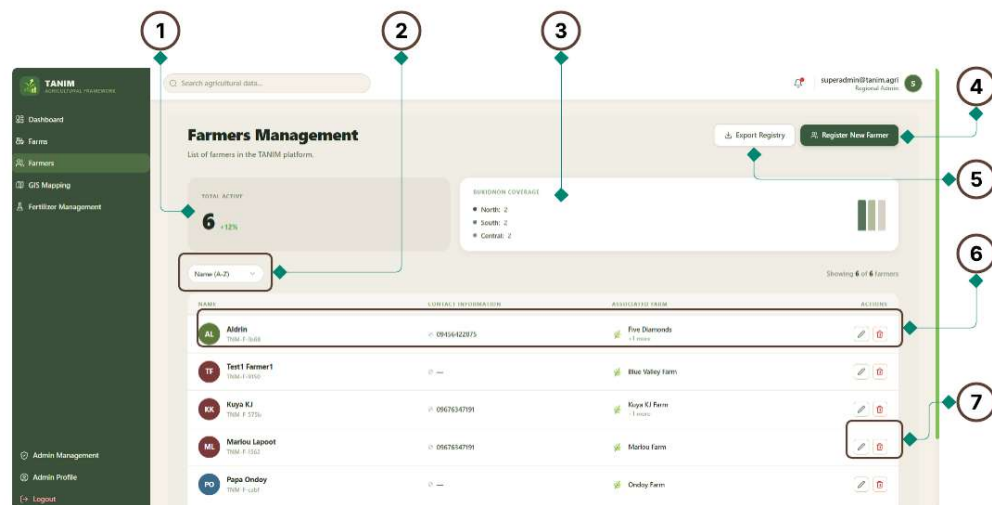
Figure 4.20*TANIM Web Application Farms Management*

The Farms Management page, shown in Figure 4.20, is designed to monitor and manage land assets across the Bukidnon region. The interface presents a page overview (1) displaying the total managed land area and growth metrics, allowing administrators to assess system scale and expansion. Table controls (2) are provided to filter and sort farm records based on status or recency. A soil health distribution indicator (3) categorizes farms into conditions such as optimal, requiring attention, critical, or with no available data. The page also includes an add farm button (4) for registering new land parcels and an export registry button (5) for downloading farm data. A registered farm data table (6) lists detailed information such as farm name, location, owner, and

soil status, with corresponding control buttons (7) for editing and deleting individual records.

Figure 4.21

TANIM Web Application Farmers Management

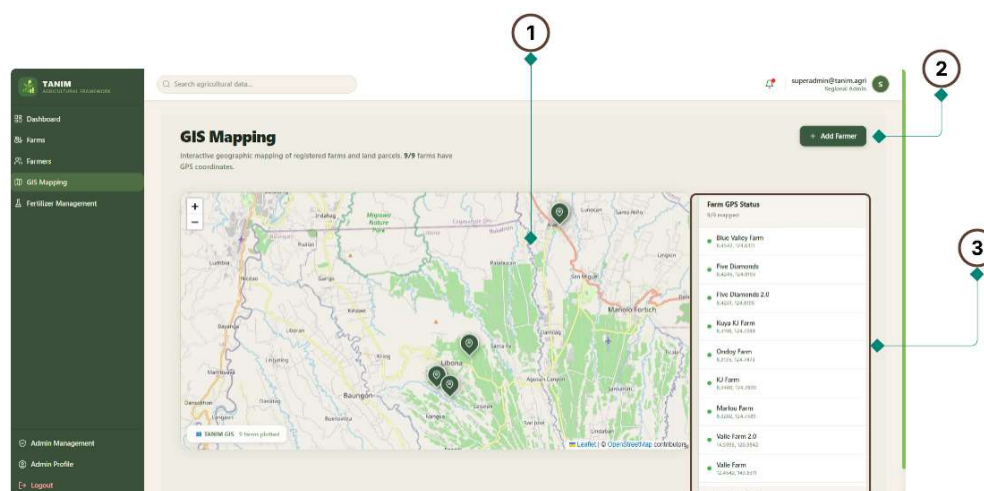


The Farmers Management page, illustrated in Figure 4.21, focuses on managing farmer profiles and associated information. The page overview (1) displays the total number of registered farmers along with growth indicators. Table controls (2) allow administrators to sort records, such as by recently added entries. A geographic coverage component (3) presents the distribution of farmers across different areas of Bukidnon. The page provides a register new farmer button (4) for adding personnel and an export registry button (5) for downloading farmer data. A registered farmer table (6) displays key in-

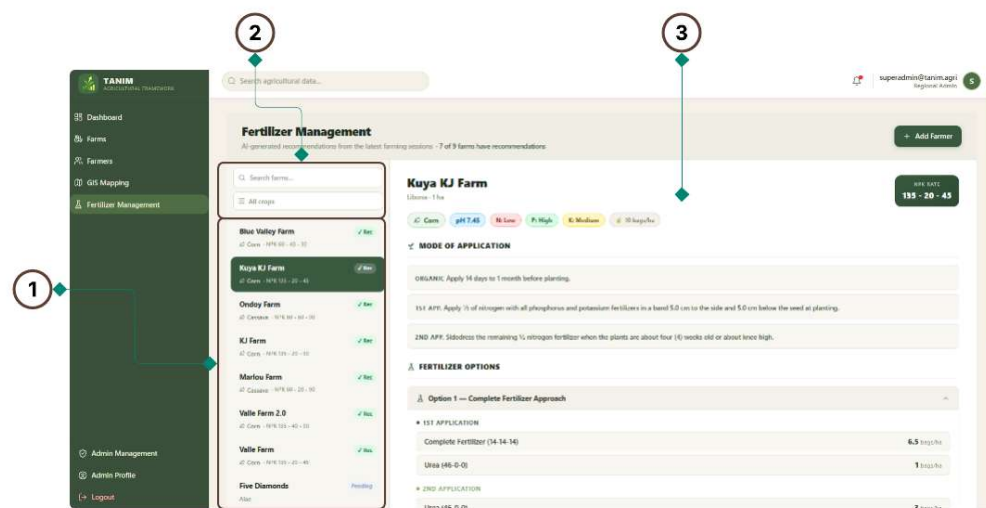
formation including identification details, contact information, and associated farms, while control buttons (7) allow for updating or removing records.

Figure 4.22

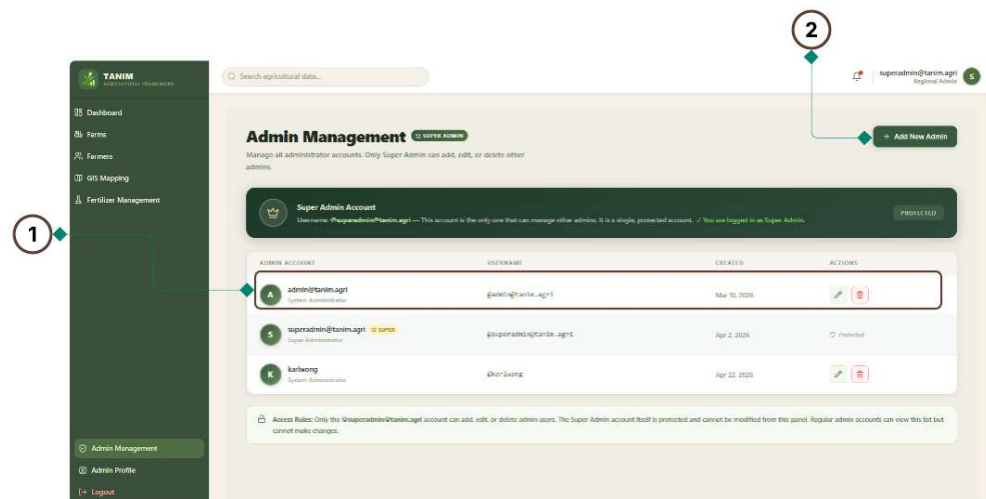
TANIM Web Application GIS Mapping



The GIS Mapping page, as shown in Figure 4.22, provides a spatial representation of farm locations through an interactive map interface. The farm mapping component (1) displays GPS-based pin locations of each farm, allowing administrators to visualize distribution and density. A quick-action add farmer button (2) is available within the interface for registering users directly from the map view. Additionally, a list of farms panel (3) presents the GPS status of each farm, confirming whether location data has been successfully recorded.

Figure 4.23*TANIM Web Application Fertilizer Management*

The Fertilizer Management page, presented in Figure 4.23, serves as a decision-support feature by providing system-generated fertilizer recommendations based on soil data. The interface includes a list of farms panel (1), which allows administrators to select specific farms and identify those with available recommendations. Sorting controls (2), including a search bar and crop filter, are used to quickly locate relevant data. The fertilizer suggestion panel (3) displays detailed information for a selected farm, including soil profile data such as pH and NPK values, recommended fertilizer types, application timing, and dosage guidelines.

Figure 4.24*TANIM Web Application Admin Management*

The Administrative Management page, illustrated in Figure 4.24, functions as the control center for system access and user management. The admin information table (1) lists all administrators along with their assigned roles and account details, including protected accounts such as the super administrator. Control actions such as edit and delete are available for authorized accounts, while protected accounts cannot be modified. An add admin button (2) is provided to allow authorized users to register new administrators, ensuring controlled expansion of system access.

These pages collectively enhance the dashboard and data visualization capabilities of the TANIM web application by presenting complex agricultural

and administrative data in a structured, interactive, and user-friendly manner. The consistent interface design, combined with clearly labeled components, supports efficient navigation and informed decision-making for administrators.

4.4.4 System Performance

The performance of the TANIM web application was evaluated to assess its responsiveness, stability, and efficiency during typical user interactions. Testing was conducted under normal operating conditions using standard internet connectivity, focusing on key system functions such as farm selection, farmer data retrieval, farming session queries, and administrative operations. Response times were measured using browser developer tools, specifically the network monitoring feature.

As presented in Table 16, the majority of API requests returned a 200 status code, indicating reliable communication between the frontend and backend. Core system functionalities demonstrated consistent performance, with response times generally ranging from 200 ms to 330 ms. Farm and farmer data requests were completed within 246 ms to 329 ms, while farming session queries averaged between 260 ms and 310 ms. Administrative requests, including authentication processes, also showed stable performance, with response times between 205 ms and 308 ms.

Table 16: TANIM System API and Machine Learning Endpoint Response Times

Endpoint Name	Method	Type	Time (ms)	Notes
/api/v1/auth/login-farmer	POST	API	1687	Farmer authentication
/api/v1/farm/2f5fc3ff-f01f-42ea-bf3f-6595a0f39150	GET	API	979	Fetch farm by ID
/api/v1/farm/farming/by-farmer/2f5fc3ff-f01f-42ea-bf3f-6595a0f39150	GET	API	1840	Fetch farming records by farmer
/api/v1/farm/farming/0b2f84db-678c-4da1-8c2c-2a6aacd01304	GET	API	968	Fetch specific farming record
/pending/soil	GET	ML	1299	Soil health data fetch
/api/v1/weather?lat=14.5995&lon=120.9842	GET	API	1365	Weather fetch for Manila coordinates
/predict/fertilizer	POST	ML	263	Fertilizer prediction

Data-intensive operations, such as forecast-related requests, exhibited higher latency, reaching up to 1.59 seconds due to increased processing requirements. Despite this, all requests were successfully completed, with no system failures observed during testing.

Overall, the results indicate that the TANIM web application delivers fast and consistent performance across its core functionalities, maintaining sub-second response times in most operations. These findings confirm that the system provides a responsive and stable user experience, with opportunities for further optimization in handling complex data processes.

4.4.5 Accessibility and Responsiveness

The responsiveness and accessibility of the TANIM web application were evaluated to determine how effectively the system adapts to different devices while ensuring ease of use for a wide range of users, particularly farmers accessing the platform in field environments. This combined evaluation focuses on the system's ability to dynamically adjust its interface across varying screen

sizes and resolutions, while also maintaining clarity, usability, and user-friendly interaction.

The testing process involved accessing the system using multiple devices, including desktop computers, laptops, tablets, and mobile phones. Screen resolutions such as 1920×1080 for desktops, 1920×1080 for laptops, 2560×1600 for tablets, 2400×1080 for android devices, and 1792×828 for IOS devices in were considered to represent a wide range of user environments. In addition, the system was tested using different web browsers, including Google Chrome, Brave, Safari, and Microsoft Edge, to ensure cross-browser compatibility. During testing, key functionalities such as admin and superadmin login, viewing soil data and crops planted, navigating between pages, and interacting with dashboard components were performed on each device.

4.5 System and Functional Evaluation

This section presents the evaluation of the developed system in terms of its functionality, performance, and overall usability. The purpose of this evaluation is to determine whether the system meets its intended objectives and satisfies user requirements.

4.5.1 Crop Recommendation Validation

The validation of the rule-based recommendation engine of the TANIM system was conducted by comparing system outputs against established agricultural standards and expert guidelines provided by the Department of Agriculture and the Bureau of Soils and Water Management. To ensure the reliability and agricultural accuracy of the system, the validation process focused on verifying the translation of qualitative Soil Test Kit results into quantitative limits. This included crop-specific fertilizer rates, modes of application protocols, and soil pH preference modeling.

The core fertilizer and crop suitability logic of the system is based entirely on protocols from the Bureau of Soils and Water Management. Validation was performed by ensuring that the programmed logic strictly follows these published standards, thereby eliminating arbitrary decision-making within the recommendation engine. The system recommendations were cross-checked against reference manuals to confirm that every combination of soil sensor data results in agriculturally sound outputs.

A critical aspect of the validation involved converting qualitative color-based results into actionable quantitative limits. The classification logic of the system was validated against official quantitative value references, as summarized in Table 17. As shown in the table, nutrient thresholds for nitrogen,

phosphorus, and potassium were categorized into low, medium, and high levels based on standardized testing conditions. These thresholds served as the basis for ensuring that the system’s interpretation of soil data aligns with scientifically accepted agricultural practices.

Table 17: Official Quantitative Values Reference for Soil Nutrients

Nutrient	Testing Standard and Condition	Low Thresh- old	Medium Thresh- old	High Thresh- old
Nitrogen from Or- ganic Matter	All Soil Types	$\leq 1.7\%$	1.7% – 3.0%	$> 3.0\%$
Phosphorus (ppm)	Bray 1 (Acidic to neutral soils, pH < 7.3)	≤ 10	10 – 20	> 20
Phosphorus (ppm)	Olsen (Alkaline soils, pH $\geq$ 7.3)	≤ 7	7 – 25	> 25
Potassium (ppm)	All Soil Types	≤ 117	117 – 235	> 235

4.5.2 Accuracy of Recommendations

The overall accuracy of the crop recommendation system was evaluated on a held-out test set comprising 15% of the augmented dataset, using stratified sampling to preserve class distribution. The evaluation results, summarized in Table 18, indicate that the model achieved a global accuracy of 82.29%, a balanced accuracy of 82.30%, and a weighted F1-score of 82.20%.

These metrics demonstrate that the model maintains consistent pre-

Table 18: Performance Metrics of the Crop Recommendation System

Metric	Score
Global Accuracy	82.29%
Balanced Accuracy	82.30%
Weighted F1-Score	82.20%

dictive performance across different crop classes while minimizing bias toward more frequent categories, thereby supporting its reliability for practical agricultural deployment.

The effectiveness of this approach is more evident when compared to the baseline model, as shown in Table 19. The original model, which was trained on an imbalanced dataset without these enhancements, yielded a global accuracy of 61.54% and a balanced accuracy of 42.44%. In contrast, the optimized pipeline—incorporating feature engineering, K-Means augmentation, and Optuna-based hyperparameter tuning—achieved a global accuracy of 82.29% and a balanced accuracy of 82.30%. This corresponds to a 20.75 percentage point increase in global accuracy and a 39.86 percentage point improvement in balanced accuracy.

Table 19: Comparison of Baseline and Optimized Model Performance

Configuration	Global curacy	Ac- Balanced Accuracy	Improvement (Bal. Acc.)
Baseline (no augmentation)	61.54%	42.44%	—
Optimized (K-Means + Op- tuna)	82.29%	82.30%	+39.86 pp

These results highlight the critical role of the refined preprocessing and optimization steps in stabilizing and improving the overall predictive performance of the system. Furthermore, the system’s weighted precision, recall, and F1-score all exceeded 82%, indicating that the model maintains consistent performance across all ten target crop classes without disproportionately favoring precision over recall or vice versa.

To systematically validate the crop recommendation system under varying field conditions, six test case scenarios were shown in Table 20. These scenarios represent soil properties encountered in Region 10, Philippines, as recorded in the Regional Soils Laboratory (RSL) dataset from 2023 to 2025. This approach ensures the system is evaluated across a spectrum ranging from nutrient-deficient acidic soils to fertile neutral soils.

The model processes four primary inputs—Organic Matter (OM%), Phosphorus (P), Potassium (K), and pH—while internally computing four interaction features: P/K ratio, $OM \times pH$, $K \times pH$, and P/OM ratio. As illus-

trated in Table 20, these scenarios cover majority and minority crop classes, nutrient-deficient profiles, and boundary edge cases to ensure recommendation accuracy and robustness.

Table 20: Validation Test Case Profiles Based on RSL Dataset (2023–2025)

Test Case	OM (%)	P (ppm)	K (ppm)	pH	Target Condition
TC-1	3.48	23.4	354.0	5.16	Typical corn-belt soil
TC-2	4.27	127.3	438.0	5.14	High-phosphorus fertile soil
TC-3	3.74	5.0	168.4	4.68	Acidic, low-nutrient soil
TC-4	3.02	46.3	459.3	6.47	Neutral, high-potassium soil
TC-5	4.50	30.0	320.0	5.10	High organic-matter soil
TC-6	1.20	2.0	50.0	8.00	Edge-case / boundary values

The practical application of the crop recommendation engine was validated by comparing expected outcomes against actual system outputs across twenty test cases. As shown in Table 21, the system analyzed four core soil parameters: Organic Matter (%), Phosphorus (ppm), Potassium (ppm), and pH. The model achieved an overall accuracy of 85.0%, successfully predicting the correct crop in 17 out of 20 instances.

Table 21: Validation Results: Expected Crop vs. System Output

OM (%)	P (ppm)	K (ppm)	pH	Expected	Output	Conf.	Match
3.36	16.2	751.4	5.38	Ampalaya	Ampalaya	96.1%	CORRECT
3.79	138.7	907.0	5.92	Ampalaya	Ampalaya	99.0%	CORRECT
3.82	15.6	240.8	5.25	Beans	Beans	88.7%	CORRECT
3.96	16.8	238.5	5.25	Beans	Beans	86.5%	CORRECT
4.64	4.0	358.0	5.01	Cassava	Cassava	96.5%	CORRECT
4.24	2.6	226.9	4.88	Cassava	Cassava	92.9%	CORRECT
3.00	6.2	107.0	5.07	Corn	Ampalaya	53.3%	INCORRECT
2.60	10.0	338.0	5.01	Corn	Corn	32.9%	CORRECT
6.50	728.1	4402.7	6.67	Eggplant	Tomato	50.9%	INCORRECT
3.25	9.9	135.9	5.52	Eggplant	Eggplant	52.3%	CORRECT
3.70	32.8	3544.5	5.65	Sorghum	Sorghum	99.9%	CORRECT
3.69	35.2	3455.1	5.81	Sorghum	Sorghum	99.8%	CORRECT
6.72	89.0	1016.5	6.10	Squash	Squash	99.7%	CORRECT
3.59	7.3	113.8	5.11	Squash	Ampalaya	46.9%	INCORRECT
15.16	1607.0	2701.3	7.03	Sweet Pepper	Sweet Pepper	99.9%	CORRECT
3.79	16.3	193.7	4.72	Sweet Pepper	Sweet Pepper	94.5%	CORRECT
2.10	12.9	198.9	5.81	Tobacco	Tobacco	91.7%	CORRECT
2.09	23.9	187.6	6.64	Tobacco	Tobacco	99.1%	CORRECT
3.68	86.5	653.6	5.80	Tomato	Tomato	99.2%	CORRECT
5.00	54.3	660.2	5.77	Tomato	Tomato	99.4%	CORRECT

The evaluation revealed a robust model calibrated for Region 10's unique agronomic conditions. Key strengths included equitable performance, where K-Means augmentation successfully mitigated minority class bias, ensuring consistent accuracy across all ten crops. The model also demonstrated strong feature efficacy, as engineered features, particularly the P/K ratio and OM×pH interaction, served as pivotal predictors. In terms of reliability, most predictions exceeded a 70% confidence threshold, while lower scores effectively flagged ambiguous profiles for further review.

Despite these strengths, misclassifications occurred in crops with overlapping nutrient requirements (e.g., Tomato vs. Eggplant). Furthermore, the

model's current focus on soil-only features lacks climatic context such as elevation and seasonality. Future iterations should integrate environmental IoT sensor data and periodic retraining with fresh laboratory results to maintain long-term precision.

4.5.3 System Testing

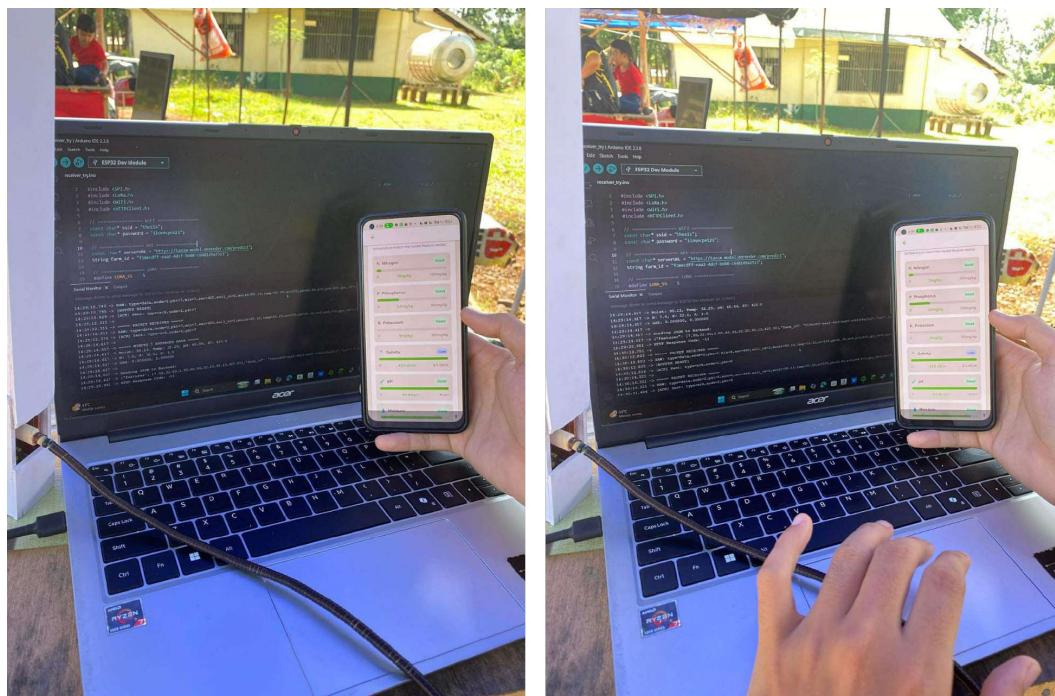
The system testing phase was conducted to evaluate the functional performance and reliability of the TANIM hardware in a real-world agricultural environment. This phase focused on the integration between the sensor probes, the processing unit, and the cloud-based data transmission system.

As shown in Figure 4.25, the hardware unit was deployed in a field setting. The sensor was inserted into a prepared soil pit to ensure direct contact with the substrate for accurate data acquisition.

Figure 4.25
System Field Testing



These results were successfully transmitted and displayed on the digital interface, confirming the end-to-end connectivity of the system. This testing phase ensures that the environmental data gathered is both consistent and reflective of the actual field conditions, providing a solid foundation for the management recommendations generated by the TANIM application as shown in Figure 4.26.

Figure 4.26*System Logs*

4.6 Usability Evaluation

This section presents a rigorous usability evaluation of the developed system, employing a structured, user-centered assessment framework to measure its effectiveness, efficiency, and user satisfaction across identified user groups. The evaluation was conducted using purposefully selected participants, consisting of a total of ten (10) respondents, specifically seven (7) farmers as primary end-users and three (3) Department of Agriculture (DA) officials as administrative users. This 7:3 distribution ensures adequate representation

of both operational and managerial perspectives in system interaction.

4.6.1 Respondent Profile

The participants possess varying levels of farming experience ranging from a few years to two decades. Instead of relying only on personal knowledge, they follow established guidelines to direct their farming methods. This practical approach is evident in their fertilizer application where they combine their own estimates with specific agricultural recommendations. Over the years, the group has successfully cultivated a variety of crops such as corn, cassava, sweet potatoes, and bananas. At present, each participant focuses their time and resources on managing a single farm.

The respondent from the Department of Agriculture (DA) has been serving the institution since 2014, reflecting over a decade of professional experience in the agricultural sector. This extensive tenure suggests a strong familiarity with the department's programs, policies, and field operations, particularly in supporting farmers and implementing agricultural initiatives.

In terms of professional role, the respondent holds the position of Senior Agriculturist, indicating a higher level of expertise and responsibility within the organization. As an agriculturist, their primary area of focus revolves around crop production, soil management, and the provision of technical as-

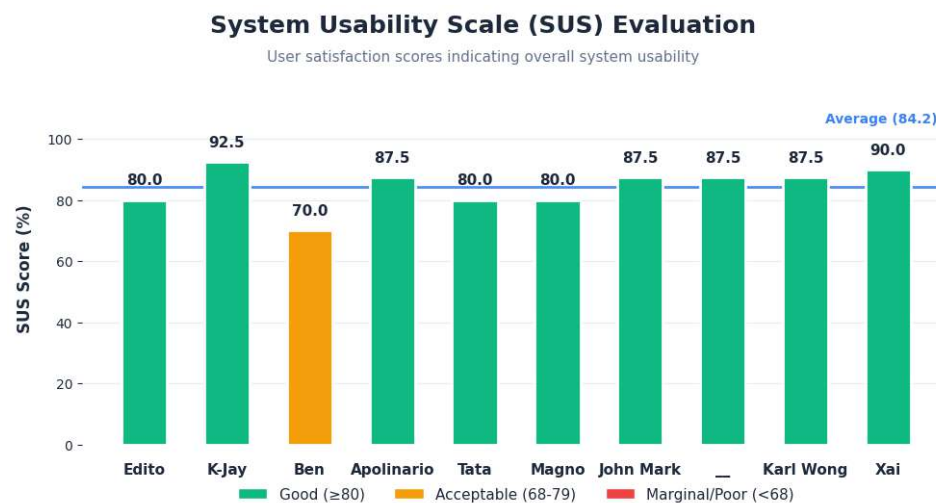
sistance to farmers. This role typically involves advising on best farming practices, evaluating soil and crop conditions, and contributing to the development and implementation of agricultural programs aimed at improving productivity and sustainability.

4.6.2 Results and Analysis

The usability of the TANIM system was evaluated using the System Usability Scale (SUS) to determine user satisfaction and the overall efficiency of the interface. As illustrated in Figure 4.27, a total of ten (10) participants evaluated the system, providing individual scores that reflect their interaction experience.

Figure 4.27

System Usability Scale (SUS) Evaluation Results



The participants recorded scores of 80.0, 92.5, 70.0, 87.5, 80.0, 80.0, 87.5, 87.5, 87.5, and 90.0. Based on these results, the system achieved an average SUS score of 84.2. According to standard SUS benchmarks, this score indicates an "Excellent" rating and a high level of usability as perceived by the respondents. Most participants provided scores categorized as "Good" (≥ 80), suggesting that the system is intuitive, well-integrated, and effective in supporting complex agricultural tasks.

While one respondent provided a relatively lower score of 70.0, this value still falls within the "Acceptable" usability range (68-79). The consistency of high scores among the majority of respondents demonstrates that users were able to interact with the TANIM system efficiently and with minimal difficulty. Overall, the evaluation confirms that the system provides a positive user experience and meets the high usability expectations required for its deployment.

4.6.3 Interpretation

Table 22: Summary of Usability Evaluation Results

Average SUS Score	Grade	Adjectival Rating	Remarks
84.25	A+	Excellent	Acceptable

As shown in the Table 22 and based on the SUS interpretation scale, the

average score of 84.25 corresponds to an “Excellent” rating and falls within the acceptable to high usability range. This score also indicates that the system is likely to be well accepted by users and demonstrates strong usability performance.

The high SUS score suggests that the TANIM system is easy to learn, efficient to use, and provides a smooth interaction experience for its users. The consistency of scores above the standard benchmark of 68 further confirms that the system exceeds the minimum usability requirements.

The usability evaluation validates that the TANIM system is user-friendly and suitable for real-world agricultural applications, supporting both technical and non-technical users in performing their tasks effectively.

CHAPTER 5

CONCLUSIONS AND RECOMMENDATIONS

This chapter provides the summary of the results obtained in this study and gives some recommendations for further investigation.

5.1 Summary of Findings

This study developed and evaluated an IoT-based crop recommender system integrated with machine learning for agricultural applications in Bukidnon. The findings demonstrate that the system successfully integrated hardware and software components, including soil sensors, a machine learning model, and web and mobile applications, to provide a functional decision-support tool for farmers. The system was able to measure key soil parameters such as pH, nitrogen (N), phosphorus (P), potassium (K), salinity, moisture, and temperature, and transmit these data to a centralized platform for processing and visualization.

The calibration results of the soil pH sensor revealed a non-linear response across different buffer solutions, indicating that a single constant offset correction is insufficient for achieving accurate readings. Validation results further showed that while the sensor outputs were generally aligned with qual-

itative classifications from Soil Test Kit (STK) results, there were significant numerical discrepancies when compared to laboratory measurements from the Department of Agriculture. Error analysis confirmed the presence of noticeable offsets, particularly in nutrient and pH values, suggesting limitations in achieving laboratory-grade precision. Despite this, the system demonstrated reliability in categorical classification (low, medium, high), which is essential for practical crop recommendation.

The machine learning model achieved an accuracy of 82.29%, with consistent performance across evaluation metrics, indicating its capability to generate reliable crop and fertilizer recommendations based on soil health data. Additionally, the developed mobile application enabled farmers to monitor soil conditions, crop predictions, and fertilizer schedules, while the web application provided administrative functionalities such as farm zoning, farmer data management, and timeline tracking. The system also achieved a System Usability Scale (SUS) score of 84.25, which corresponds to an “excellent” rating, indicating that users found the system intuitive and effective for agricultural use.

Overall, the results confirm that all the objectives of the study were successfully achieved. The implementation of the IoT system, along with the development of both mobile and web applications and the integration of a

machine learning model, demonstrates the feasibility of the proposed system as a comprehensive agricultural decision-support platform.

5.2 Conclusion

Based on the findings of the study, it can be concluded that the developed IoT-based crop recommender system is a functional and effective tool for supporting data-driven agricultural practices in Bukidnon. The system successfully achieved its primary objective of integrating IoT, machine learning, and software platforms to provide real-time soil monitoring and crop recommendation. Each of the specific objectives was attained, including the implementation of a machine learning model for crop and fertilizer recommendation, the development of a mobile application for farmer use, the creation of a web-based monitoring system for administrative purposes, and the evaluation of system accuracy and usability in actual agricultural settings.

The results highlight that while the system performs well in terms of usability, system integration, and predictive capability, there are limitations in the numerical accuracy of the soil sensor readings. The presence of significant offsets when compared to laboratory readings from the Department of Agriculture - Region 10 farm data indicates that the system cannot fully replace traditional laboratory testing for precise soil analysis. However, the

system remains effective as a field-level decision-support tool, particularly due to its ability to provide reliable categorical interpretations of soil health data.

In comparison to existing studies, this research addresses several gaps by providing an end-to-end integrated system that combines real-time IoT sensing, machine learning-based recommendation, and dual-platform accessibility through both web and mobile applications. While many related studies focus only on simulation models, isolated sensor systems, or standalone applications, this study contributes a more comprehensive and deployable solution. Nevertheless, similar to other studies utilizing low-cost sensors, limitations in sensor accuracy remain a challenge, indicating an area that requires further improvement.

Overall, the TANIM system demonstrates strong potential in advancing precision agriculture by enabling accessible, real-time, and data-driven decision-making for farmers, especially in regions with limited access to laboratory facilities.

5.3 Recommendations

Based on the conclusions of the study, several recommendations are proposed to further enhance the system and guide future research. First, improvements in sensor calibration are strongly recommended, particularly

through the use of multi-point or non-linear calibration techniques to address the observed inaccuracies and offsets in sensor readings. The implementation of polynomial or regression-based calibration models, along with periodic recalibration, may significantly improve measurement accuracy. Additionally, software-based correction mechanisms may be applied to minimize systematic errors and improve alignment with laboratory results.

For the machine learning model, further enhancements are recommended to improve the accuracy and robustness of crop and fertilizer recommendations. It is suggested to expand the training dataset by incorporating more diverse and laboratory-validated soil samples to ensure better generalization across different soil conditions. The integration of calibration-adjusted sensor data into the training process is also important to reduce the impact of sensor inaccuracies on prediction outcomes. Furthermore, exploring advanced techniques such as ensemble learning, hybrid models, or deep learning approaches may help improve classification performance and reduce misclassification among crops with similar nutrient requirements. Continuous retraining of the model using newly collected field data is also recommended to maintain its accuracy and adaptability over time.

In terms of hardware improvements, it is recommended to utilize higher-precision sensors and expand the range of measurable soil parameters, includ-

ing micronutrients and soil organic matter, to provide a more comprehensive soil analysis. Integrating additional environmental factors, such as weather conditions, rainfall patterns, and seasonal variations, may further improve both the sensor system and the predictive capability of the machine learning model.

For practical implementation, the system should be positioned as a preliminary soil assessment and decision-support tool rather than a replacement for laboratory testing. Collaboration with agricultural agencies may support standardization and wider adoption, particularly in rural areas where access to laboratory facilities is limited. Training programs should also be conducted to help users effectively interpret both numerical and categorical outputs of the system.

Finally, future research may explore the integration of artificial intelligence techniques for real-time sensor calibration and adaptive learning systems. Further studies may also evaluate the long-term impact of the system on crop yield, fertilizer efficiency, and sustainability to fully establish its effectiveness in real-world agricultural applications.

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APPENDIX B

FUNCTIONAL TESTING FORM

A. Hardware and Data Capture

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	Sensor powers on and initializes correctly.	Sensor activates and displays readiness.		
2	The sensor captures soil parameters (N, P, K, pH, salinity, temperature, moisture).	Accurate readings displayed on serial monitor or app.		
3	Multiple readings under the same conditions are consistent.	Readings show minimal variance ($\pm 5\%$).		
4	ESP32 collects and processes sensor data properly.	Data packets successfully generated.		
5	The LoRa module transmits data to the gateway.	Data successfully received at gateway.		
6	Data stored correctly in the Firebase database.	Database shows complete and valid entries.		

B. Mobile Application (Farmer Side)

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	User login and authentication.	Users can log in using valid credentials.		
2	Soil health readings display correctly.	Dashboard shows latest sensor readings.		
3	Crop recommendation generation.	App suggests appropriate crops based on soil data.		
4	Fertilizer recommendation with dosage and schedule.	Correct fertilizer type and amount displayed.		
5	Multiple crop recommendations work without error.	The system handles multiple queries smoothly.		
6	Offline mode and data synchronization.	Data syncs automatically once reconnected.		
7	Farm information update.	Farmers can add/edit farm details successfully.		

C. Web Application (DA / Administrator Side)

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	Admin login and authentication.	DA officials can securely log in.		
2	Real-time soil health dashboard.	Data from sensors updates in real time.		
3	GIS mapping visualization.	Farm locations shown accurately on the map.		
4	Farmer profile management.	Officials can view and edit farmer data.		
5	Crop rules and recommendation management.	Admins can modify rules and save updates.		
6	Web responsiveness.	The system is accessible and functional on all devices.		

D. System-Wide Features

Test Case No.	Test Description	Expected Result	Does it achieve the expected result?	
			Yes	No
1	End-to-end data flow.	Data moves seamlessly from sensor → gateway → Firebase → apps.		
2	Error handling for missing/corrupt data.	System displays proper error messages or retries.		
3	Security and access control.	Only authorized users can access restricted features.		
4	Multi-user capability.	Multiple users can operate simultaneously without system crash.		

APPENDIX C

USABILITY TESTING SCALE

System Usability Scale (Farmers)						
No.	Item	Frequency				
		SD	D	N	A	SA
1	I think that I would like to use the TANIM mobile app frequently for my farming activities.					
2	I found the TANIM mobile app unnecessarily complex to use.					
3	I thought the app was easy to use for checking my soil’s NPK and pH levels.					
4	I think that I would need the support of a technical person to be able to understand the soil parameter and the crop recommendations.					
5	I found the various functions (weather, soil health, and fertilizer schedules) were well integrated.					
6	I thought there was too much inconsistency in how the soil health data (NPK, pH, moisture, temperature, and salinity) and recommendations were displayed.					
7	I would imagine that most farmer would learn to use this app very quickly.					
8	I found the process of viewing my farm very complex.					
9	I felt very confident using the app to make decisions about which crop and fertilizer to apply.					
10	I needed to learn a lot of things before I can use this system.					

System Usability Scale (Farmers)						
No.	Item	Frequency				
		SD	D	N	A	SA
1	I think that I would like to use the TANIM mobile app frequently for my farming activities.					
2	I found the TANIM mobile app unnecessarily complex to use.					
3	I thought the app was easy to use for checking my soil's NPK and pH levels.					
4	I think that I would need the support of a technical person to be able to understand the soil parameter and the crop recommendations.					
5	I found the various functions (weather, soil health, and fertilizer schedules) were well integrated.					
6	I thought there was too much inconsistency in how the soil health data (NPK, pH, moisture, temperature, and salinity) and recommendations were displayed.					
7	I would imagine that most farmer would learn to use this app very quickly.					
8	I found the process of viewing my farm very complex.					
9	I felt very confident using the app to make decisions about which crop and fertilizer to apply.					
10	I needed to learn a lot of things before I can use this system.					

APPENDIX D

TESTING CONDUCTED







APPENDIX E

LETTER OF CONSENT FOR INTERVIEW

November 28, 2025

Department of Agriculture – Regional Field Office 10
Antonio Luna St., Cagayan de Oro City

Subject: Request to Conduct an Interview for Thesis Purposes

To Whom It May Concern,

Good day!

We, the undersigned students of Bachelor of Science in Computer Engineering – 4th Year from the University of Science and Technology of Southern Philippines – Cagayan de Oro Campus, are currently conducting our thesis entitled “**TANIM: An Intelligent Agricultural Framework for Crop Recommendation and Soil Health Monitoring.**”

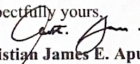
As part of our data-gathering process, we would like to request permission to conduct an interview with an official or staff member from the **Department of Agriculture Region 10** who is knowledgeable in areas related to soil health, crop management, or agricultural technology. **We are hoping to conduct the interview on December 1, 2025 at 3:30 PM.** The purpose of the interview is to gather expert insights that will support and strengthen the findings of our study.

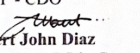
We assure you that all information obtained will be used solely for academic purposes and will be treated with the utmost confidentiality. Should you require further details, please feel free to contact us through apuyachristianjames@ustp.edu.ph or 09161513497.

We sincerely hope for your kind approval of our request. Your cooperation and support would greatly contribute to the success of our research.

Thank you very much for your time and consideration.

Respectfully yours,


Christian James E. Apuya
Researcher, BS in Computer in Engineering
USTP - CDO


Albert John Diaz
Researcher, BS in Computer in Engineering
USTP - CDO


Robert Roy Salvo
Researcher, BS in Computer in Engineering
USTP - CDO

Noted by:


Engr. Jogie Rey D. Fernandez
Research Adviser, Faculty in Computer Engineering
USTP - CDO


Ray Simon L. Bantaculo
Researcher, BS in Computer in Engineering
USTP - CDO


Alex U. Guadaluquil Jr.
Researcher, BS in Computer in Engineering
USTP - CDO



APPENDIX F

LETTER OF REQUEST FOR SOIL TESTING KIT

February 3, 2026

Jose Apollo Y. Pacamalan
Regional Executive Director, Department of Agriculture - Region 10
Antonio Luna Street, Cagayan de Oro City, Misamis Oriental, 9000, Philippines

Re: **Engr. Elena C. Susaya**
RSL Chief Engineer, Department of Agriculture - Region 10

Subject: Request for Borrowing Soil Test Kit (STK) for Calibration

Good day, Ma'am/ Sir!

We, the 4th Year students of Bachelor of Science in Computer Engineering from the University of Science and Technology of Southern Philippines – Cagayan de Oro Campus, are currently conducting our thesis titled **“TANIM: An Intelligent Agricultural Framework for Crop Recommendation and Soil Health Monitoring.”**

As part of our thesis, we would like to **borrow a Soil Test Kit** from the Department of Agriculture Region 10 this February, for the recalibration of our hardware components to obtain the desired accuracy of our hardware components. The purpose of this recalibration is to ensure that we can provide the accurate data to be analyzed on our machine learning model so that our system is reliable in real-world scenarios.

We assure you that all information obtained will be used solely for academic purposes and will be treated with the utmost confidentiality. Should you require further details, please feel free to contact us through apuya.christianjames@ustp.edu.ph or **09161513497**.

We sincerely hope for your kind approval of our request. Your cooperation and support would greatly contribute to the success of our research. The Department of Agriculture - Region 10 will be acknowledged on our paper and we will be glad to share the output of this research to the Department of Agriculture - Region 10

Thank you very much for your time and consideration.

Respectfully yours,


Christian James E. Apuya
Researcher, BS in Computer in Engineering
USTP - CDO


Albert John C. Diaz
Researcher, BS in Computer in Engineering
USTP - CDO


Robert Roy P. Salvo
Researcher, BS in Computer in Engineering
USTP - CDO

Noted by:

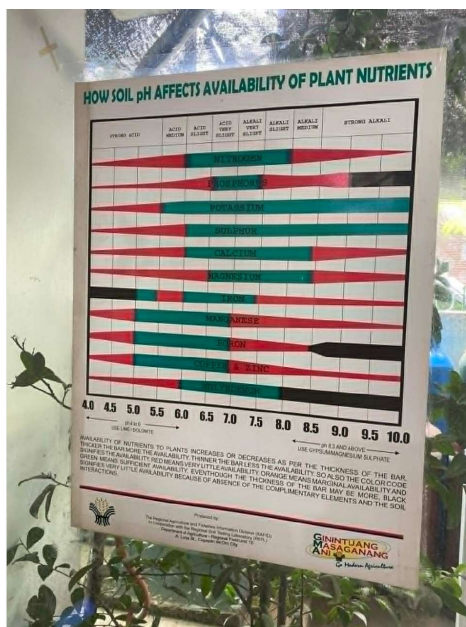

Engr. Jodie Rey D. Fernandez
Research Adviser, Faculty in Computer Engineering
USTP - CDO


Ray Simon L. Bantaculo
Researcher, BS in Computer in Engineering
USTP - CDO


Alex U. Guardaquivil Jr.
Researcher, BS in Computer in Engineering
USTP - CDO

APPENDIX G

DOCUMENTS REVIEWED



NITROGEN TEST

- In a clean and dry test tube, transfer soil sample up to first scratch mark (approximately 0.5 gram soil sample).
- Fill the test tube with N reagent up to the 2nd scratch mark (approximately 1 mL).
- Mix well by gently tapping the test tube into palm for 1 minute.
- Stand the test tube for 5 minutes. Repeat step 3.
- Heat the bottom of the test tube for about 30 seconds or until the solution starts bubbling.
- Stand the test tube for 2 minutes.
- Using a clean and white background, incline the test tube and match the color of the resulting solution with the colors in Figure 2.

Figure 2. Color chart for Nitrogen (% Organic Matter)

PHOSPHORUS TEST

- In a clean and dry test tube, transfer soil sample up to first scratch mark (approximately 0.5 gram soil sample).
- Fill the test tube with P reagent up to the second scratch mark (approximately 1 mL) then add 4 drops of P1 reagent.
- Mix well by gently tapping the test tube into your palm for 1 minute.
- Stand the test tube for 3 minutes. Repeat step 3.
- Stand the test tube for 5 minutes.
- Attach the tin foil to the stirring stick. Without disturbing the soil, stir solution for 1 minute.
- Stand the test tube for 2 minutes then stir for another 1 minute using tin foil.
- Using a clean and white background, incline the test tube and match the color of the resulting solution with the colors in Figure 3.

NOTE: The tin foil attached to the stick can be used up to four (4) times, provided that the analyses are done on the same day. Rinse the tin foil with distilled water after each analysis.

Figure 3. Color chart for Phosphorus

POTASSIUM TEST

- In a clean and dry test tube, transfer soil sample up to first scratch mark (approximately 0.5 gram soil sample).
- Fill the test tube with K reagent up to the 2nd scratch mark (approximately 1 mL).
- Mix well by gently tapping the test tube into your palm for about 1 minute.
- Stand the test tube for 1 minute. Repeat step 3.
- Stand the test tube for 5 minutes.
- In a separate test tube, fill with K1 reagent up to the first scratch mark (approximately 0.5 mL) and K2 up to the second scratch mark (approximately 1 mL).
- Mix well by gently tapping the test tube into your palm for 30 seconds.
- Transfer aliquot from first test tube up to the third scratch mark (approximately 1 mL).
- Mix well by gently tapping the test tube into your palm for 1 minute.
- Using the provided purple background, observe the formation of a turbid orange solution. Match the turbidity of the solution in Figure 4.

Figure 4. Chart for Potassium

SOIL pH

- In a clean and dry test tube, transfer soil sample up to first scratch mark (approximately 0.5 gram soil sample).
- Fill the test tube with CPR reagent up to the 2nd scratch mark (approximately 1 mL).
- Mix well by gently tapping the test tube into your palm for 1 minute.
- Stand the test tube for 2 minutes. Repeat step 3.
- Stand the test tube for 5 minutes.
- Using a clean and white background, incline the test tube and match the resulting solution with the colors for CPR in Figure 1.
- If soil pH is greater than 5.8, repeat steps 1 to 6 using BTB instead of CPR. However, if soil pH is less than or equal to 5.4, repeat steps 1 to 6 using BCG instead of CPR.

NOTES:

- If the resulting solution matches one of the BTB or BCG colors, record this as the final pH reading. If the resulting solution does not match any of the BTB or BCG colors, refer to the initial result of the CPR. Record this as the final pH reading.
- If the color obtained is in between two succeeding values in the color chart, get the average and record this as the final pH reading.

Figure 1. Color chart for Soil pH

CROP AND/OR VARIETY	NITROGEN			PHOSPHORUS			POTASSIUM		
	L	M	H	L	M	H	L	M	H
VEGETABLES									
AMPALAYA/UPO	10	50	40	80	50	20	60	45	30
BATAT	45	40	35	60	35	20	60	45	30
BEANS									
String bean	50	40	35	90	45	20	50	40	30
Bugo	45	40	35	60	35	20	60	45	30
Lima (Peanut)	45	40	35	60	35	20	60	45	30
Winged (egg-bean)									
Dwarf variety	40	30	20	40	25	10	45	30	30
Tree type	100	40	30	30	20	10	45	30	20
CHAYOTE	60	50	40	100	50	20	60	45	30
CUCUMBER	100	80	60	100	60	20	60	45	30
PANRIP	100	80	60	100	60	20	60	45	30
PATANI	90	60	45	60	35	20	60	45	30
PATOLA	45	40	35	60	35	20	60	45	30
PINE	90	60	30	80	50	20	60	45	30
PEANUT	20	10	5	40	20	10	45	30	20
CABBAGE	90	80	70	100	60	20	120	75	30

L: Low 20 Medium 40 High: 60

[illegible]

CROP AND THE VARIETY		BEE		BEE		BEE		BEE		BEE		BEE	
		L	M	P	L	M	P	L	M	P	L	M	P
GENERAL													
WINTER													
W. D. D. D.	125	90	60	70	50	30	20	10	10	10	10	10	10
W. D. D. D.	140	110	80	70	50	30	20	10	10	10	10	10	10
WINTER													
W. D. D. D.													
W. D. D. D.	110	80	50	70	50	30	20	10	10	10	10	10	10
W. D. D. D.	130	90	60	70	50	30	20	10	10	10	10	10	10
PLANS													
Light Duty	90	60	70	50	30	20	10	10	10	10	10	10	10
Mid Heavy Duty	100	70	60	70	50	30	20	10	10	10	10	10	10

1. 1000 2. 1000 3. 1000 4. 1000 5. 1000 6. 1000 7. 1000 8. 1000 9. 1000 10. 1000 11. 1000 12. 1000 13. 1000 14. 1000 15. 1000 16. 1000 17. 1000 18. 1000 19. 1000 20. 1000 21. 1000 22. 1000 23. 1000 24. 1000 25. 1000 26. 1000 27. 1000 28. 1000 29. 1000 30. 1000 31. 1000 32. 1000 33. 1000 34. 1000 35. 1000 36. 1000 37. 1000 38. 1000 39. 1000 40. 1000 41. 1000 42. 1000 43. 1000 44. 1000 45. 1000 46. 1000 47. 1000 48. 1000 49. 1000 50. 1000 51. 1000 52. 1000 53. 1000 54. 1000 55. 1000 56. 1000 57. 1000 58. 1000 59. 1000 60. 1000 61. 1000 62. 1000 63. 1000 64. 1000 65. 1000 66. 1000 67. 1000 68. 1000 69. 1000 70. 1000 71. 1000 72. 1000 73. 1000 74. 1000 75. 1000 76. 1000 77. 1000 78. 1000 79. 1000 80. 1000 81. 1000 82. 1000 83. 1000 84. 1000 85. 1000 86. 1000 87. 1000 88. 1000 89. 1000 90. 1000 91. 1000 92. 1000 93. 1000 94. 1000 95. 1000 96. 1000 97. 1000 98. 1000 99. 1000 100. 1000

[illegible]

Further Recommendations for Various Crops - continued

CROP AND GRADE/VEGETY		METROCK			PROMPERGUE			PUT KEEVIN				
FEED TYPE		L	M	H	L	M	H	L	M	H		
HAY/ALFALFA												
Hay	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (2nd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (3rd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (4th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
CORN												
Barnyard Corn (2nd year)												
Hay	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (2nd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (3rd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (4th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (5th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (6th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (7th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (8th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (9th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (10th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (11th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (12th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (13th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (14th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (15th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (16th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (17th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (18th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (19th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (20th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (21st year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (22nd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (23rd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (24th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (25th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (26th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (27th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (28th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (29th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (30th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (31st year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (32nd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (33rd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (34th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (35th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (36th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (37th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (38th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (39th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (40th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (41st year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (42nd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (43rd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (44th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (45th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (46th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (47th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (48th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (49th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (50th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (51st year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (52nd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (53rd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (54th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (55th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (56th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (57th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (58th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (59th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (60th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (61st year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (62nd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (63rd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (64th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (65th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (66th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (67th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (68th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (69th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (70th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (71st year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (72nd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (73rd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (74th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (75th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (76th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (77th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (78th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (79th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (80th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (81st year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (82nd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (83rd year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (84th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (85th year)	0.01	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
Hay (86th year)	0.01	0.001	0.0									

[illegible][illegible][illegible]

APPENDIX H

INTERVIEW QUESTIONS

Table 23: Interview with DA Senior Agriculturist

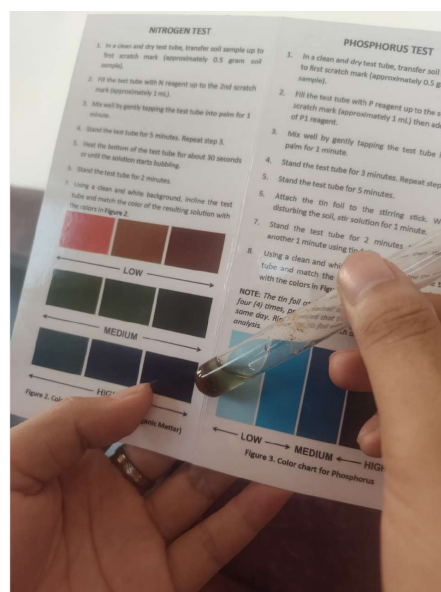
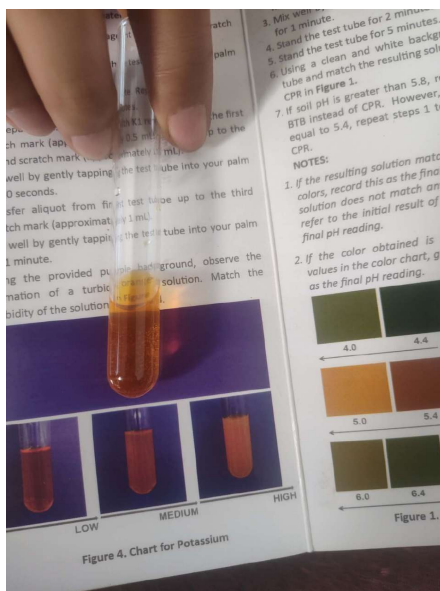
Category	Item	Details
Interview Information		
Info	Interviewers	Apuya, Christian James E.; Bantaculo, Ray Simon; Diaz, Albert Jhon; Guardaquivil, Alex Jr; Salvo, Robert Roy
Info	Location	Department of Agriculture, Regional Field Office 10
Info	Date	January 27, 2026; February 23, 2026; March 13, 2026
Info	Name	Karl Roman Wong
Info	Address	DA-10
Info	Sex	Male
Info	Occupation	DA Senior Agriculturist
Interview Questions and Responses		
Q1	How do farmers know the soil nutrients in their agricultural lands?	Ang pag test sa yuta naa tay duha ka methods which is ang Laboratory Results Testing and the Soil Test Kits (STK) for testing, and the output lang ga differ. Sa Laboratory is naa gyud didto ang quantitative nga result sa pH, and NPK, then sa STK purely qualitative to based from the color nga result which is LOW, MEDIUM, HIGH.
Q2	If ever magpa test ang farmer, unsa ilang method for both ways?	Pag STK lang ang gusto sa farmer, need lang nila ug agreement with the DA to borrow the STK. Meanwhile, sa laboratory results, they will need to transport or bring ilang soil sample diri sa atung laboratory para ma test ilang soil pero naa lang payment since ga involve na siya ug instruments.
<i>Continued on next page</i>		

Category	Item	Details
Q3	What is the proper method in order to test a soil sample?	Per hectare mana sila gapa test then need lang nila mag kalut ug atleast 3 ka bangag for atleast 12 inches or 1 foot, and didto kuha sila ug soil sample then mix it with the other soil sample from other holes. Pero ga depende ang ka lalom sa crops nga itanom nila kay naa man tay classification sa crops which is shallow rooted and deep rooted nga mga crops or plants. Pero sa part ninyo focus raman mo sa mga shallow rooted. Also, naa pud tay classification if low-land or highland ang lugar, but since you mentioned nga bukidnon mo so most of it is highland lang siya.
Q5	How often should our system monitor soil conditions to provide reliable crop recommendations?	Kung laboratory testing, okay ra siya for 1–2 years, pero kung gusto gyud updated ang data, mas maayo every cropping cycle, which is pre and post planting.
Q6	What type of data should be collected to improve its crop recommendation accuracy?	Kinahanglan naay baseline data sa soil ug crops. Pareha sa among ginagamit nga apps, ang data gikan sa field trials ug techno demo. Importante pud ang historical data para mas accurate ang recommendations.
Q7	Which crops should TANIM prioritize when recommending based on soil conditions?	Sa among region, priority ang rice ug corn ug cassave, pero dapat various crops maapil kay lain-lain man silag nutrient requirements.
Q8	What challenges might be face in terms of farmer adoption of soil monitoring systems?	Bisan naa nay mga apps karon, gamay ra ang nag adopt. Lisod pud usahay kay ang farmers naanad na sa ilang sariling pamaagi. Kinahanglan ang system kay simple ug dali masabtan.
<i>Continued on next page</i>		

Category	Item	Details
Q9	How do farmers currently decide on fertilizer application, and how can Fertright improve this process?	Kasagaran sa farmers kay experience-based ra ilang pag apply ug fertilizer. Usahay dili sila musunod sa recommendations kay naanad na sila sa ilang style. Makatabang ang Fertright kung makahatag siya ug klaro, sakto, ug masaligan nga guide.
Q10	How can Fertright ensure that its fertilizer recommendations are practical for farmers?	Dapat ang recommendations kay precise ug accurate. Dili dapat komplikado kay ang farmers ganahan ug simple nga instructions nga dali masunod.
Q11	What features should be included to be usable for farmers in rural areas?	Importante nga naa siyay offline functionality kay daghan farmers walay internet. Dapat pud user-friendly ug dili lisod gamiton.
Q12	What improvements in fertilizer management to promote sustainable farming?	Pwede siya makatabang sa balanced fertilization, paggamit ug organic fertilizers, ug soil conservation practices. Importante ni para magdugay ang kaayo sa yuta ug mudako ang ani.

APPENDIX I

CALIBRATION OF SOIL SENSOR USING SOIL TESTING KIT



APPENDIX J

DOCUMENTATION OF THE IMPLEMENTATION



APPENDIX K

FUNCTIONAL AND USABILITY TESTING CONDUCTED

SYSTEM USABILITY SCALE FORM

Name(optional): Wilson
 Date: April 23, 2024

System Usability Scale (Department of Agriculture)						
No.	Item	Strongly Disagree			Strongly Agree	
		1	2	3	4	5
1	I think that I would like to use the TANIM web frequently for regional soil health monitoring.					/
2	I found the dashboard unnecessarily complex for managing farms, farmer profiles and zoning maps.	/				
3	I thought the system was easy to use for visualizing nutrient trends across Bukidnon.				/	
4	I think that I would need the support of an IT specialist to monitor and update the web application.	/				
5	I found the integration between the IoT sensor data and the GIS mapping to be very effective.					/
6	I thought there was too much inconsistency in the data reporting between different farm zones.	/				
7	I would imagine that agricultural technicians would learn to use this system very quickly.				/	
8	I found the web interface difficult to navigate when searching for specific farmer data.		/			
9	I felt very confident using the system to generate regional agricultural reports.				/	
10	I needed to learn a lot of things before I could effectively manage the system's administrative features.	/				

SYSTEM USABILITY SCALE FORM

Name(optional): Xai Gobitang
 Date: April 23, 2024

System Usability Scale (Department of Agriculture)						
No.	Item	Strongly Disagree			Strongly Agree	
		1	2	3	4	5
1	I think that I would like to use the TANIM web frequently for regional soil health monitoring.					/
2	I found the dashboard unnecessarily complex for managing farms, farmer profiles and zoning maps.	/				
3	I thought the system was easy to use for visualizing nutrient trends across Bukidnon.					/
4	I think that I would need the support of an IT specialist to monitor and update the web application.		/			
5	I found the integration between the IoT sensor data and the GIS mapping to be very effective.				/	
6	I thought there was too much inconsistency in the data reporting between different farm zones.		/			
7	I would imagine that agricultural technicians would learn to use this system very quickly.					/
8	I found the web interface difficult to navigate when searching for specific farmer data.		/			
9	I felt very confident using the system to generate regional agricultural reports.				/	
10	I needed to learn a lot of things before I could effectively manage the system's administrative features.	/				

SYSTEM USABILITY SCALE FORM

Name(optional): Karl Wong
 Date: April 23, 2020

System Usability Scale (Department of Agriculture)						
No.	Item	Strongly Disagree			Strongly Agree	
		1	2	3	4	5
1	I think that I would like to use the TANIM web frequently for regional soil health monitoring.					/
2	I found the dashboard unnecessarily complex for managing farms, farmer profiles and zoning maps.		/			
3	I thought the system was easy to use for visualizing nutrient trends across Bukidnon.				/	
4	I think that I would need the support of an IT specialist to monitor and update the web application.	/				
5	I found the integration between the IoT sensor data and the GIS mapping to be very effective.					/
6	I thought there was too much inconsistency in the data reporting between different farm zones.		/			
7	I would imagine that agricultural technicians would learn to use this system very quickly.				/	
8	I found the web interface difficult to navigate when searching for specific farmer data.		/			
9	I felt very confident using the system to generate regional agricultural reports.					/
10	I needed to learn a lot of things before I could effectively manage the system's administrative features.	/				

SYSTEM USABILITY SCALE FORM

Name(optional): Ben
 Date: April 21, 2024

System Usability Scale (Farmers)						
No.	Item	Strongly Disagree Strongly Agree				
		1	2	3	4	5
1	I think that I would like to use the TANIM mobile app frequently for my farming activities.				/	
2	I found the TANIM mobile app unnecessarily complex to use.		/			
3	I thought the app was easy to use for checking my soil's NPK and pH levels.					/
4	I think that I would need the support of a technical person to understand the soil parameters and the crop recommendations.			/		
5	I found the various functions (weather, soil health, and fertilizer schedules) were well integrated.				/	
6	I thought there was too much inconsistency in how the soil health data (NPK, pH, moisture, temperature, and salinity) and recommendations were displayed.		/			
7	I would imagine that most farmers would learn to use this app very quickly.			/		
8	I found the process of viewing my farm very complex.	/				
9	I felt very confident using the app to make decisions about which crop and fertilizer to apply.			/		
10	I need to learn a lot of things before I can use this system.			/		

SYSTEM USABILITY SCALE FORM

Name(optional): Apolinario
 Date: April 20, 2026

System Usability Scale (Farmers)						
No.	Item	Strongly Disagree			Strongly Agree	
		1	2	3	4	5
1	I think that I would like to use the TANIM mobile app frequently for my farming activities.					/
2	I found the TANIM mobile app unnecessarily complex to use.	/				
3	I thought the app was easy to use for checking my soil's NPK and pH levels.					/
4	I think that I would need the support of a technical person to understand the soil parameters and the crop recommendations.		/			
5	I found the various functions (weather, soil health, and fertilizer schedules) were well integrated.				/	
6	I thought there was too much inconsistency in how the soil health data (NPK,pH, moisture, temperature, and salinity) and recommendations were displayed.		/			
7	I would imagine that most farmers would learn to use this app very quickly.				/	
8	I found the process of viewing my farm very complex.	/				
9	I felt very confident using the app to make decisions about which crop and fertilizer to apply.					/
10	I need to learn a lot of things before I can use this system.		/			

SYSTEM USABILITY SCALE FORM

Name(optional): John Mark
 Date: April 21, 2020

System Usability Scale (Farmers)						
No.	Item	Strongly Disagree			Strongly Agree	
		1	2	3	4	5
1	I think that I would like to use the TANIM mobile app frequently for my daily farming activities.					/
2	I found the mobile app unnecessarily complex to navigate.	/				
3	I thought the app was easy to use for checking my soil's NPK and pH levels.				/	
4	I think that I would need the support of a technical person to understand the crop recommendations.		/			
5	I found the various functions (weather, soil health, and fertilizer schedules) were well integrated.				/	
6	I thought there was too much inconsistency in how the recommendations were displayed.	/				
7	I would imagine that most farmers would learn to use this app very quickly.					/
8	I found the process of viewing my farm's		/			
9	I felt very confident using the app to make decisions about which fertilizer to apply.					/
10	I needed to learn a lot of things before I could get going with this system.		/			

SYSTEM USABILITY SCALE FORM

Name(optional): Magnu
 Date: April 21, 2026

System Usability Scale (Farmers)						
No.	Item	Strongly Disagree			Strongly Agree	
		1	2	3	4	5
1	I think that I would like to use the TANIM mobile app frequently for my farming activities.				/	
2	I found the TANIM mobile app unnecessarily complex to use.		/			
3	I thought the app was easy to use for checking my soil's NPK and pH levels.				/	
4	I think that I would need the support of a technical person to understand the soil parameters and the crop recommendations.		/			
5	I found the various functions (weather, soil health, and fertilizer schedules) were well integrated.			/		
6	I thought there was too much inconsistency in how the soil health data (NPK, pH, moisture, temperature, and salinity) and recommendations were displayed.	/				
7	I would imagine that most farmers would learn to use this app very quickly.				/	
8	I found the process of viewing my farm very complex.	/				
9	I felt very confident using the app to make decisions about which crop and fertilizer to apply.				/	
10	I need to learn a lot of things before I can use this system.	/				

SYSTEM USABILITY SCALE FORM

Name(optional): Tata
 Date: April 20, 2020

System Usability Scale (Farmers)						
No.	Item	Strongly Disagree			Strongly Agree	
		1	2	3	4	5
1	I think that I would like to use the TANIM mobile app frequently for my farming activities.				/	
2	I found the TANIM mobile app unnecessarily complex to use.		/			
3	I thought the app was easy to use for checking my soil's NPK and pH levels.				/	
4	I think that I would need the support of a technical person to understand the soil parameters and the crop recommendations.		/			
5	I found the various functions (weather, soil health, and fertilizer schedules) were well integrated.					/
6	I thought there was too much inconsistency in how the soil health data (NPK, pH, moisture, temperature, and salinity) and recommendations were displayed.		/			
7	I would imagine that most farmers would learn to use this app very quickly.				/	
8	I found the process of viewing my farm very complex.		/			
9	I felt very confident using the app to make decisions about which crop and fertilizer to apply.				/	
10	I need to learn a lot of things before I can use this system.	/				

SYSTEM USABILITY SCALE FORM

Name(optional): K - Jay
 Date: April 21, 2024

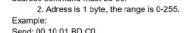
System Usability Scale (Farmers)						
No.	Item	Strongly Disagree			Strongly Agree	
		1	2	3	4	5
1	I think that I would like to use the TANIM mobile app frequently for my farming activities.					/
2	I found the TANIM mobile app unnecessarily complex to use.		/			
3	I thought the app was easy to use for checking my soil's NPK and pH levels.					/
4	I think that I would need the support of a technical person to understand the soil parameters and the crop recommendations.	/				
5	I found the various functions (weather, soil health, and fertilizer schedules) were well integrated.					/
6	I thought there was too much inconsistency in how the soil health data (NPK,pH, moisture, temperature, and salinity) and recommendations were displayed.	/				
7	I would imagine that most farmers would learn to use this app very quickly.					/
8	I found the process of viewing my farm very complex.					
9	I felt very confident using the app to make decisions about which crop and fertilizer to apply.					/
10	I need to learn a lot of things before I can use this system.			/		

SYSTEM USABILITY SCALE FORM

Name(optional): Eldito P. Laguring
 Date: April 21, 2020

System Usability Scale (Farmers)						
No.	Item	Strongly Disagree Strongly Agree				
		1	2	3	4	5
1	I think that I would like to use the TANIM mobile app frequently for my farming activities.					/
2	I found the TANIM mobile app unnecessarily complex to use.		/			
3	I thought the app was easy to use for checking my soil's NPK and pH levels.				/	
4	I think that I would need the support of a technical person to understand the soil parameters and the crop recommendations.		/			
5	I found the various functions (weather, soil health, and fertilizer schedules) were well integrated.				/	
6	I thought there was too much inconsistency in how the soil health data (NPK, pH, moisture, temperature, and salinity) and recommendations were displayed.		/			
7	I would imagine that most farmers would learn to use this app very quickly.				/	
8	I found the process of viewing my farm very complex.		/			
9	I felt very confident using the app to make decisions about which crop and fertilizer to apply.					/
10	I need to learn a lot of things before I can use this system.		/			

NiuBo



NiuBo

the wiring method, then insert the sensor probe

the testing procedure, chest inserts are drilled probe

APPENDIX M
TURNITIN RESULTS

APPENDIX N
GRAMMARIAN RESULTS